Information Processing and Management 62 (2025) 104287

Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/infoproman

A systematic mapping study of information retrieval-based requirements traceability methods

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ARTICLE INFO

Keywords:
Requirements traceability
Systematic mapping study
Information retrieval

ABSTRACT

Requirements traceability (RT) is critical for ensuring consistency, quality, and maintainability in software development. While learning-based approaches have gained increasing attention, traditional information retrieval (IR) methods remain widely used in practice. However, existing literature lacks a systematic synthesis of their best practices and recent advancements. To address this gap, we conducted a systematic mapping study (SMS) of 40 primary studies published between 2014 and 2024, selected from an initial pool of 2,052 publications. Our review examines widely adopted IR models, enhancement strategies, evaluation datasets, performance metrics, and baseline methods. Specifically, we identify and categorize 32 representative enhancement strategies into four methodological types: (1) artifact text information, (2) artifact structural information, (3) model-based optimization, and (4) human intervention. Furthermore, we analyze 53 commonly used datasets and 9 evaluation metrics for validation. Our findings indicate that among various IR models, the Vector Space Model (VSM) and Latent Semantic Indexing (LSI) typically achieve stronger performance in RT tasks. This study provides a comprehensive synthesis of IR-based RT research and offers practical insights to advance traceability in software engineering.

1. Introduction

Requirements traceability (RT) refers to the ability to trace and meticulously document the entire lifecycle of requirements, including their origin, development, specification, deployment, usage, and any subsequent refinements throughout these phases (Gotel & Finkelstein, 1994). As shown in Fig. 1, it is a specialized subset of software traceability (ST) that defines the process of establishing and maintaining links between source and target artifacts through both forward and backward tracing across the software lifecycle (Gotel & Finkelstein, 1994). If RT is neglected or incomplete and inconsistent RT links are used in the software development lifecycle, it may lead to degradation of system quality and iterative modifications (Cleland-Huang et al., 2014). High-quality trace links are crucial for supporting various activities in the software development lifecycle (Cleland-Huang et al., 2014), such as analyzing the impact of requirement changes (Wang et al., 2018), ensuring system functional integrity, verifying that product designs meet their intended purposes (verification and validation, V&V) (Antoniol et al., 2002), evaluating requirements coverage, and assisting developers in reusing system components (Lucia et al., 2007), among others.

Various technologies have been developed to address RT, including ontology-based methods, machine learning-based (ML-based)

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https://doi.org/10.1016/j.ipm.2025.104287
Received 5 March 2025; Received in revised form 29 June 2025; Accepted 30 June 2025
Available online 11 July 2025
0306-4573/

methods, deep learning-based (DL-based) methods and information retrieval-based (IR-based) methods (Oliveto et al., 2010). While ontology-based methods excel in semantic analysis, they heavily depend on human expertise and project-specific ontologies, making them difficult to generalize across projects (Wang et al., 2018). In contrast, ML-based methods rely on large-scale training datasets, which are often unavailable in real-world software projects due to insufficient traceability links (Wang et al., 2018). Additionally, in agile development environments, the shrinking size of artifacts further limits the effectiveness of ML methods. With the rapid advancement of natural language processing (NLP) technologies, DL-based tracing methods, such as large language models (LLMs) and pre-trained models, have gained significant attention. These models hold the potential to automatically learn and recover traceability links with minimal manual intervention, offering improvements in both accuracy and automation.

Despite the emergence of these advanced techniques, traditional IR-based methods remain vital in RT (Gao et al., 2023), particularly in the era of large models. Their advantages in efficiency, interpretability, and structured data processing make them indispensable in requirements tracing. To further investigate the ongoing relevance of traditional IR methods in RT, we conducted a systematic mapping study (SMS) that analyzes the synergies between RT and IR. Our SMS generates a comprehensive research roadmap, synthesized from prior studies on the application of IR techniques in RT. This roadmap aims to outline and guide future research at the intersection of RT and IR. The key contributions of this study are as follows:

- An in-depth review of IR-based RT models, enhancement strategies, datasets, and evaluation measures, highlighting key advancements in the field between 2014 and 2024.
- Directly usable baseline data that serves as a foundation for future research and practice, providing researchers and practitioners with essential insights into the evolving field of RT.
- Identification of critical limitations and gaps in current IR-based RT methods, along with an outline of key directions for future work.

The organization of this paper is as follows: Section 2 examines related work, Section 3 outlines the research methodology, Section 4 presents the essential findings from our SMS, and Section 5 discusses potential threats to validity and provides an overview of future research directions. Finally, Section 6 concludes the study.

2. Related work

Table 1 shows that numerous surveys have been conducted over the past decades to investigate advancements in both general RT methods and IR-based traceability approaches. However, a closer examination reveals that these studies exhibit several limitations and fail to fully capture the latest developments or practical demands of the field.

Earlier studies such as Borg et al. (2014) established important foundations by examining pre-2013 IR techniques, but their findings have become outdated in light of recent advancements in IR. Similarly, while Saleem and Minhas (2018) provided specialized insights into term mismatch resolution, their narrow technical focus left other critical aspects of IR-based RT unexplored. Both studies do not reflect recent advances in IR techniques, benchmark datasets, or evaluation measures, which are essential for understanding the current state of the field. More recent work by Li et al. (2023) demonstrates a growing interest in ML approaches but notably excludes traditional IR methods, creating an incomplete picture of contemporary RT solutions.

Furthermore, broader surveys including Nair et al. (2013), Torkar et al. (2012), and Wang et al. (2018) suffer from both temporal limitations (ending in 2012, 2007, and 2016, respectively) and a lack of focus on IR-specific techniques, while generally failing to provide the standardized benchmarks and empirical baselines needed for practical implementation.

These limitations collectively highlight the need for a more comprehensive and up-to-date synthesis that addresses both theoretical and practical dimensions of IR-based RT. Our study directly addresses these gaps through a systematic examination of literature published between 2014 and 2024, offering not only a structured taxonomy of enhancement strategies and a comprehensive analysis of evaluation datasets and metrics, but also reproducible experimental baselines and implementation guidelines that bridge the gap between research and practice. By combining rigorous academic analysis with reusable experimental resources, this work significantly advances the field's understanding while providing immediately applicable tools for both researchers and practitioners.

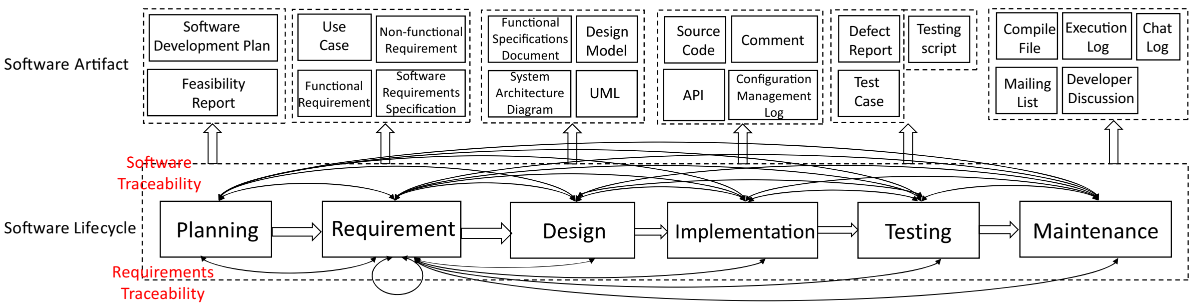

Fig. 1. Overview of ST and RT across the software lifecycle (Gotel & Finkelstein, 1994).

Table 1
Related work.

	Contribution	Scope	Type	Literature delimitation	Scale	Interval
Saleem and Minhas (2018)	(1) Examining the IR-based RT methods and presenting a synthesis of them; (2) Identifying potential barriers hindering the adoption of RT methods based on IR; (3) Categorizing the methods to address the issue of term mismatch.	IR-based RT	Review	(1) ACM Digital Library (2) ScienceDirect (3) Elsevier (4) Springer Link (5) IEEE Xplore	57 papers	2001–2018
Borg et al. (2014)	(1) Presenting an overview of IR models and strategies for enhancing performance; (2) Recognizing linked software artifacts within IR models; (3) Organizing the datasets utilized in the original publications for evaluation purposes.	IR-based ST	SMS	(1) EI Compendex (2) Web of Science (3) Sci Verse hub beta (4) ACM Digital Library (5) Inspec (6) IEEE Xplore	79 papers	2003–2013
Torkar et al. (2012)	(1) Organizing definitions related to RT; (2) Investigating the challenges, tools, and methods associated with RT; (3) Uncovering the factors that impede the adoption of RT.	RT	Systematic Literature Review (SLR)	(1) ACM Digital Library (2) Inspec (3) Springer Link (4) IEEE Xplorer (5) EI Compendex	52 papers	1997–2007
Nair et al. (2013)	(1) Organizing research topics and challenges related to RT; (2) Grouping the contributions of the papers based on the evaluation methods and their efforts in tackling challenges; (3) Identifying the types of trace artifacts, features of trace tools, and the systems.	RT	Review	RE Conference	70 papers	1993–2012
Li et al. (2023)	(1) Identification of ML models and strategies for improvement; (2) Evaluation of the quality of primary studies; (3) Identification of requirement artifacts connected by ML models.	ML-based RT	SMS	(1) Springer Link (2) ACM Digital Library (3) ScienceDirect (4) IEEE Xplore (5) EI Compendex	26 papers	2013–2022
Wang et al. (2018)	(1) Organizing and categorizing the challenges related to RT; (2) Identifying the various RT technologies available; (3) Evaluating the extent of support provided for technology transfer of these RT technologies.	RT	SLR	(1) ACM Digital Library (2) ScienceDirect (3) IEEE Xplorer (4) Springer Link (5) EI Compendex	114 papers	2006–2016
This work	(1) Sorting out models and strategies used for IR-based RT methods; (2) Identifying datasets and measures utilized to evaluate IR-based RT methods; (3) Providing the baselines applied to verify IR-based RT methods or other automated RT methods.	IR-based RT	SMS	(1) EI Compendex (2) Springer Link (3) ACM Digital Library (4) Google Scholar (5) IEEE Xplore (6) ScienceDirect	40 papers	2014–2024

3. Research method

SMS provide a rigorous methodology for conducting comprehensive research domain surveys through systematic and verifiable procedures (Petersen et al., 2015). While SLRs primarily synthesize evidence to address specific research questions (RQs), such as evaluating technique efficacy under controlled conditions, a SMS is better suited for providing a structured overview of broad research domains. Given our objective to systematically categorize literature on IR-based RT methods and outline advancements at the intersection of IR and RT, we adopt SMS as our foundational methodology. This approach facilitates the systematic classification of IR models, enhancement strategies, datasets, and evaluation methodologies, while also identifying emerging trends and research gaps in this rapidly evolving field.

To implement this methodology, we adhere to two primary guidelines: (1) the SMS-specific framework outlined in (Petersen et al., 2015), and (2) the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines for reporting systematic reviews (Page et al., 2021). By synthesizing the research landscape cohesively, our study delivers actionable insights to guide future research and advance collective knowledge in the field.

3.1. Research questions

Numerous studies on IR-based RT methods have been conducted between 2014 and 2024. However, a comprehensive summary of their advancements is still lacking. To address this gap, we formulate eight RQs to investigate and understand the current state of IR-based RT. These RQs are outlined in Table 2.

3.2. Search process

Following the SMS guidelines (Page et al., 2021; Petersen et al., 2015), we establish a protocol to provide guidance during the search process. This protocol ensures a systematic and comprehensive method during the retrieval process. The automatic retrieval of relevant papers is conducted from the databases.

3.2.1. Eligibility criteria

After obtaining potentially relevant studies, it is necessary to assess their actual relevance. Following the SMS guidelines (Page et al., 2021; Petersen et al., 2015), we establish criteria for inclusion and exclusion.

Inclusion criteria:

- I1: The study period is from January 2014 to December 2024.
- I2: The study needs to be written in English, and the topic of study must be IR-based RT.
- I3: Only papers published in conferences or journals listed in CCF, CORE, and QUALIS are considered.

Exclusion criteria:

- E1: The studies that are not relevant to the focus of this research.
- E2: The study which is a review paper or a thesis.
- E3: If the same author presents two papers on the same technology and topic, the paper with the less comprehensive description will be excluded.

3.2.2. Search scope

Period. The scope of this SMS is limited to studies published between January 2014 and December 2024.

Electronic databases. In alignment with the recommendations from (Petersen et al., 2015) and considering the access permissions of our institution, the databases listed in Table 3 are chosen as the principal sources for this study.

3.2.3. Search strategy

Based on the SMS guidelines (Page et al., 2021; Petersen et al., 2015), we employ the Population, Intervention, Comparison, and Outcome (PICO) criteria to establish the search terms used in database queries. Table 4 presents the search terms employed for the population and intervention components. Boolean operators are applied to link the key terms and their synonyms when constructing the search string. The final search string is as follows:

("requirements traceability" OR "requirements trace" OR "requirements tracing" OR "traceability recovery" OR "trace assessment" OR "trace maintenance") AND ("information retrieval" OR "IR" OR "semantic")

Population: The population for this SMS primarily focuses on 'Requirements Traceability'. Synonymous terms such as

Table 2
Research questions of this review.

Research Question	Rationale
RQ1: What are the venues/years of the primary studies?	The study of this issue provides an overview of the trends in IR-based RT methods over the last eleven years, an understanding of the publications in the field.
RQ2: What are the IR models that have been employed in RT?	Exploring this RQ can equip researchers and practitioners with a holistic view of the diverse IR models that are readily applicable for generating trace links.
RQ3: Which enhancement strategies have been applied to IR-based RT?	Enhancement strategies can help achieve higher-quality trace links. This RQ investigates all kinds of enhancement strategies in IR-based RT.
RQ4: Which datasets have been utilized to validate IR-based RT methods?	Datasets are an essential part of the experiments used to validate the performance of the proposed methodology. The response to this RQ can serve as a valuable guideline for researchers when selecting datasets that align with their specific research requirements.
RQ5: Which experimental evaluation methods have been utilized to assess IR-based RT methods?	Different evaluation methods will present different experimental phenomena. This RQ can help select suitable experiment evaluation methods.
RQ6: Which measures have been utilized to assess the performance of IR-based RT methods?	The answer to this RQ provides standard measures for evaluating the accuracy of generating trace links.
RQ7: How do the commonly used IR models perform on RT tasks?	The answer to this RQ provides baselines for trace link generation for commonly used IR models and experience for researchers.
RQ8: What evidence has been documented to validate the performance evaluation of IR-based RT methods?	This RQ aims to assess the credibility and evaluate the quality of the initial analyses by examining the level of evidence presented in each survey regarding the application of IR-based RT techniques.

Table 3
Electronic databases for the automatic search.

No.	Electronic database	Search terms used in
ED1	IEEE Xplore	Title, Abstract, Keywords
ED2	Engineering Village	Title, Abstract, Keywords
ED3	Science Direct	Title, Abstract, Keywords
ED4	Springer	Title, Abstract, Keywords
ED5	ACM Digital Library	Title, Abstract, Full text
ED6	Google Scholar	Title, Abstract, Full text

Table 4
List of search terms in population, intervention.

PI	Search terms
Population (P)	requirements traceability, requirements trace, requirements tracing, traceability recovery, trace assessment, trace maintenance
Intervention (I)	information retrieval, IR, semantic

'Requirements Trace', 'Requirements Tracing', 'Requirements Traceability', 'Traceability Recovery', 'Trace Assessment', and 'Trace Maintenance' are used to define the population in the search terms.

Intervention: The intervention in this SMS is centered around 'information retrieval'. Terms such as 'information retrieval', 'IR', and 'semantic' represent the intervention in the search terms.

Comparison: According to the SMS guidelines (Petersen et al., 2015), since no specific comparative method is defined for this review, the comparison component of the PICO framework is not considered when generating the search terms.

Outcome: In line with the SMS guidelines (Petersen et al., 2015), since no specified outcome is defined for this review, the outcome component of the PICO framework is not included in the search terms.

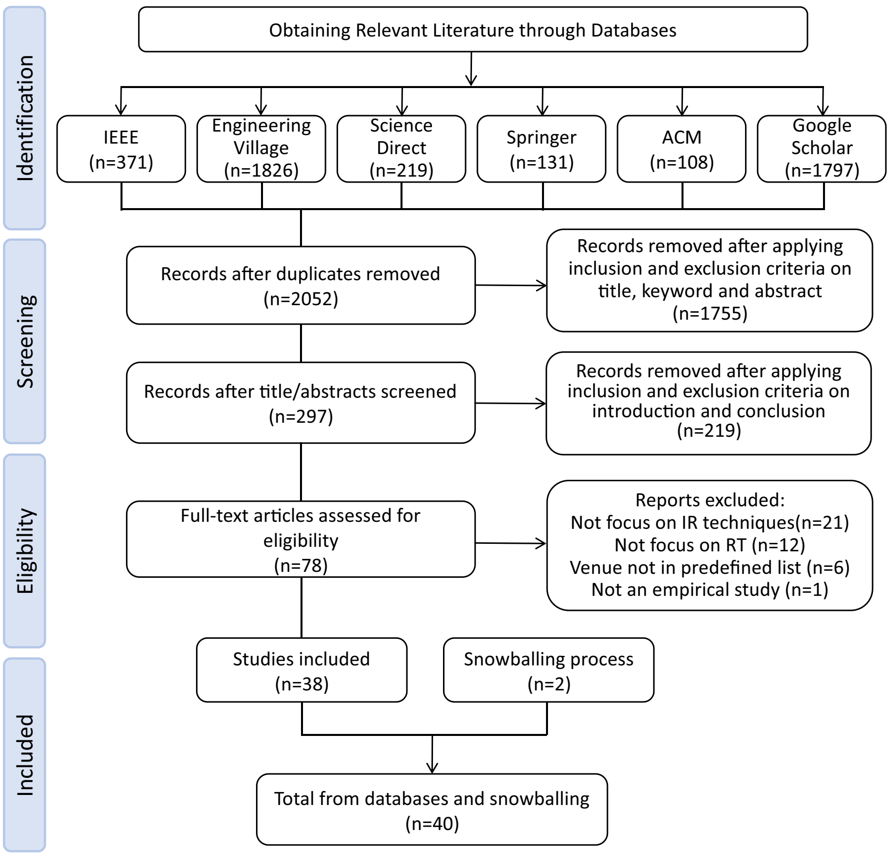

Fig. 2. The process of literature retrieval and selection.

3.2.4. Literature retrieval and selection

Fig. 2 provides an overview of the process of literature retrieval and selection, following the PRISMA flow diagram (Page et al., 2021). The process includes the following stages:

Phase 1: Identification

In this phase, the search terms (as defined in Table 4) are applied exclusively to the title, keywords, and abstract of each paper to minimize irrelevant results. Initial searches retrieve 4,452 records. After removing duplicates, 2,052 unique papers remain for screening.

Phase 2: Screening

The screening process evaluates the 2,052 unique papers based on titles, abstracts and keywords using predefined inclusion and exclusion criteria. This initial screening excludes 1,755 clearly irrelevant papers leaving 297 papers for further assessment. The remaining papers undergo evaluation of their introduction and conclusion sections to verify alignment with the study scope. After detailed evaluation, 219 papers are excluded, resulting in 78 papers advancing to the next phase. To ensure rigorous screening, the first and second authors independently conduct this process. When inclusion/exclusion decisions are unclear based on title, abstract, or keywords, the papers are retained for full-text assessment in the subsequent phase to avoid premature exclusion.

Phase 3: Eligibility

During the full-text assessment, studies are excluded for the following reasons: 1 non-empirical study, 21 papers lacking a significant focus on IR techniques, 12 papers not specifically addressing RT, and 6 papers published in venues not listed in CCF, CORE, or QUALIS rankings. A detailed record of exclusions is provided in Appendix A. This phase identified 38 papers meeting all eligibility criteria. To ensure methodological rigor, the first and second authors independently conduct all full-text assessments using the predefined criteria. Any disagreements or ambiguous cases are resolved through research team discussions until consensus is reached. This rigorous review process enhances the transparency and reliability of study selection.

Phase 4: Included

The final selection comprises 38 papers. Additionally, we employ snowballing techniques (Page et al., 2021) to examine references and citations of these selected papers, which yields 2 additional relevant papers. In total, these 40 papers constitute the primary study set for this SMS, as listed in Appendix B.

3.3. Quality assessment

To ensure the quality of the included studies, a checklist based on the IMRAD structure (Saleem and Minhas, 2018) (Introduction, Methods, Results, Analysis, and Discussion) is employed, as summarized in Table 5. Each primary study is evaluated according to five quality assessment items. For each item, a score of 1 is assigned if the criterion is fully met, and 0 otherwise. Studies that achieve a total score of 3 or more (out of 5) are considered to meet the minimum quality threshold and are included in the final study pool. Based on this procedure, all 40 studies meet the quality criteria and are retained for further analysis.

3.4. Data extraction and synthesis

Fourteen statistics objects have been described to be extracted from the principal research to supply demographic statistics and reply to the RQs introduced in Section 4. Among these, five data items (D1–D5) extract the demographic details of the primary studies, while the remaining nine data items (D6–D14) address the eight RQs. Table 6 provides a comprehensive overview of the data items along with their corresponding RQs. The data extraction process is conducted using a predefined set of RQs to guide the review of each study. Before proceeding with full extraction, the first and second authors perform a pilot extraction on a sample of studies to align their understanding of data definitions and prevent potential misinterpretations. They then independently extract data from all studies using the established protocol. To ensure consistency and reliability, all discrepancies or uncertainties in the extraction process are resolved through iterative team discussions until full consensus is achieved. All extracted data are systematically documented in a structured spreadsheet to facilitate subsequent analysis. For transparency and reproducibility, the full technical report is publicly available at: https://zenodo.org/records/15448946.

Both qualitative and quantitative data are employed in the synthesis of the collected information. Quantitative data are analyzed using descriptive statistics (e.g., frequencies and percentages) to identify trends, which are then visualized through charts. For qualitative data, thematic analysis is applied to identify and categorize themes related to the RQs, such as IR models, enhancement strategies, and evaluation measures. After separately analyzing the qualitative and quantitative data, both sets are integrated to cross-validate the findings and provide a comprehensive perspective. For example, the quantitative analysis identifies the frequency of IR

Table 5
Study quality assessment checklist.

Step	Quality Assessment Item	Evaluation Criterion
Q1	Introduction	Does the study discuss the topic of IR-based RT?
Q2	Methods	Does the study propose methods, models, or experiments for IR-based RT?
Q3	Results	Does the study present novel findings or useful results related to IR-based RT?
Q4	Analysis	Does the study analyze related works or compare different approaches?
Q5	Discussion	Does the study discuss strengths, limitations, or suggest future research directions for IR-based RT?

Table 6
Data items extracted from the primary studies with the relevant RQs.

#	Data item name	Description	Relevant RQs
D1	Index	The ID of the paper	Overview
D2	Title	The title of the paper	Overview
D3	Publication type	Conference, journal, workshop, or book chapter	Overview
D4	Year	Year the paper was published	Overview, RQ1
D5	Venue	Place of publication	Overview, RQ1
D6	IR model	IR models applied to perform RT	RQ2
D7	Enhancement strategy	The strategy employed to enhance the efficiency and effectiveness of IR-based RT methods	RQ3
D8	Dataset	Datasets used to verify the IR-based RT methods	RQ4
D9	Source Artifact	Source entity of the trace	RQ4
D10	Target Artifact	Destination entity of the trace	RQ4
D11	Cut Points	The method by which intercepts are selected in the IR model	RQ5
D12	Measure	Evaluation measures employed for assessing the effectiveness of IR-based RT methods	RQ6
D13	Baseline	Baseline of paper	RQ7
D14	Evidence level	The evidence level of the performance evaluation of IR-based RT methods	RQ8

models used, while the qualitative analysis explains why these models perform better in certain contexts. Additionally, the results across multiple RQs are combined to create an evidence map, offering a holistic view of research trends and identifying gaps by considering various perspectives.

It is important to note that data item (D14) is specifically extracted to address RQ8, which focuses on gathering evidence regarding the performance evaluation of IR-based RT methods. The evidence level reflects the quality and credibility of the selected studies. Following Petersen et al. (2015), we adopt a six-level framework to assess and classify the gathered evidence:

Level 0. Absence of evidence. (No evidence)

Level 1. Evidence derived from demonstrations or working out with toy examples. (Toy examples)

Level 2. Evidence based on expert opinions or observations, such as surveys or interviews. (Expert opinions)

Level 3. Evidence obtained through academic studies, including controlled laboratory experiments. (Academic studies)

Level 4. Evidence derived from industrial studies, such as causal case studies conducted in real-world industrial settings. (Industrial studies)

Level 5. Evidence obtained from practical implementation in industry, reflecting the solutions adopted by practitioners. (Industrial implementation)

4. Results

The following section showcases the outcomes of the SMS with the order of the RQs introduced in Section 3.1.

4.1. What are the venues/years of the primary studies (RQ1)?

As depicted in Fig. 3, a total of forty primary studies have been published across twenty-eight conferences and journals. Among these venues, ICSE and RE stand out as the top two publication platforms, showcasing the highest number of papers.

Fig. 3. Conferences and Journals in which the studies are published.

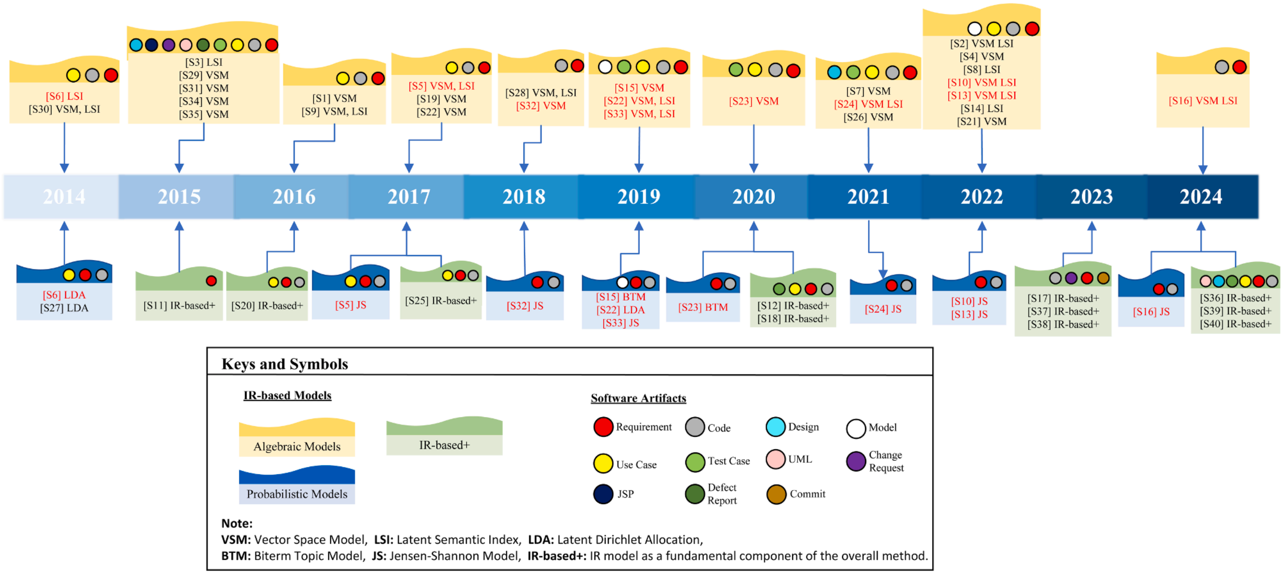

Fig. 4. Yearly distribution of the studies.

Fig. 4 presents the annual distribution of the primary studies published between 2014 and 2024. Notably, the VSM and LSI models have gained significant popularity due to their effectiveness and efficiency. In 2015, 2017, 2019, 2021 and 2022, much of the research focus on VSM and LSI. In contrast, there has been a relatively lukewarm trend in the use of other IR models. Regarding artifacts, requirements and code are the most commonly used, followed by use cases.

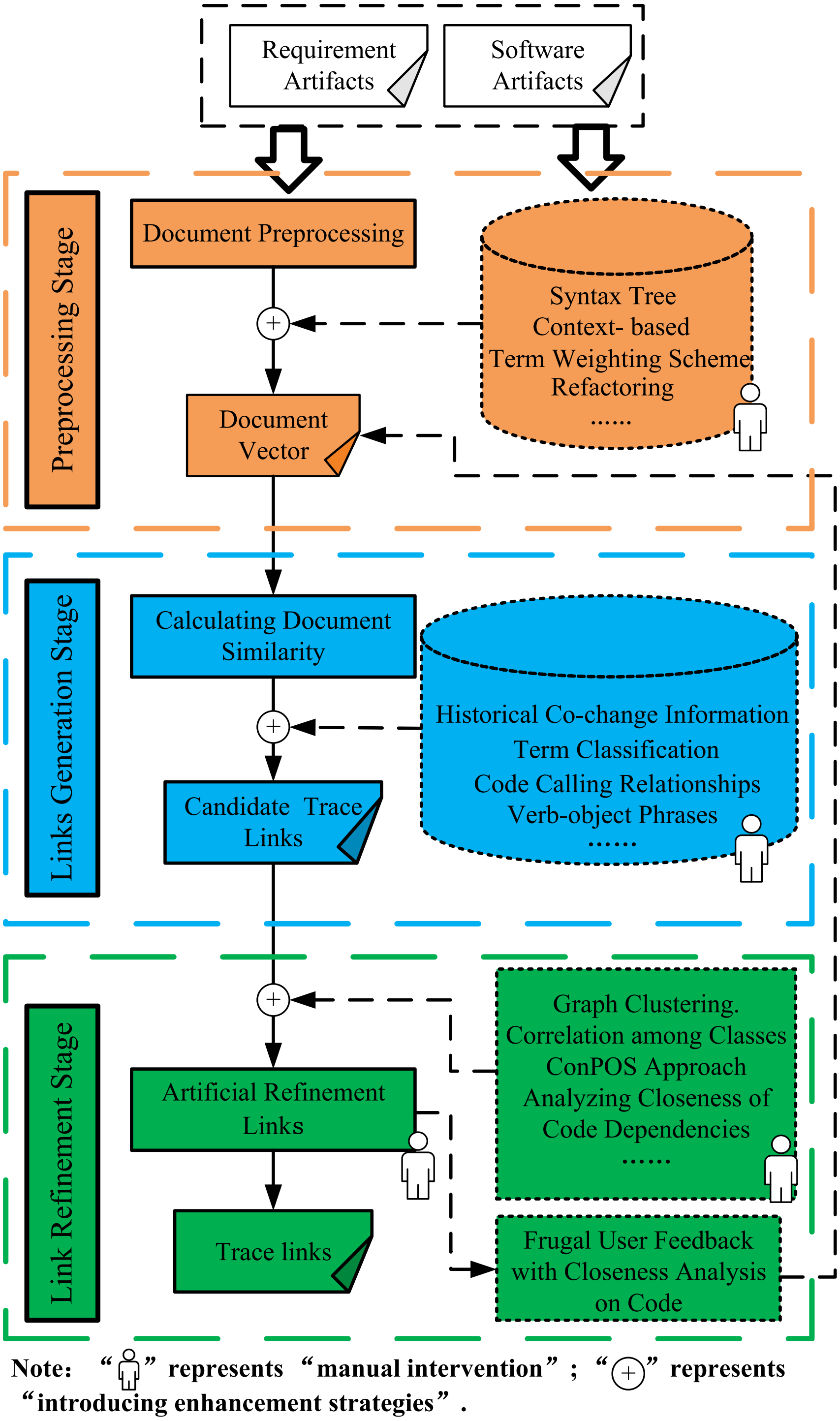

Fig. 5. The general process of IR-based RT generation.

4.2. What are the IR models that have been employed in RT (RQ2)?

Fig. 5 illustrates the general traceability recovery process through IR techniques. This process typically involves three stages: (1) **Preprocessing Stage:** Textual treatment is applied to both the source artifacts (Requirement Artifacts) and the target artifacts (Software Artifacts). To facilitate subsequent processing, specific document preprocessing methods are employed to remove noise and generate a suitable document representation for further analysis. (2) **Trace Link Recovery (Generation) Stage:** Various methods for calculating document similarity are employed to assess the similarity between the two artifacts. Subsequently, the artifacts are organized based on their similarity scores, and potential trace links are selected based on a predefined threshold. (3) **Trace Link Refinement (Vetting) Stage:** The candidate trace links are refined using manual or semi-automated methods. The analyst responsible for the trace recovery process verifies and confirms the final trace links.

Typically, once the preprocessing stage is completed, various IR-based models can automatically establish trace links. Table 7 demonstrates that LSI and VSM are the two most frequently utilized IR models in trace link generation. Notably, some studies have integrated the IR model as a fundamental component of the overall method. Additionally, it is worth noting that most primary studies have demonstrated, utilized, or verified the effectiveness of multiple IR models. To improve clarity and enable easier comparison, we summarized the typical advantages, limitations, and application scenarios of these IR models, as shown in Table 8.

VSM is an algebraic model that employs N-dimensional vectors to represent and compare objects. In this model, each dimension corresponds to a distinct orthogonal feature of the object (Mahmoud & Niu, 2015). The primary idea behind VSM is that a document's unique phrases represent specific, distinct topics. The model then sorts documents by calculating the similarities between the query and the document.

LSI is a statistical technique used to extract and represent the semantic aspects of words and passages based on their usage. It operates on the assumption that phrases have an underlying structure, which may be partially concealed by the variability in the ways different phrases describe a particular concept (Paul et al., 2022). To capture this latent structure, LSI performs Singular Value Decomposition (SVD) on the term-document matrix, reducing its dimensionality to reveal hidden semantic relationships (e.g., synonymy) and thereby overcome vocabulary mismatch.

LDA is a probabilistic model employed to estimate the distribution of topics within a document (Hindle et al., 2012). It assumes that each document is a mixture of topics, with each topic represented by a set of words. By analyzing multiple documents and extracting their topic distributions, further tasks such as topic clustering or text categorization can be performed based on these distributions.

BTM (Yan et al., 2013) is an unsupervised text topic model that captures co-occurrence relationships between topics based on biterms in short texts, ignoring the positional information of words in the document. It processes short texts more efficiently than traditional topic models and can uncover deeper structure.

JS is a probabilistic hypothesis-testing technique that utilizes probability distributions to represent each document (Nielsen, 2020). By calculating the divergence between these probability distributions, a ranked list of links can be generated.

4.3. Which enhancement strategies have been applied to IR-based RT (RQ3)?

Research in the field of RT can be categorized into the following key scenarios (Wang et al., 2018):

- **Traceability Link Generation (G):** Utilizing techniques such as NLP, IR, DL, and ML, traceability links are established between source artifacts (e.g., requirements) and target artifacts (e.g., software artifacts) to maintain their relationships.
- **Traceability Link Maintenance (M):** The process of maintaining established traceability links to ensure their continued accuracy and relevance.
- **Traceability Link Validation (V):** The process of validating traceability links to assess their correctness, consistency, and completeness.
- **Traceability Link Application (A):** The use of traceability links in software development activities such as requirements evaluation, program comprehension, and software maintenance, demonstrating the practical value of traceability.

As summarized in Table 9, various enhancement strategies have been proposed to improve the efficiency of IR-based RT, particularly focusing on the preprocessing, link recovery, and link refinement stages. Among these, the majority aim to enhance the performance of VSM and LSI models. Notably, 63.88% (23/36) of the strategies are applied during the trace link recovery stage. Fig. 6 presents a visual summary of the analysis: the left panel maps selected studies to their respective enhancement stages, while the right panel indicates the number of datasets used for evaluating different IR models. It further confirms that VSM and LSI are the most

Table 7
IR models applied in RT.

Algebraic Models	VSM	[S1] [S2] [S4] [S5] [S7] [S9] [S10] [S13] [S15] [S16] [S19] [S21] [S22] [S23] [S24] [S26] [S28] [S29] [S30] [S31] [S32] [S33] [S34] [S35]
	LSI	[S2] [S3] [S5] [S6] [S8] [S9] [S10] [S13] [S14] [S16] [S22] [S24] [S28] [S30] [S33]
Probabilistic Models	LDA	[S6] [S22] [S27]
	BTM	[S15] [S23]
	JS	[S5] [S10] [S13] [S16] [S24] [S32] [S33]
IR-based+	[S11] [S12] [S17] [S18] [S20] [S25] [S36] [S37] [S38] [S39] [S40]	

Table 8
Summary of IR models.

Models	Advantages	Limitations	Typical Applications
VSM	• Simple implementation • Interpretable • Effective for exact term matching	• Cannot handle synonymy/polysemy • Poor at capturing context	Traceability tasks involving consistent terminology in requirements and code (Mahmoud & Niu, 2015)
LSI	• Captures latent semantics • enables dimensionality reduction • tolerant to term variation	• huge storage requirements • Limited interpretability • Parameter sensitive	Link recovery in heterogeneous or terminology-diverse documents (Paul et al., 2022)
LDA	• Models topic distributions • Suitable for large-scale text	• Performs poorly on short texts • Topics depend on parameter tuning	Topic modeling in large-scale requirement specifications (Hindle et al., 2012)
BTM	• Designed for short texts • Captures word co-occurrence	• Requires parameter tuning • unsuitable for long-form documents	Traceability in short texts like comments, commits (Yan et al., 2013)
JS	• Measures distribution similarity • Robust to vocabulary variation	• Requires accurate distribution estimation • Limited interpretability	Ranking traceability links using semantic distribution similarity (Nielsen, 2020)

frequently used models, and that most enhancement efforts focus primarily on the recovery phase.

According to the characteristics of these enhancement strategies, they can be broadly classified into four categories: artifact text information, artifact structural information, model-based methods, and human intervention-based methods. These categories are discussed in detail below.

4.3.1. *Enhancement strategies using artifact text information*

Principle: Text-based enhancement strategies improve trace link generation through enrichment or restructuring of software artifacts' textual content. These approaches leverage linguistic patterns (e.g., verb-object structures), dynamic artifact characteristics (e.g., co-changing files), and supplementary data sources to address fundamental limitations of traditional IR models in traceability tasks.

Methods: A variety of techniques have been developed to enhance traceability using artifact text information. These strategies include using Verb-object Phrases (Zhang et al., 2016), Trace Link Evolver (Rahimi & Cleland-Huang, 2019), ConPOS method (Ali et al., 2019), and so on. For instance, Faiz et al. (2016) and Mahmoud and Niu (2014) improves code quality and discovers more implicit relationships between requirements and code. Ali et al. (2015) propose an improved term weighting scheme that captures developers' eye movements using an eye-tracking device to weight identifiers during RT validation. Similarly, Gao et al. (2023) enhance the efficacy of IR techniques by enriching the corpus with consensual biterms extracted from the data, and then using global and local weight to modify the ranking of candidate lists. Additionally, Mahmood et al. (2015) uses Formal Concept Analysis (FCA) to recover and visualize traceability links between requirements and code. Finally, Hey et al. (2024) filtered irrelevant requirements using classification results before applying a fine-grained word embedding-based method for link recovery.

Other information in the code-level software artifacts, such as code comments (Shen et al., 2021), configuration management logs (Tsuchiya, Washizaki, Fukazawa, Kato, et al., 2015; Tsuchiya, Washizaki, Fukazawa, Oshima, et al., 2015), etc., can be used to assist in the requirements-code traceability relationship. Shen et al. (2021) uses different comments to compensate for the vocabulary mismatch. If two or more files change frequently and simultaneously, there may be a traceability relationship between them. There are also strategies that incorporate the particularities of other artifacts. For example, Lapeña et al. (2022) leveraged the specific characteristics of BPMN to enhance the link recovery process between requirements and BPMN models.

Discussion: Text-based strategies offer lightweight and broadly applicable solutions, particularly effective in early-stage projects with well-structured artifacts. However, their reliance on lexical similarity limits effectiveness against semantic variations, ambiguous terminology, and domain-specific expressions (Mahmoud & Niu, 2015). These constraints are particularly evident when processing sparse or unstructured text requiring contextual understanding. While valuable as baseline methods, they prove inadequate as standalone solutions for complex traceability tasks (Kicsi et al., 2021).

4.3.2. *Enhancement strategies using artifact structural information*

Principle: Structural-based enhancement strategies improve trace link generation by incorporating structural information from software artifacts that traditional IR models typically ignore. These methods often utilize code dependencies, class hierarchies, or call graphs to capture relationships beyond textual similarity. By leveraging structural proximity or co-change patterns, they can reveal trace links that are not evident from text.

Methods: These strategies are typically applied to code artifacts. For example, some researchers extract structural information in artifacts by semantic similarity. Wang et al. (2021) indicated whether there is a close relationship between products by calculating the similarity. And Peng et al. (2023) addressed link recovery by combining fine-grained correlation analysis and query expansion to analyze semantic similarity and establish valid links beyond the original query. Other researchers build connections between artifacts using external information. For example, Zhang et al. (2022) used the trace recommendation-based code class structure (TRCCS) to

Table 9
List of enhancement strategies for IR-based requirements trace recovery methods.

Category	Strategy	IR model						Applying Phrase	Scenarios	Strategy Characteristics
		VSM	LSI	JS	LDA	BTM	IR-based+			
Artifact Text	Verb-object Phrases [S1]	●						**P**	**G**	Extracting verb-object phrases as main information and essential meaning.
	Improved Term Weighting Scheme [S6]		●		●			**P**	**V**	Proposing an improved term weighting scheme, namely, Developers Preferred Term Frequency/Inverse Document Frequency (DPTF/IDF).
	Refactoring [S9] [S30]	●	●					**P**	**G, M**	Solving the problem of missing symbols, misplaced symbols and repeated symbols.
	Annotation extraction [S2]	●	●					**P**	**G**	Introducing different types of comments to some extent compensate for vocabulary mismatches between requirements and source code to improve the accuracy of tracing links.
	Consensual Biterms [S10]	●	●	●				**P**	**G**	Extracting consensual biterms to first enrich the corpus for IR techniques.
	Code Comments [S26]	●						**P**	**M**	Introducing different types of comments to some extent compensate for vocabulary mismatches between requirements and source code to improve the accuracy of tracing links.
	Semantic Augmentation [S35]	●						**R**	**G**	By utilizing the VSM with additional support from domain-specific thesauri and general-purpose thesauri such as WordNet.
	Trace Link Evolver (TLE) [S28]	●	●					**R**	**M**	Proposing a TLE, which relies on a set of heuristics combined with refactoring detection tools and IR algorithms, to detect predefined change scenarios between successive software versions.
	Configuration Management Log [S29] [S31]	●						**R**	**M**	Restoring links by finding revisions in the configuration management log that contain words related to requirements.
	Integrating semantic similarity [S11]						●	**R**	**G**	This strategy leverages external knowledge bases, such as DBpedia and BabelNet, to enrich the textual information of requirements artifacts.
	Global and Local Weight [S10]	●	●	●				**F**	**G**	Using consensual biterms to adjust global and local weight to adjust the ranking of candidate lists.
	ConPOS Approach [S32]	●		●				**F**	**G**	Pruning trace links using the primary POS classification and apply constraints to recovery as a filtering process.
	Leveraging BPMN Particularities [S14]		●					**R**	**G**	Incorporating the particularities of BPMN into link recovery process.
	filtering requirements[S36]						●	**R**	**G**	Using results of requirements classification to filter parts of requirements.
Artifact Structural	Traceability Rules [S8]		●					**P**	**G**	Defining traceability rules to determine correspondences between the requirement modeled with the use case diagram based on the enriched textual description and design diagrams modeled
	Commonality and Variability Analysis (CVA)[S31]	●						**P**	**M**	Analyzing to which products elements (e.g., requirements, code elements) belong.
	Analyzing Closeness of Code Dependencies [S5] [S13]	●	●	●				**P, F**	**G, V**	Quantifying the interaction degree of call dependency and data dependency between two code classes.
	Biterm Extraction Strategies [S16]	●	●	●				**R**	**G**	Extracting pairs of terms (biterms) from documents, which are then used to capture co-occurring concepts within short texts.
	Analyzing Close Relations [S7]	●						**R**	**G**	Calculating the close relations (semantic similarity) between target artifacts
	Code class Structure [S4]	●						**F**	**G**	Utilizing the hierarchical and relational structure of code classes, such as inheritance or associations between classes.
	Correlation Among Classes [S19]	●						**F**	**V**	Using structural or co-changing dependencies or both to find correlations between classes and use these dependencies to verify traceability links.
	Fine-grained Query Expansion [S37]						●	**R**	**G**	Combining fine-grained correlation analysis and query expansion.
Model-based	Artificial Bee Colony Algorithm [S12]						●	**R**	**G**	An optimization technique inspired by honey bee foraging behavior, used to efficiently explore and improve potential solutions in trace link recovery.

(*continued on next page*)

Table 9 (*continued*)

Category	Strategy	IR model						Applying Phrase	Scenarios	Strategy Characteristics
		VSM	LSI	JS	LDA	BTM	IR-based+			
	Genetic Algorithm [S18] [S20] [S23] [S25]	●				●	●	**R**	**G, M**	A search heuristic inspired by the process of natural selection.
	Heuristic Measures [S3]		●					**R**	**M**	A fully automated technique to determine appropriate configurations for LSI to recover links between requirements artifacts.
	Hybrid Method [S15] [S23]	●				●		**R**	**G**	Combing VSM and BTM which can help relieve data sparsity caused by short text.
	Semi-Supervised Techniques [S22]	●	●		●			**R**	**V**	Utilizing a combination of semi-supervised learning techniques and NLP to enhance IR-based traceability recovery.
	Classification [S31]	●						**F**	**M**	Classifying traceability links into 5 five types using the CVA results, then using the classification to refine links.
	IR Features as Input to ML Models [S38] [S39] [S40]						●	**R**	**G**	Combines IR similarity scores with structural and semantic features as input to ML models for enhanced trace link prediction
human intervention-based	Adaptive User Feedback [S34]	●						**F**	**V**	Determining whether and how to apply relevant feedback based on the verbosity of the software artifacts and the number of correct links and false positives that have been categorized.
	User Feedback [S29]	●						**F**	**M**	Log-based traceability recovery and user feedback link recommendation.
	Frugal User Feedback with Closeness Analysis on Code [S13] [S33]	●	●	●				**F**	**V, G**	Introducing only a small amount of user feedback into the closeness analysis on call and data dependencies in code.

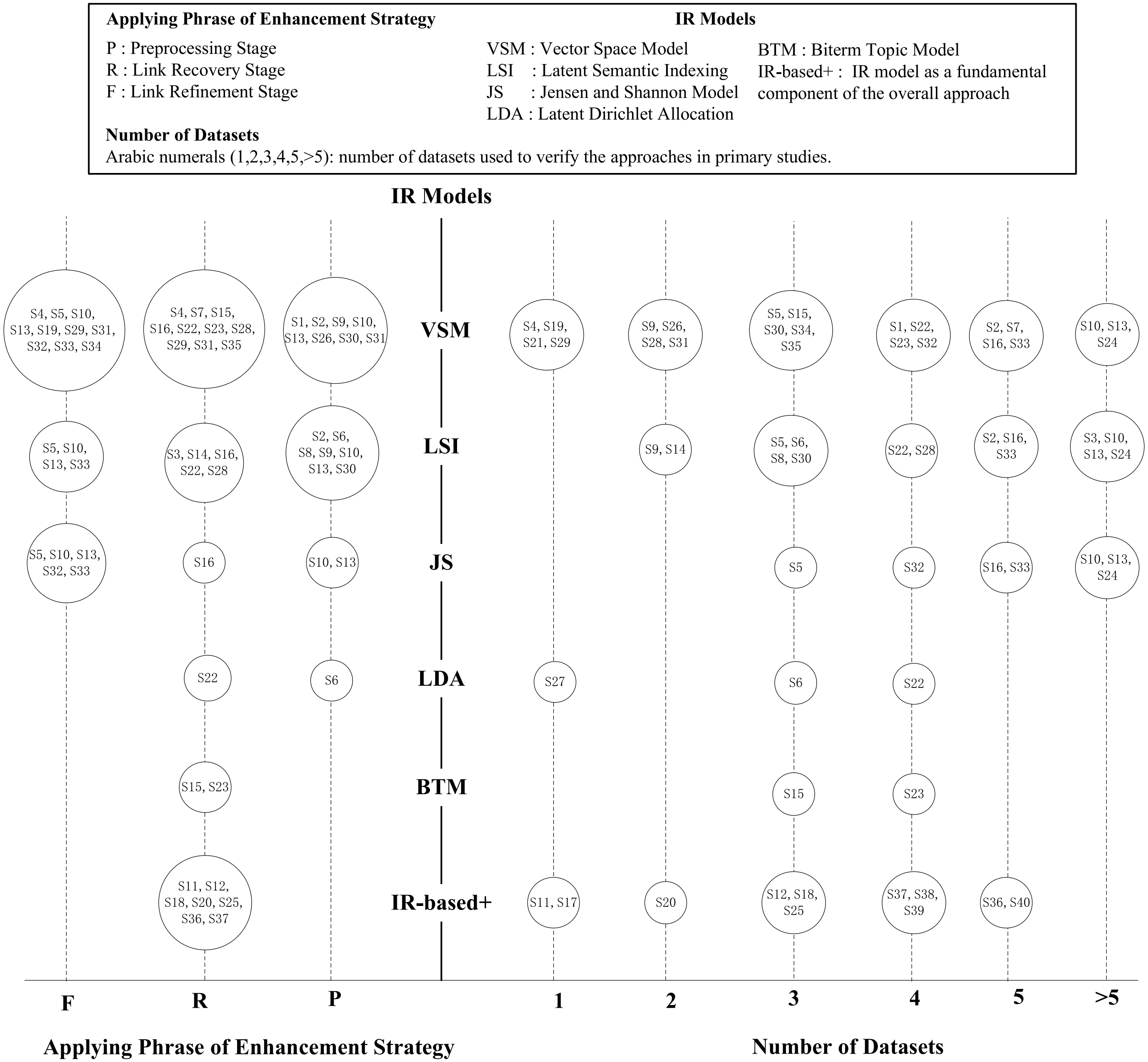

Fig. 6. Bubble chart over IR models, applying phrase of enhancement strategy, and the number of datasets used to verify the methods.

refine further the links generated by the VSM. In addition, there are some other methods to extract structural information. Kuang et al. (2017) introduced a method for analyzing the proximity of code dependencies, quantifying the level of interaction between call dependencies and data dependencies among two code classes. They used a dynamic analysis tool to capture the code dependencies. Gao et al. (2022) extended their earlier work (Kuang et al., 2019) by introducing a two-phase closeness analysis on code dependencies, aimed at identifying candidate regions that could collaborate with frugal user feedback.

Discussion: Structural-based strategies enhance IR by capturing relationships embedded in system architecture and change history. They have shown improved performance in requirement-to-code traceability tasks where structural information is leveraged. However, their utility diminishes when artifacts exhibit sparse structure (Zou et al., 2024). These methods rely on external program analysis tools and structural extraction (Kuang et al., 2017), which may introduce additional preprocessing steps. When applied to structurally rich artifacts, they can significantly improve the accuracy of trace link generation (Zou et al., 2024).

4.3.3. Enhancement strategies using model-based methods

Principle: Model-based enhancement strategies improve traceability link generation through automated parameter optimization and multi-model integration, effectively handling diverse and sparse data. These methods often incorporate topic modeling, optimization algorithms, or ML techniques to overcome the rigidity of manually configured IR models.

Methods: These strategies combine different models or configure the optimal model parameters. Wang et al. (2019, 2020) proposed the combination of VSM and BTM. This combination proves helpful in mitigating the issue of data sparsity that arises from working with short texts. Wang et al. (2020) used the genetic algorithm for the parameter configuration of BTM. The literature investigates the RT recovery problem as an optimization problem, using an artificial bee colony (ABC) algorithm (Rodriguez & Carver, 2020a), a non-dominated sorting genetic algorithm (Ersoy et al., 2023; Ghannem et al., 2017; Hubner, 2016), etc., to find the maximum solution of the objective function. More recently, hybrid approaches have emerged that use IR model outputs as input features for ML models (Li et al., 2024; Wang et al., 2023; Zou et al., 2024), highlighting the potential of IR as a feature extraction component.

Discussion: Model-based strategies significantly enhance traceability automation by minimizing manual parameter configuration while effectively adapting to complex artifact characteristics (Wang et al., 2020). These approaches prove particularly valuable for large-scale projects demanding flexible solutions. However, their implementation requires substantial computational resources and extensive labeled training data (Effa Bella et al., 2019). In addition, the interpretability limitations of ML-based methods remain a notable concern for their practical adoption (Dalpiaz et al., 2019).

4.3.4. Enhancement strategies using human intervention-based methods

Principle: Human intervention-based strategies improve trace link generation by integrating user feedback to validate and refine candidate links. These approaches leverage expert judgment to adjust similarity metrics, optimize ranking orders, and verify critical traceability decisions, thereby combining automated analysis with human expertise.

Methods: A common approach involves interactive trace recovery, where users iteratively evaluate and adjust candidate links produced by IR models. For example, Panichella et al. (2015) utilized adaptive user feedback to refine the similarity calculation scheme and reorder unvalidated candidate trace links. Similarly, Tsuchiya et al. (2015) employed user feedback to directly validate and reorder trace links based on call relationships. In contrast, Kuang et al. (2019) and Gao et al. (2022) introduces minimal feedback for efficient dependency analysis in code.

Discussion: Human intervention-based strategies significantly enhance trace link accuracy by leveraging expert practitioners' insights to capture nuanced, domain-specific relationships beyond automated methods' capabilities. While highly effective for precision-critical applications like safety-sensitive development or final-stage trace validation (Kuang et al., 2019), these approaches face inherent scalability limitations due to their manual nature (Zou et al., 2024), particularly in large-scale or rapidly evolving systems (Wang et al., 2018).

4.4. Which datasets have been utilized to validate IR-based RT methods (RQ4)?

The response to this RQ provides valuable guidance for researchers in selecting datasets that align with their specific research requirements and dataset characteristics. Tables 10 and 11 provide comprehensive overviews of both open-source and non-open-source datasets, including metadata such as artifact types and dataset scale, to support researchers in selecting suitable datasets.

From all the primary studies, fifty-three datasets are extracted for experimental validation. The datasets provided by the Center of Excellence for Software and Systems Traceability (CoEST), including iTrust, eTOUR, and EasyClinic, have attained widespread recognition and are considered the three most favored datasets among researchers. Furthermore, of the thirty-five open-source datasets available, CoEST offers ten, which can be accessed at http://www.coest.org/. This prominence stems from their unique advantages: being freely accessible, well-documented, and providing uniformly structured artifacts (e.g., requirements, use cases, and source code) that are both compatible with IR-based RT tasks and immediately usable without requiring data scraping or manual annotation. Additionally, these datasets span diverse domains like healthcare, tourism, and education, enabling varied evaluation contexts. Their widespread adoption facilitates direct comparability across different studies and helps establish informal benchmarking standards. However, excessive reliance on CoEST datasets may introduce biases. They primarily consist of well-structured, research-oriented systems which, while suitable for validating IR methods, may not fully represent the complexity or domain-specific challenges of large-scale industrial applications.

Table 11 provides information on non-open-source datasets used in the primary studies. More than 30% of the primary studies (15/

Table 10
Open-source datasets information used in the primary studies.

Dataset Name	Source Artifacts (Number)	Target Artifacts (Number)	Space	True Links	Scale	Domain	Freq.	Resource Links	Primary Studies
iTrust	Use cases (34) Requirements (50) Use cases (33)	Code (243) Code (299) JSP (47)	8,262 14,950 1,551	603 314 58	Large Small	Healthcare	21	http://www.coest.org/	[S2] [S4] [S5] [S6] [S7] [S9] [S10] [S13] [S19] [S20] [S26] [S30] [S32] [S33] [S34] [S35] [S36] [S37] [S38] [S39] [S40]
eTour	Use cases (58) Requirements (58)	Code (116) Code (116)	6,728 6,728	308 366	Large	Tourism	15	http://www.coest.org/	[S1] [S2] [S9] [S12] [S18] [S23] [S25] [S26] [S30] [S35] [S36] [S37] [S38] [S39] [S40]
EasyClinic	Requirements (30) Use cases (30) UML interaction diagram (20)	Code (47) Test cases (63) Code classes (47)	1,410 1,890 940	83 63 69	Small	Healthcare	10	http://www.coest.org/	[S1] [S2] [S3] [S7] [S15] [S16] [S23] [S34] [S38] [S40]
Gantt	Requirements (17) Requirements (16) High-level Requirements (17) Use cases (67)	Code (55) Code (124) Low-level Requirements (69) Code (100)	935 1,984 1,173 6,700	54 315 – 1,044	Small Large	Software Engineering	7	http://www.ganttproject.biz https://github.com/bardsoftware/ganttproject http://www.coest.org/	[S5] [S6] [S7] [S10] [S13] [S33] [S40]
EBT	Requirements (40) Requirements (16)	Code (50) Code (124)	2,000 1,173	98 315	Small	Event-Based Traceability	5	http://www.coest.org/	[S12] [S15] [S16] [S18] [S23]
Pig	Requirements (87) Requirements (58)	Code (289) Code (754)	25,143 43,732	547 –	Large	Data Analysis	5	https://github.com/apache/pig	[S10] [S13] [S24] [S34] [S38]
Albergate	Requirements (82)	Code (1771)	145,222	871	Large	Hotel	4	http://www.coest.org/	[S2] [S12] [S18] [S25]
SMOS	Requirements (67) Use cases (67)	Code (100) Code (99)	6,700 6,633	1,044 1,027	Large Large	Education	4	http://www.coest.org/	[S2] [S36] [S37] [S39]
CM-1	High-level Requirements (235) Requirement (298) Requirement (22)	Design (220) Code (90) Design (53)	51,700 26,820 1,166	361 546 –	Large Large Small	Aerospace	4	http://www.coest.org/	[S3] [S7] [S35] [S40]
WARC	Functional requirements (43) High-level Requirements (17)	Requirements specification (89) Low-level Requirements (69)	3,827 1,173	78 68	Small Small	Web Archiving	3	http://www.coest.org/	[S15] [S16] [S23]
Derby	Requirements (390) Requirements (133)	Code (611) Code (2184)	238,290 290,472	2,315 –	Large Large	Relational Database	3	https://github.com/apache/derby	[S10] [S13] [S24]
Infinispan	Requirements (116) Requirements (232)	Code (413) Code (319)	47,908 74,008	744 1,116	Large Large	Distributed database	3	http://infinispan.org/ https://github.com/infinispan/infinispan	[S10] [S13] [S33]
Maven	Requirements (68) Requirements (36)	Code (236) Code (82)	16,048 2,880	356 151	Large Small	Project Management	3	http://maven.apache.org/ https://github.com/apache/maven	[S10] [S13] [S33]
eAnci	Requirements (139)	Code (55)	7,4645	567	Large	Governance	3	http://www.coest.org/	[S36] [S37] [S39]

(continued on next page)

Table 10 (*continued*)

Dataset Name	Source Artifacts (Number)	Target Artifacts (Number)	Space	True Links	Scale	Domain	Freq.	Resource Links	Primary Studies
Libest	Requirements (52)	Code (14)	728	204	Small	Certificate	2	https://gitlab.com/SEMERU-Code-Public/Data/icse20-comet-data-replication-package/-/tree/main/LibEST	[S16] [S36]
Drools	Requirements (183)	Code (248)	45,384	841	Large	Business Rule	2	https://github.com/kiegroup/drools	[S10] [S13]
Groovy	Requirements (104)	Code (100)	10,400	180	Large	Programming Language	2	https://github.com/apache/groovy	[S10] [S13]
MODIS	Requirements (26)	Code (521)	13,546	229	Large	Imaging Radiometer	2	http://www.coest.org/index.php/resources/dat-sets	[S3] [S34]
Seam	Requirements (189)	Code (150)	28350	463	Large	Java Web Framework	2	http://www.seamframework.org/Seam2.html	[S10] [S13]
Pooka	Requirements (90)	Code (298)	26,820	507	Large	email	2	http://www.suberic.net/pooka/	[S6] [S32]
Dronology	Requirements (58)	Code (184)	8,584	–	Large	Unmanned Aerial Vehicle	2	https://dronology.info/datasets/	[S16] [S28]
Cassandra	Requests (65)	Code (328)	21,320	–	Large	Database	2	http://tinyurl.com/TLEArtifacts	[S24] [S28]

Note: There are other open-source datasets with one frequency, i.e., Lucene, Lynx, jHotDraw (JHD), SMS, Hive, Mina, Solr, Synapse, Tika, ActiveMQ, ibooks, SIP Communicator, Software Engineering Course

Table 11
Non-open-source datasets information used in the primary studies.

Dataset Name	Source Artifacts (Number)	Target Artifacts (Number)	Space	True Links	Scale	Primary Studies
MR0	Use cases (24)	Test Cases (60)	1,440	–	Small	[S3]
MR1	Defect reports (135)	Use Cases (28)	3,780	–	Small	[S3]
MR2	Change Requests (28)	Use Cases (21)	588	–	Small	[S3]
Car rental system	Use cases (9)	Classes (98) Methods (252)	882 2,268	–	Small	[S8]
Customer relationships system	Use cases (7)	Classes (65) Methods (124)	455 868	–	Small	[S8]
trains	Requirements (100)	Model (850)	85,000	–	Large	[S14]
LEDA	Requirements (88)	Classes (208) Methods (283)	18,304 24,904	–	Large	[S25]
Insurance project	URS document (69)	Regulatory documents (16)	1,104	–	Small	[S27]
Dronology v0.01	Requirements (34)	Classes (28)	952	69	Small	[S28]
Dronology v0.02	Requirements (39)	Classes (32)	1,248	79	Small	[S28]
Dronology v0.03	Requirements (43)	Classes (34)	1,462	86	Small	[S28]
Dronology v0.04	Requirements (46)	Classes (36)	1,656	92	Small	[S28]
Dronology v0.05	Requirements (49)	Classes (38)	1,862	98	Small	[S28]
Dronology v0.06	Requirements (51)	Classes (39)	1,989	105	Small	[S28]
Enterprise system	Requirements (192)	Code (694)	133,248	–	Large	[S29]
WDS	Requirements (26)	Code (521)	13,546	229	Large	[S30]

Note: This table lists only non-open-source datasets with sufficient details reported in the original papers. Datasets lacking information due to privacy restrictions are not listed.

40) rely on these non-open-source collections, whose restricted accessibility poses significant challenges: reduced reproducibility, limited cross-study comparability, and hindered establishment of consistent baselines. To address these limitations, researchers should actively pursue data-sharing agreements to improve accessibility where feasible. Furthermore, to ensure robust methodological validation, future research should validate proposed methods using diverse publicly available datasets that vary in scale, domain, and system characteristics.

Fig. 7 provides another perspective on the types of trace links. The most frequent trace link is between requirements and source code (22 evaluations), followed by the linkage between use cases and source code (14 evaluations). It is important to note that use cases and UML interaction diagrams are considered refined requirements, which explains why the source code is the most frequently studied target artifact for IR-based RT. Specifically, in over 56% of cases (39/69, 56.52%), source code is the destination of a trace.

Table 12 presents the distribution of the number of datasets used across the reviewed primary studies. According to the data presented in Table 12, and the right section of Fig. 6, it is concerning that a significant proportion of primary studies, approximately one-fifth (7/40, 17.5%), validated their methods using fewer than two datasets. This threatens the external validity of their findings

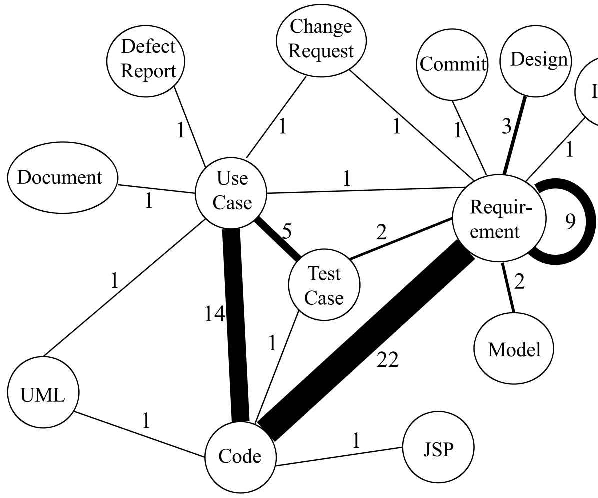

Fig. 7. Types of links recovered in IR-based RT.

Table 12
Distribution of the number of datasets used in primary studies.

Number of datasets used in primary studies	Number of primary studies	Proportion
1	7	17.5%
2	6	15%
3	10	25%
4	8	20%
5	5	12.5%
>5	4	10%
Total	40	100.00%

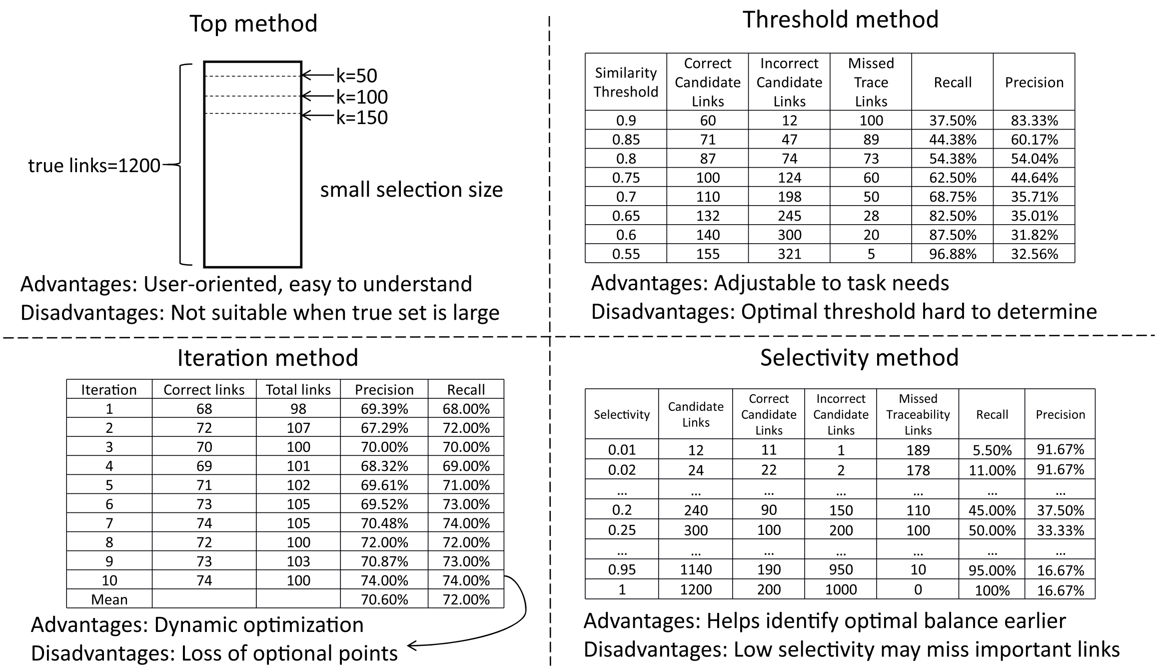

Fig. 8. Comparison of evaluation methods for IR-based RT.

(Lucia et al., 2007). It is widely recognized that the more datasets a method is tested on, the greater its generalizability. To mitigate this threat to external validity, it is strongly recommended that researchers incorporate a diverse range of datasets in their studies.

4.5. Which experimental evaluation methods have been utilized to assess IR-based RT methods (RQ5)?

There are four primary methods for determining cut points in IR-based traceability recovery: the Top method, Threshold method, Iteration method, and Selectivity method. Each method has distinct characteristics, trade-offs, and suitability for different scenarios. Fig. 8 provides a comparative analysis of these methods.

(1) **Top method:** As shown in Fig. 8, this method sorts the similarity values calculated by the model in descending order and then selects the top 50, 100, or 150 items as candidate trace links. When applied to a dataset with 1200 true trace links and selecting the top 10, the method correctly identifies 8 candidate links, but misses the remaining 1192 true links. This results in high precision but low recall, as the small selection size is insufficient for large datasets. Consequently, this method is not suitable for large datasets, as it may only capture a small portion of the true links.
(2) **Threshold method:** The threshold method involves setting a threshold value and selecting trace links with a similarity score above this threshold as candidates. Alternatively, the threshold with the best results can be chosen directly. According to Fig. 8, using this method with a true set of 160 trace links and a threshold of 0.9 yields 72 candidate trace links, of which 60 are correct, while 100 true links are missed. This results in high precision but low recall. However, recall is more critical, and a threshold that maximizes recall should be selected. The optimal threshold value for different datasets is uncertain and may require adjustment to achieve the best results (Faiz et al., 2016; Gao et al., 2023; Khlif et al., 2022).
(3) **Iteration method:** Evolutionary algorithms such as the NSGA-II algorithm (Ghannem et al., 2017; Rodriguez & Carver, 2020b) have been introduced to address the issue of not knowing how to select the optimal intercept point in (2). As shown in Fig. 8, the total of the true set is 100. This algorithm uses the result of each iteration as the link recommendation result. However, this method tends to get an average value and does not give the best results. This results in some loss of Recall or Precision.
(4) **Selectivity method:** Selectivity is used to represent the length of the candidate links taken at each point within the graph. It is computed using the following equation:

$$Selectivity = \frac{Number\ of\ Candidate\ Links}{Total\ links} \tag{1}$$

As the number of candidate links increases, the rate of change in the Precision and Recall of the results decreases. Therefore, in this paper, we choose the "dense and sparse" method to take points. When Selectivity∈[0, 0.2], 0.01 is taken as the interval. When the Selectivity∈[0.2, 1], 0.05 is taken as the interval because we hope to achieve good results after 20% of the links have been viewed. As shown in Fig. 8, there are a total of 1200 trace links, of which 200 are in the true set. As the Selectivity increases, the Recall rises. Selectivity can help the researcher to find the best results faster.

4.6. Which measures have been utilized to assess the performance of IR-based RT methods (RQ6)?

Once trace links have been generated, evaluating their quality becomes an essential task. Evaluation approaches that rely solely on individual performance metrics may result in biased or incomplete conclusions. Therefore, most studies adopt Precision, Recall, and F-Measure together to evaluate the effectiveness of IR-based RT methods, as demonstrated in Table 13. In addition, recent research emphasizes evaluating traceability from multiple dimensions to capture diverse perspectives on traceability quality, such as link ranking effectiveness, robustness to class imbalance, and the level of analyst effort required. Metrics such as Mean Average Precision (MAP), Average Precision (AP), Difference in Average Rank (DiffAR), and Selectivity have been introduced to provide a more nuanced understanding of system performance. These measures offer a more detailed interpretation of the capability and effectiveness of the RT methods being assessed.

Table 13
Measures of performance employed in primary studies.

Categories	Measures	Primary Studies
Primary Measure (Quality)	Recall Precision	[S1] [S2] [S4] [S5] [S6] [S7] [S8] [S10] [S11] [S12] [S13] [S14] [S15] [S16] [S17] [S18] [S19] [S20] [S21] [S22] [S23] [S24] [S25] [S26] [S28] [S29] [S30] [S31] [S32] [S33] [S34] [S35] [S36] [S37] [S38] [S39] [S40]
	F-Measure	[S1] [S2] [S4] [S5] [S6] [S10] [S12] [S13] [S14] [S16] [S17] [S18] [S19] [S21] [S22] [S23] [S24] [S28] [S29] [S31] [S33] [S36] [S37] [S38] [S39] [S40]
	MCC	[S14]
	AUC	[S14]
Secondary Measure (Browsability)	MAP	[S3] [S5] [S7] [S10] [S13] [S16] [S26] [S27] [S30] [S32] [S33] [S37]
	AP	[S3] [S5] [S7] [S10] [S13] [S16] [S26] [S27] [S32] [S33] [S37]
	DiffAR	[S30]
	Selectivity	[S23]

Table 14
Verification results for trace link classification.

Actual Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

Table 14 presents four distinct classification types, key to comprehending the verification outcomes in trace link classification. The table outlines the categories: True Positive (TP), False Negative (FN), False Positive (FP), and True Negative (TN).

Based on their function, these measures can be classified into two distinct categories as outlined below:

(1) Primary Measure (Quality)

Precision, Recall: The primary IR measures utilized to evaluate the effectiveness of the various trace tools are **Precision (P)** and **Recall (R)** (Zou et al., 2024). Precision measures the proportion of retrieved trace links that are relevant to the true trace links, while Recall indicates the proportion of relevant trace links successfully retrieved. The two evaluation metrics exhibit an inherent trade-off relationship: high precision means fewer false positives but may miss relevant trace links, leading to low recall. Conversely, high recall ensures more relevant links are retrieved but may introduce irrelevant links, reducing precision. In RT, it is often important to balance precision and recall, as a system with high precision may miss many relevant trace links. In contrast, a system with high recall may retrieve many irrelevant trace links. According to Dong et al. (2022), automatic tracing algorithms consider recall to be more important than precision. The definitions for Recall and Precision are as follows:

$$R\ (Recall) = \frac{TP}{TP + FN} \tag{2}$$

$$P\ (Precision) = \frac{TP}{TP + FP} \tag{3}$$

F-measure: It is the harmonic mean of Precision and Recall, providing a single value that balances both. A higher F-measure indicates better overall performance, offering a more comprehensive evaluation than either Precision or Recall alone. The F-measure is determined by the following calculation:

$$F - measure = \frac{2 \times P(Precision) \times R(Recall)}{P(Precision) + R(Recall)} \tag{4}$$

Matthew Correlation Coefficient (MCC): It measures the correlation between true positive and true negative results. Its value ranges from −1 to 1, with 1 indicating perfect correlation, 0 representing no better performance than random guessing, and −1 indicating an inverse correlation. MCC is a more reliable metric than Precision for imbalanced datasets, as it considers both TP and TN, which Precision can overlook. Moreover, MCC is symmetric and unaffected by the choice of class to label as positive. The MCC is calculated as follows:

$$MCC = \frac{(TP * TN - FP * FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{5}$$

AUC, or the Area Under the Receiver Operating Characteristic (ROC) Curve: It is a measure used to assess the performance of binary classifiers. The ROC curve is a graphical representation of the relationship between the true positive rate (TPR) and the false positive rate (FPR) at various threshold values. The AUC is a measure of the separability of the classes in the ROC curve. A perfect classifier would have an AUC of 1.0, while a classifier that performs no better than random guessing would have an AUC of 0.5. The AUC is a commonly used measure because it is not affected by class imbalance and does not depend on the choice of threshold value (Jubair et al., 2025). The equation for false positive rate FPR is:

$$FPR = \frac{FP}{(FP + TN)} \tag{6}$$

Table 15 presents an example of experimental results, showing the performance metrics under varying selectivity thresholds. As shown in Fig. 8 and Table 15, when the Selectivity is 0.01, the set of correct links is 200, and the set of retrieved candidate links is 12.

Table 15
An example of experimental results with Recall, Precision, F-measure and MCC.

Selectivity	TP	TN	FP	FN	Recall	Precision	F-measure	MCC
0.01	11	999	1	189	5.50%	91.67%	10.38%	20.23%
0.02	22	998	2	178	11.00%	91.67%	19.64%	28.75%
…	…	…	…	…	…	…	…	…
0.2	90	850	150	110	45.00%	37.50%	40.91%	27.95%
0.25	100	800	200	100	50.00%	33.33%	40.00%	25.82%
…	…	…	…	…	…	…	…	…
0.95	190	50	950	10	95.00%	16.67%	28.36%	0.00%

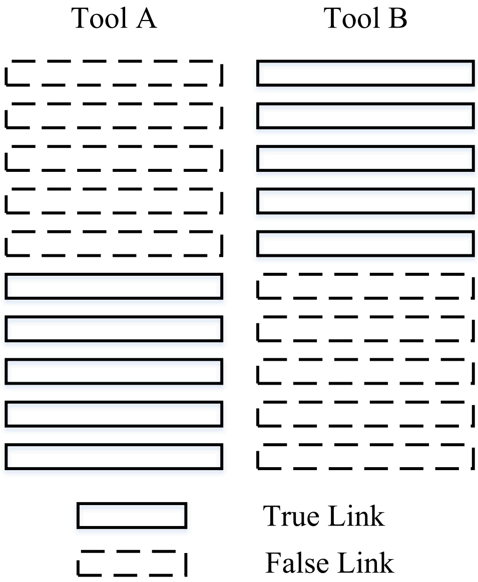

Fig. 9. An example for browsability.

Therefore, *TP* is 11, which is the intersection of the set of correct links and the set of retrieved candidate links. *FP* is 1 and *FN* is 189. The Precision is 91.57%, the Recall is 5.50%, the F-measure is 10.38%, and the MCC is 20.23%.

As shown in Fig. 9, two tools (methods/models) are used to generate trace links, each with 100% Recall and 50% Precision. The list of trace links generated by Tool A has false positive links before true links, while the list generated by Tool B has the opposite order. Both tools performed equally well for Precision and Recall. However, for the analyst, the effect of the B tool is better than that of the A tool. At this point, this difference can be measured by several independent retrieval rate indicators. Accordingly, browsability is introduced to assess the effectiveness of IR-based RT methods.

(2) Secondary Measure (Browsability)

Browsability can significantly affect the effectiveness and efficiency of analysts when evaluating the generated trace links. For an efficient and swift comprehension process, it is crucial to retrieve and present the correct trace links in an organized manner. This is particularly important for RT tools or methods that use ranked lists to display the results. While Precision, Recall, and F-measure focus on set-based evaluation, they do not provide insights into the browsability of the listed items (Shen et al., 2021). Browsability addresses this limitation by considering the order and organization of trace links, offering a deeper understanding of how easily an analyst can navigate the list and assess the relevance of the retrieved links (Mahmoud & Niu, 2014).

AP: It is a measure that reflects the average precision of a tool (method/model) when it is used to retrieve trace links. Eq. (7) is used to calculate this measure, which assesses how effectively trace links from requirements are prioritized at the top of the retrieved link.

$$AP = \frac{1}{|R|} \cdot \sum\nolimits_{i=1}^{n} (\mathrm{Prec}(i) \cdot \mathrm{relevance}(i)) \tag{7}$$

Where $|R|$represents the number of relevant documents, $Prec(i)$ represents Precision at top *i* documents, relevance(i) represents the relevance of the document at rank *i*, equaling 1 if the document is relevant, and 0 otherwise.

Table 16
An example of experimental results with the mean average Precision and average Precision.

Tool A				Tool B			
ID	Whether is Correct Link	The Calculation Process of AP	AP	ID	Whether is Correct Link	The Calculation Process of AP	AP
1	×	0	0.00	3	√	1/1	1.00
2	×	0	0.00	7	√	2/2	1.00
3	√	0	0.00	6	×	0	0.00
4	×	0	0.00	8	√	3/4	0.75
5	×	0	0.00	9	√	4/5	0.80
6	×	1/6	0.17	10	√	5/6	0.83
7	√	2/7	0.29	1	×	0	0.00
8	√	3/8	0.38	2	×	0	0.00
9	√	4/8	0.50	4	×	0	0.00
10	√	5/8	0.63	5	×	0	0.00
MAP	0.20			**MAP**	0.44		

MAP: MAP has good discrimination and stability. It is calculated as the mean of the AP for each true trace link. MAP considers the precision at each trace link, and provides a summary measure of the overall performance of the tool. A higher MAP indicates a significantly better quality of the tool (method/model). The equation for MAP is shown as follows:

$$MAP = \frac{AP}{Q} \tag{8}$$

Where Q is the number of queries.

Table 16 shows the results of trace links generated by different tools (methods/models). The ID is used to number each trace link. For tool A, the true positives are in a more backward order, resulting in a lower AP for true positives with the same ID. The MAP value of Tool A is also lower compared to Tool B, which demonstrates that Tool B is superior to Tool A.

DiffAR: This measure indicates the level of differentiation within the list, with a higher DiffAR indicating a clearer distinction between correct and incorrect links. Consequently, a list with a higher DiffAR is considered superior (Shen et al., 2021). Table 17 shows the similarity scores generated by Tool A and Tool B. The higher DiffAR for Tool A indicates that its false positives are ranked further forward, indicating worse performance. The equation for DiffAR is shown as follows:

$$DiffAR = \frac{\sum_{(h,d)} \mathrm{sim}(h,d)}{|L_T|} - \frac{\sum_{(h',d')} \mathrm{sim}(h',d')}{|L_F|} \tag{9}$$

Where $\mathrm{sim}(h,d)$ represents the similarity of true positive links and $\mathrm{sim}(h',d')$ represents the similarity of false positive links in a ranked list of trace links. $|L_T|$ and $|L_F|$ denote the number of true positive and false positive links, respectively.

Selectivity: It quantifies the time and effort an analyst saves when they review a list generated by an automated process instead of conducting a meticulous pairwise comparison of every element. A lower Selectivity value signifies greater efficiency for the analyst. Typically, when an analyst engages in a manual sub-component matching (tracing) task, they must examine numerous potential matches: every component from one artifact must be compared to another. As previously noted, an automated mining method generates a catalog of potential matches. The example of Selectivity is shown in Fig. 8 and its calculation method is defined in Eq. (1).

4.7. How do the commonly used IR models perform on RT tasks (RQ7)?

4.7.1. Selection of datasets

The selection criteria for the datasets are divided into two types: (1) The eight most commonly used datasets for IR-based RT over the past eleven years are selected, comprising 18 sets of artifacts for which trace links can be established; (2) All datasets are open-source and can be freely downloaded from the RT research community via the CoEST website (http://www.coest.org/).

Additionally, the datasets are primarily categorized into code-level software artifacts and document-level software artifacts, based on the classification of software artifacts, as outlined in Table 18. Code-level software artifacts include datasets such as iTrust, eTOUR, and Albergate. These datasets primarily consist of textual content and source code in programming languages, which makes it challenging to establish traceability links between the code and other artifacts. On the other hand, document-level software artifacts include datasets such as EasyClinic, CM-1, and GANNT. These datasets mainly contain requirements, design documents, and test cases, which are typically easier to trace due to their structured format and textual similarity.

To improve readability, Table 18 has been reorganized to group datasets by their primary artifact types (code-level vs. document-level). By categorizing the datasets this way, we aim to provide a clearer understanding of the distinct challenges associated with traceability in different types of software artifacts. To address these challenges, this experiment treats code text separately and investigates code-level and document-level software artifacts independently.

4.7.2. Quality evaluation standard

This experiment uses Precision and Recall to evaluate the results. However, for the code segmentation experiment, Precision and Recall alone are insufficient to fully assess the effectiveness of code segmentation. Based on industry practices, human analysts are

Table 17
An example of experimental results with the DiffAR.

	Tool A			Tool B	
ID	Whether is Correct Link	Similarity	ID	Whether is Correct Link	Similarity
1	×	0.98	3	√	0.98
2	×	0.97	7	√	0.97
3	√	0.95	6	×	0.95
4	×	0.94	8	√	0.94
5	×	0.92	9	√	0.92
6	×	0.89	10	√	0.89
7	√	0.88	1	×	0.88
8	√	0.85	2	×	0.85
9	√	0.82	4	×	0.82
10	√	0.8	5	×	0.8
DiffAR	0.19		**DiffAR**	−0.19	

Table 18
The details of CoEST datasets.

Category	Dataset Name	Source Artifacts (Number)	Target Artifacts (Number)	Space	True Links
Code-Level Artifacts	iTrust	Use Cases (131)	Java Code (226)	29606	418
	ETOUR	Requirements (58)	Code (116)	6728	308
	Albergate	Requirements (17)	Code (55)	935	53
Document-Level Artifacts	EasyClinic	Class Description (47)	Class Description (47)	2209	69
		Class Description (47)	Test Cases (63)	2961	204
		Interaction Diagrams (20)	Class Description (47)	940	69
		Interaction Diagrams (20)	Interaction Diagrams (20)	400	59
		Interaction Diagrams (20)	Test Cases (63)	1260	83
		Interaction Diagrams (20)	Use Cases (30)	600	26
		Use Cases (30)	Class Description (47)	1410	93
		Use Cases (30)	Interaction Diagrams (20)	600	26
		Use Cases (30)	Test Cases (63)	1890	63
		Use Cases (30)	Use Cases (30)	900	144
	CM-1	High-level Requirements (22)	Low-level Requirements (53)	1166	45
	EBT	Requirements (41)	Test Cases (25)	1025	51
	WARC	FRS (43)	SRS (89)	3827	78
		NFR (21)	SRS (89)	1869	58
	GANNT	High-level Requirements (17)	Low-level Requirements (69)	1173	68

generally more efficient at determining whether a candidate link is correct rather than evaluating its completeness, with a higher success rate. Therefore, in addition to Precision and Recall, this experiment also introduces the F-measure as an evaluation metric.

Regarding quality assessment criteria for automated RT techniques, this experiment adopts a measure based on Precision and Recall rates proposed by Hayes et al. (2006), which is grounded in industrial practice. The specific criteria are outlined in Table 19.

4.7.3. Data pre-processing

To ensure consistent and valid evaluation, the experimental datasets undergo standardized preprocessing through the following three stages:

- **Partitioning Source Code:** For Java source code files, method names, class names, and variable names are extracted separately using JavaParser, while code and block structures are identified using Python regular expressions. As a result, each Java source code file is divided into four sections, as shown in Table 20. These sections are then saved into four distinct files, with the corresponding file name for each line stored separately in another file.
- **Merging Software Artifacts:** Software artifacts in different data formats are collated. A software product typically consists of multiple text files representing various requirements, test cases, and other elements. First, the content and file names of these text files within the software product folder are integrated into two separate text files. Each line of text content corresponds to its respective file name.
- **Requirements and Source Code Pre-processing:** The following pre-processing steps are performed using the NLTK library to normalize the requirements and source code sections: (1) Text Splitting: The text is split into tokens. (2) Stop Word Removal: Stop words (e.g., punctuation, numbers, etc.) are removed. (3) Lexical Annotation: Words are filtered to retain only nouns and verbs using Python scripts. (4) Stemming: The Porter Stemmer algorithm is applied to reduce words to their root forms.

4.7.4. Results and analysis

Tables 21-25 illustrate the impact of code-level software artifacts in CoEST on the IR-based RT. The findings are summarized as follows: (1) The performance of the IR model is unsatisfactory and does not meet the criteria outlined in Table 19; (2) Different combinations of code segments affect RT recovery results. Specifically, the combination of the eTOUR and iTrust datasets with annotations yields better RT recovery results, while VN shows competitive performance in multiple models in the Albergate dataset; (3) When comparing the best-performing combinations, the eTOUR and iTrust datasets outperform the Albergate dataset in most models.

In conclusion, this experiment further compares the annotations across datasets and reveals that the eTOUR and iTrust datasets have more complete annotations, while nearly half of the code files in Albergate are unannotated. These findings highlight the critical importance of standardized, high-quality annotations during software development to enable effective automated traceability.

The effect of RT for both IR models on document-level artifacts is shown in Fig. 10, and detailed experimental results are presented in Appendix C. The results show that among the five models (VSM, LSI, LDA, BTM, and JS), VSM and LSI consistently exhibit stable and relatively strong performance across most tasks. Specifically, VSM maintains more stable performance in datasets such as EasyClinic

Table 19
Standards from Hayes et al. (2006).

Measure	Acceptable	Good	Excellent
Recall	60% — 69%	70% — 79%	80% — 100%
Precision	20% — 29%	30% — 49%	50% — 100%

Table 20
Source code sections utilized in experimentation.

Acronym	Identifier Type
CN	Class Names
MN	All Public and Private Method Names of a Class
VN	Class and Method Variable Names of a Class
CMT	All Block and Single Line Comments of a Class

Table 21
The results of source-code partitions' combination for VSM.

	Albergate			eTOUR			iTrust		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
CN	0.6604	0.0750	0.1346	0.4578	0.0524	0.0940	0.2105	0.0166	0.0307
MN	0.7359	0.0759	0.1376	0.1169	0.1071	0.1118	0.0550	0.0777	0.0644
VN	**0.1509**	**0.4444**	**0.2253**	0.2143	0.1964	0.2050	0.4641	0.0164	0.0316
CMT	0.1698	0.1385	0.1525	0.2468	0.3781	0.2986	0.2943	0.4155	0.3445
CN+MN	0.6981	0.0720	0.1305	0.1591	0.0810	0.1073	0.0550	0.0777	0.0644
CN+VN	0.3019	0.1600	0.2092	0.2013	0.1845	0.1925	0.4641	0.0164	0.0316
CN+CMT	0.1698	0.1216	0.1417	0.3377	0.2581	0.2925	0.2943	0.4155	0.3445
MN+VN	0.1698	0.2432	0.2000	0.2597	0.1322	0.1752	0.1005	0.0284	0.0443
MN+CMT	0.1321	0.1250	0.1284	0.2468	0.3781	0.2986	**0.3038**	**0.4291**	**0.3557**
VN+CMT	0.2642	0.1373	0.1806	**0.3604**	**0.2754**	**0.3122**	0.2943	0.4155	0.3445
CN+MN+VN	0.2076	0.1964	0.2018	0.2305	0.1320	0.1678	0.1005	0.0284	0.0443
CN+MN+CMT	0.1321	0.1250	0.1284	0.3084	0.2827	0.2950	0.3014	0.4257	0.3529
CN+VN+CMT	0.2264	0.1200	0.1569	0.3507	0.2680	0.3038	0.2990	0.4223	0.3501
MN+VN+CMT	0.3019	0.1074	0.1584	0.2857	0.3271	0.3050	0.2990	0.4223	0.3501
CN+MN+VN+CMT	0.1509	0.1081	0.1260	0.3214	0.2946	0.3075	0.2990	0.4223	0.3501

Table 22
The results of source-code partitions' combination for LSI.

LSI	Albergate			eTOUR			iTrust		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
CN	0.6604	0.0728	0.1308	0.5812	0.0532	0.0975	0.2105	0.0166	0.0307
MN	0.7359	0.0759	0.1376	0.1136	0.1301	0.1213	0.0861	0.1216	0.1008
VN	**0.1509**	**0.4444**	**0.2253**	0.2078	0.1905	0.1988	0.3636	0.0205	0.0389
CMT	0.1698	0.1385	0.1525	0.2403	0.3682	0.2908	0.5120	0.1807	0.2672
CN+MN	0.6981	0.0720	0.1305	0.1136	0.1301	0.1213	0.0813	0.1149	0.0952
CN+VN	**0.1509**	**0.4444**	**0.2253**	0.2630	0.1506	0.1915	0.2703	0.0294	0.0530
CN+CMT	0.1698	0.1216	0.1417	0.3117	0.2857	0.2981	0.3110	0.4392	0.3641
MN+VN	0.1698	0.2432	0.2000	0.1786	0.1637	0.1708	0.2727	0.0227	0.0418
MN+CMT	0.1509	0.1081	0.1260	0.2468	0.3781	0.2986	0.3158	0.4460	0.3697
VN+CMT	0.1321	0.2500	0.1728	**0.2825**	**0.3234**	**0.3016**	0.3038	0.4291	0.3557
CN+MN+VN	0.2264	0.1846	0.2034	0.2110	0.1383	0.1671	0.1842	0.0325	0.0553
CN+MN+CMT	0.1321	0.1250	0.1284	0.2468	0.3781	0.2986	0.3344	0.2556	0.2897
CN+VN+CMT	0.2264	0.1200	0.1569	0.3409	0.2606	0.2954	0.3086	0.4358	0.3613
MN+VN+CMT	0.3019	0.1074	0.1584	0.2435	0.3731	0.2947	0.3158	0.4460	0.3697
CN+MN+VN+CMT	0.1509	0.1081	0.1260	0.2792	0.3197	0.2981	**0.3182**	**0.4493**	**0.3725**

(CC–TC) and WARC, highlighting its strength in capturing lexical relationships, while LSI shows competitive results in tasks like CM-1 and EBT (RE–TC), reflecting its ability to model latent semantic similarity. BTM also performs competitively, particularly in tasks like EasyClinic (UC–TC), where its use of biterms proves effective in capturing meaningful document relationships. However, its performance is notably weaker in tasks like EasyClinic (CC–CC) and EasyClinic (UC–CC), suggesting limitations in capturing detailed contextual relationships necessary for effective document-level traceability. The JS divergence model consistently underperforms in most tasks, likely due to its sensitivity to document length and vocabulary sparsity. In comparison, LDA performs the worst overall, showing poor results across nearly all datasets. This poor performance stems from LDA's coarse-grained topic representation, which fails to capture the finer nuances required for effective requirement tracing.

Overall, these results suggest that VSM and LSI are robust across both code-level and document-level artifacts, making them strong baseline models for requirement-to-code traceability. Future research can adopt them as reference points for evaluating the effectiveness of more advanced or hybrid approaches.

Table 23
The results of source-code partitions' combination for LDA.

	Albergate			eTOUR			iTrust		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
CN	**0.5283**	**0.1**	**0.1682**	0.3896	0.0595	0.1032	0.1579	0.0248	0.0428
MN	**0.5283**	**0.1**	**0.1682**	1	0.0458	0.0875	**0.1388**	**0.098**	**0.1149**
VN	0.2075	0.0786	0.114	**0.4253**	**0.0649**	**0.1126**	0.122	0.0345	0.0537
CMT	0.6981	0.066	0.1205	0.1591	0.0662	0.0935	0.0861	0.0608	0.0713
CN+MN	**0.5283**	**0.1**	**0.1682**	1	0.0458	0.0875	0.1794	0.0633	0.0936
CN+VN	0.8302	0.0588	0.1099	0.4026	0.0614	0.1066	0.1268	0.0358	0.0558
CN+CMT	0.6981	0.066	0.1205	0.2208	0.0594	0.0937	0.1459	0.0343	0.0556
MN+VN	0.2264	0.08	0.1182	0.5195	0.0528	0.0959	0.0646	0.0456	0.0535
MN+CMT	0.6792	0.07	0.127	0.2078	0.0559	0.0882	0.1196	0.0282	0.0456
VN+CMT	1	0.0567	0.1073	0.1558	0.0649	0.0916	0.189	0.0334	0.0567
CN+MN+VN	0.1887	0.082	0.1143	0.5584	0.0511	0.0937	0.0885	0.0417	0.0567
CN+MN+CMT	0.6226	0.0642	0.1164	1	0.0458	0.0875	0.0502	0.0709	0.0588
CN+VN+CMT	1	0.0567	0.1073	0.4156	0.0544	0.0961	0.1794	0.0362	0.0602
MN+VN+CMT	0.1887	0.0763	0.1087	1	0.0458	0.0875	0.2225	0.0314	0.055
CN+MN+VN+CMT	1	0.0567	0.1073	1	0.0458	0.0875	0.0694	0.049	0.0574

Table 24
The results of source-code partitions' combination for BTM.

	Albergate			eTOUR			iTrust		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
CN	0.5094	0.0841	0.1444	0.6169	0.0557	0.1021	0.1579	0.0326	0.054
MN	0.3208	0.0726	0.1185	0.8701	0.049	0.0927	0.7464	0.0192	0.0374
VN	0.3208	0.0955	0.1472	0.539	0.0511	0.0933	0.9976	0.0143	0.0281
CMT	0.5472	0.0609	0.1096	0.4545	0.0616	0.1084	0.2751	0.045	0.0773
CN+MN	0.4528	0.0734	0.1263	0.1623	0.0743	0.1019	0.3612	0.0255	0.0476
CN+VN	0.3019	0.0899	0.1385	0.6916	0.0528	0.098	1	0.0141	0.0278
CN+CMT	**0.2264**	**0.1165**	**0.1538**	0.2208	0.1444	0.1746	0.4139	0.1461	0.216
MN+VN	0.5094	0.0826	0.1421	0.5065	0.0595	0.1065	0.1364	0.0214	0.037
MN+CMT	0.1509	0.1231	0.1356	0.2305	0.1507	0.1823	**0.2775**	**0.3919**	**0.3249**
VN+CMT	0.7736	0.0731	0.1336	0.4708	0.0627	0.1106	0.3373	0.2382	0.2792
CN+MN+VN	0.283	0.1	0.1478	0.2987	0.076	0.1211	0.0718	0.0338	0.0459
CN+MN+CMT	0.2075	0.0902	0.1257	0.2403	0.1571	0.19	0.2656	0.375	0.3109
CN+VN+CMT	0.1698	0.1385	0.1525	0.2435	0.1394	0.1773	0.3254	0.2297	0.2693
MN+VN+CMT	0.2453	0.0992	0.1413	0.2305	0.1507	0.1823	0.3589	0.2534	0.297
CN+MN+VN+CMT	0.3019	0.0856	0.1333	**0.2468**	**0.1614**	**0.1951**	0.2464	0.348	0.2885

Table 25
The results of source-code partitions' combination for JS.

	Albergate			eTOUR			iTrust		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
CN	0.9811	0.0586	0.1105	0.5812	0.0532	0.0975	1	0.0141	0.0278
MN	0.717	0.0581	0.1075	1	0.0458	0.0875	0.0478	0.0676	0.056
VN	0.9434	0.0594	0.1117	0.6688	0.051	0.0948	0.2679	0.0223	0.0411
CMT	0.4906	0.0695	0.1218	0.1883	0.0862	0.1182	0.1818	0.0428	0.0693
CN+MN	0.8113	0.0575	0.1074	0.8831	0.0476	0.0903	0.0455	0.0642	0.0532
CN+VN	1	0.0567	0.1073	0.2662	0.0642	0.1034	0.0526	0.0372	0.0436
CN+CMT	**0.1132**	**0.2143**	**0.1481**	0.1136	0.2593	0.158	0.1148	0.1622	0.1345
MN+VN	0.1509	0.0851	0.1088	0.8312	0.0476	0.09	0.6603	0.017	0.0331
MN+CMT	**0.1132**	**0.2143**	**0.1481**	0.1104	0.2519	0.1535	0.1148	0.1622	0.1345
VN+CMT	0.1132	0.1622	0.1333	0.1786	0.1022	0.13	**0.1196**	**0.1689**	**0.1401**
CN+MN+VN	0.3962	0.0642	0.1105	0.9026	0.0486	0.0923	0.8325	0.0157	0.0308
CN+MN+CMT	**0.1132**	**0.2143**	**0.1481**	0.1136	0.1733	0.1373	0.1148	0.1622	0.1345
CN+VN+CMT	0.0943	0.1786	0.1235	0.1201	0.1832	0.1451	**0.1196**	**0.1689**	**0.1401**
MN+VN+CMT	0.4528	0.0642	0.1124	**0.1299**	**0.198**	**0.1569**	0.1005	0.1419	0.1176
CN+MN+VN+CMT	0.4528	0.0642	0.1124	0.1104	0.1683	0.1333	0.1005	0.1419	0.1176

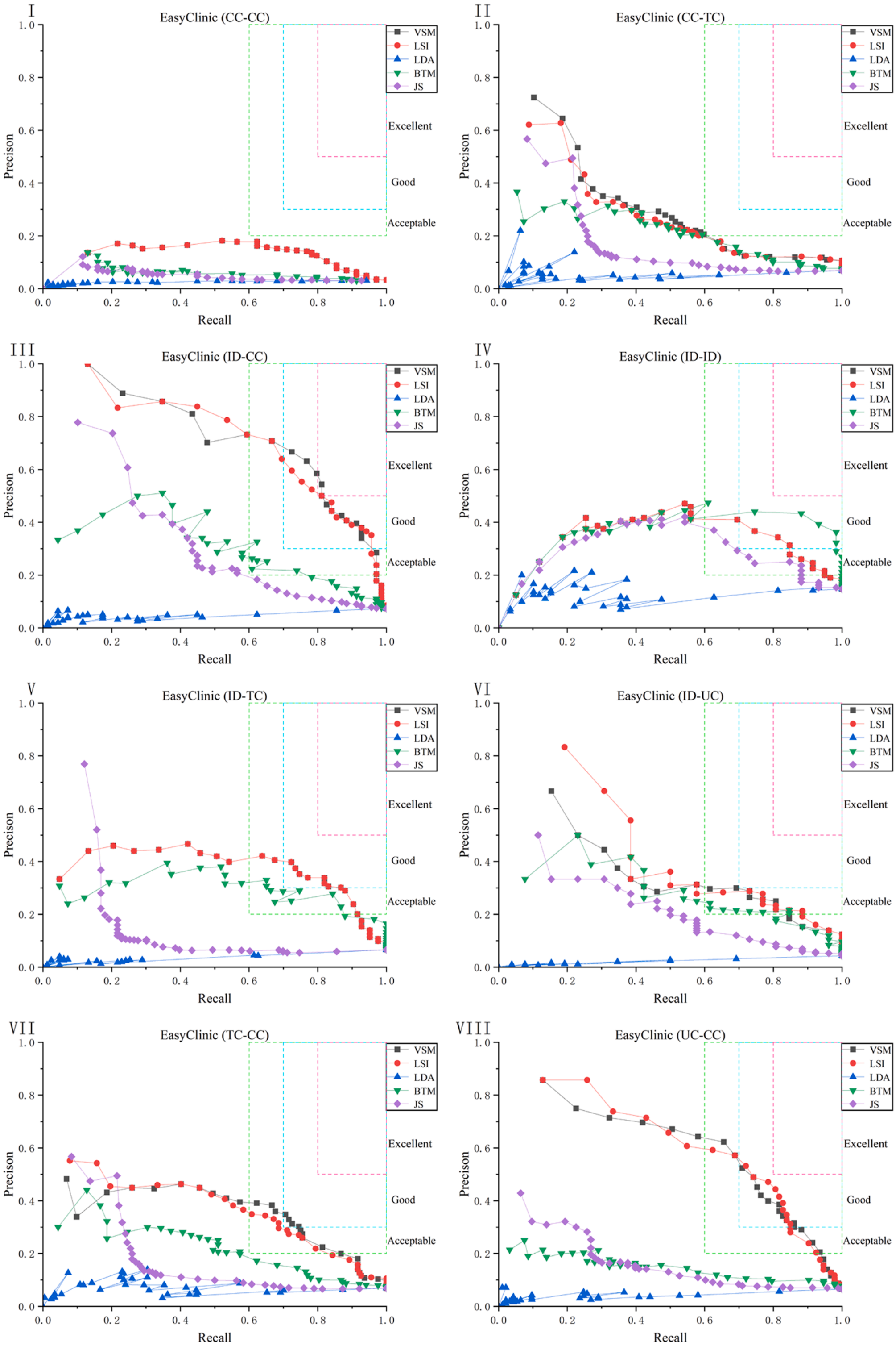

Fig. 10. Recall-Precision for CoEST datasets (baselines).

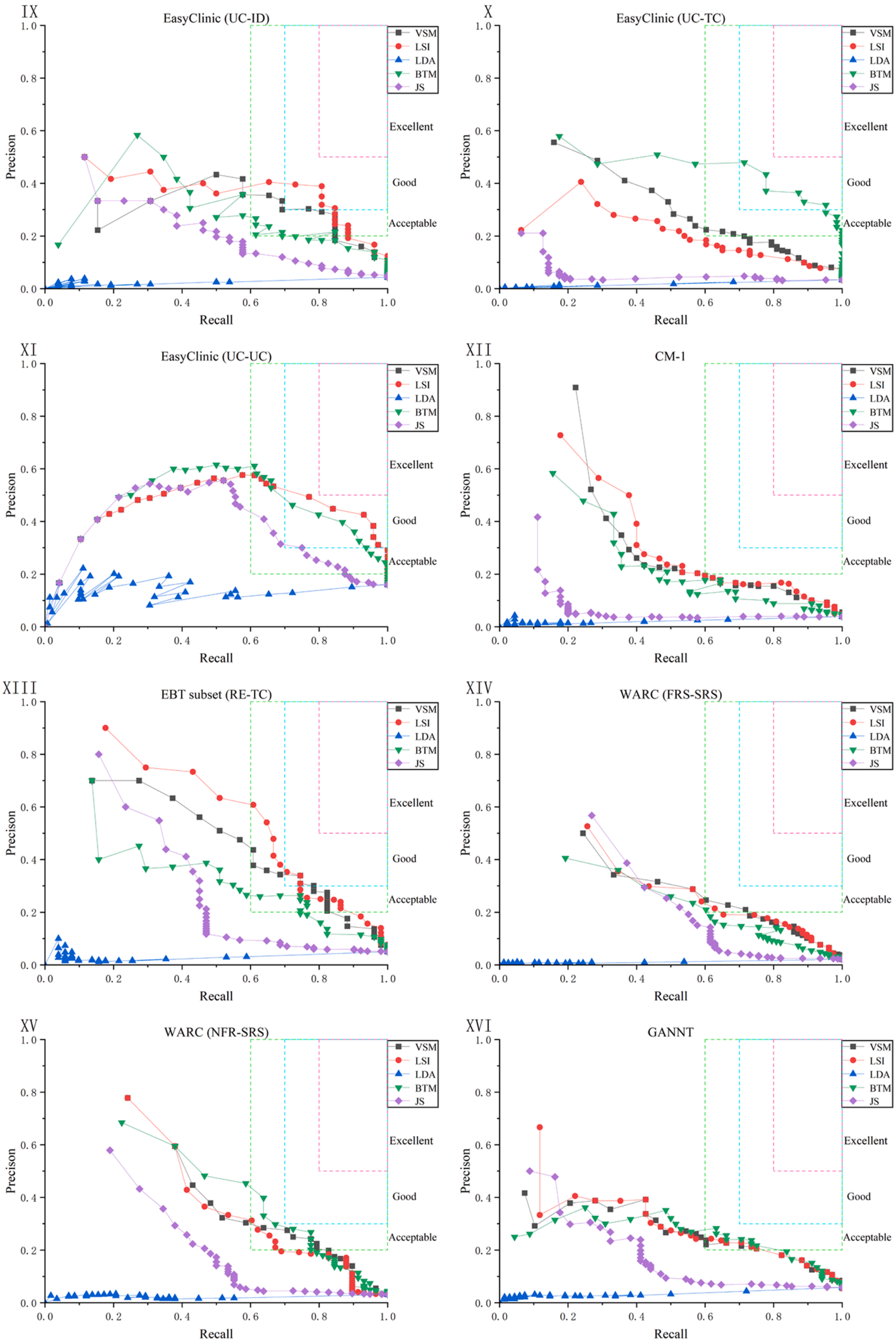

Fig. 10. (*continued*).

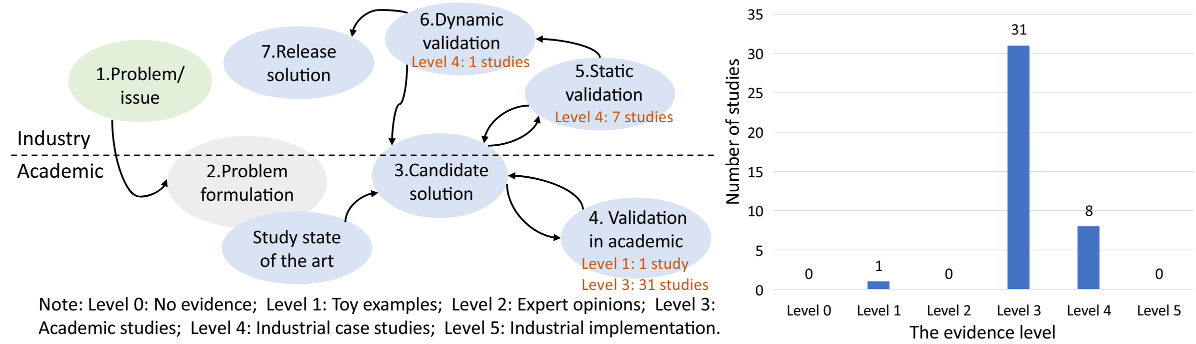

Fig. 11. Distribution of studies across technology transfer stages and levels of evidence.

4.8. *What evidence has been documented to validate the performance evaluation of IR-based RT methods (RQ8)?*

In Table 6, the data item (D14) is used to address RQ8. The evidence level plays a crucial role in evaluating the reliability of a study. As outlined in Section 3.4, the six-tier evidence hierarchy reflects the extent to which the study's findings have been validated in real-world industrial contexts. To further contextualize this, we employ Gorschek et al.'s technology transfer model (Gorschek et al., 2006), which describes seven stages for translating research into industrial practice through academia–industry collaboration.

Fig. 11 illustrates the distribution of selected studies by technology transfer stages and evidence levels (Levels 0 to 5). Following the classification criteria proposed by Ivarsson and Gorschek (2009), we map the 40 selected studies to the corresponding stages of the technology transfer process. A total of 80% of the studies (32/40) are positioned at the "Validation in Academia" stage and correspond to Level 1 (1 study) and Level 3 (31 studies) evidence. Conversely, 20% of the studies (8/40) offer Level 4 evidence using industrial datasets, with most being static validation and one achieving dynamic validation. Notably, while all studies provide some empirical evidence (excluding Level 0), none demonstrate Level 5 evidence characterized by practical implementation and adoption in industrial settings.

This distribution highlights a heavy concentration of research within academia, with limited advancement toward dynamic validation or real-world deployment. The over-reliance on academic studies restricts the generalizability and practical impact of current findings, largely due to several persistent challenges. First, industrial datasets are often inaccessible because of confidentiality concerns, intellectual property limitations, and the absence of standardized data-sharing mechanisms. Moreover, the rapid evolution of industrial requirements and development practices makes it difficult to evaluate traceability methods under realistic and dynamic conditions (Ivarsson & Gorschek, 2009). Achieving dynamic validation requires sustained collaboration with industrial partners and the integration of traceability approaches into active development workflows. These efforts demand significant resources, long-term commitment, and organizational support, which are often difficult to secure within academic research environments. Future work should prioritize stronger academia-industry collaboration and focus on developing portable, adaptable traceability solutions that can be effectively integrated into real-world software engineering processes.

5. Discussion

5.1. *Threats to validity and limitations*

This section discusses the study's limitations and potential threats to validity. Following the recommended guidelines (Petersen et al., 2015; Zhou et al., 2016), we discuss four aspects: conclusion validity, construct validity, internal validity, and external validity.

Conclusion validity: A primary concern regarding conclusion validity is the risk of excluding relevant studies due to stringent inclusion criteria, which may compromise the comprehensiveness of our review. Although these criteria were carefully defined and applied (as detailed in Section 3.2) to ensure consistency and focus, some marginally relevant studies may have been inadvertently omitted. In addition, Section 4.7 outlines the experimental setup, including dataset selection, preprocessing steps, threshold configuration, and evaluation metrics, all designed to enhance the credibility of the results. However, the experiments in Section 4.7 focus on specific IR models, including VSM, LSI, LDA, BTM, and JS, limiting the generalizability of findings to other IR techniques not evaluated. Additionally, our restriction to peer-reviewed publications indexed in CCF, CORE, or QUALIS rankings, while ensuring academic rigor, may have excluded pertinent studies from non-ranked venues, introducing potential publication bias.

Construct validity: While our initial database search may not have captured all relevant studies, we took steps to improve its coverage. Through iterative query refinement informed by insights from the literature, we developed an effective search strategy that captured a substantial proportion of relevant studies. Although we queried six major digital libraries, their combined coverage remains partial, and some relevant works may have used uncommon terminology or appeared in unindexed venues. To address these limitations, we further employed both backward and forward snowballing techniques to supplement the initial search, thereby ensuring a more comprehensive search process.

Internal validity: A primary threat to internal validity arises from potential subjectivity in data extraction and literature classification. Manual selection and exclusion of studies inherently involve a degree of subjectivity, especially when evaluating borderline cases. Although two authors independently conducted study selection and data extraction, with all discrepancies resolved through iterative team discussions to achieve consensus, some residual interpretive bias may remain unavoidable. To mitigate this risk, we implemented rigorous measures, including independent data extraction by the first and second authors using predefined criteria. Discrepancies or ambiguities were resolved through team discussions, with final decisions made by consensus to ensure methodological rigor. Nevertheless, the interpretive nature of classifying literature, particularly in cases of ambiguous studies, introduces unavoidable subjectivity.

External validity: External validity concerns the extent to which the results of this SMS can be generalized, depending on how well the primary studies reflect the review topics. To mitigate external threats, the search process outlined in Section 3.2 was developed after conducting several trial searches. In addition, the experiments in Section 4.7 are limited by their reliance on CoEST's open-source datasets. Although these datasets are commonly used to evaluate IR methods, their limited complexity and scale, as well as their lack of domain-specific coverage, may reduce the generalizability of our findings in real-world industrial contexts. Furthermore, approximately 40% of the reviewed studies used non-open-source datasets, which may also compromise the external validity of the overall results.

5.2. *Future research direction*

Our systematic review reveals several critical yet understudied research directions that hold significant potential for advancing the field of RT. As visually synthesized in Fig. 12, these key opportunities include:

(1) **Enhancing VSM through Adaptive Integration of Structural Features:** Experimental results demonstrate that VSM exhibits poor performance on code-level datasets like Albergate, primarily because it struggles to capture the complex relationships between requirements and code. While existing studies (Kuang et al., 2017; Zhang et al., 2022) have attempted to improve VSM by incorporating structural information, these approaches remain limited. Their dependence on static rules for structural feature extraction makes them unable to adapt to continuous code evolution (Zhou et al., 2024), resulting in traceability links that quickly become outdated after code modifications. Future research should explore dynamic approaches, such as real-time code change analysis using code evolution graphs (Stein et al., 2025), to enable automatic adjustment of structural features. This would facilitate automated maintenance and updating of traceability links in continuous integration environments.

(2) **Hybrid IR Models:** The SMS results show that while various IR-based methods have been used to recover traceability links between software artifacts, no single method consistently outperforms others in all scenarios. Prior studies (Gethers et al., 2011; Oliveto et al., 2010) suggest that different IR models capture distinct aspects of artifacts, indicating both orthogonality and potential complementarity among them. For example, VSM's direct lexical matching could be enhanced by LSI's ability to model latent semantic relationships. Future research should explore hybrid IR models that combine the strengths of multiple approaches to improve traceability recovery across diverse datasets.

(3) **Advancing Multimodal Traceability Recovery:** RQ7 shows that current RT research predominantly focuses on textual artifacts like code and documentation. Although studies have examined artifacts such as UML diagrams and architectural models, these are typically converted to textual representations, losing valuable multimodal information. Recent advances in multimodal learning (Shi et al., 2025) present opportunities to better leverage diverse artifact types including text, images, and models. Such approaches could significantly enhance traceability robustness and enable unified linking across artifact modalities. However, the scarcity of multimodal datasets poses a significant barrier to advancing this research direction.

(4) **Diversification of Datasets:** As evidenced in Section 4.4, current research exhibits excessive dependence on CoEST datasets. While this practice enables cross-study comparability, it potentially limits the generalizability and robustness of findings. To address this limitation, future work should incorporate more diverse datasets that vary in scale, domain, and system characteristics, while expanding validation across multiple publicly available datasets to enhance reproducibility. When confidentiality constraints allow, researchers should also make non-open-source datasets available to improve accessibility and enable more comprehensive evaluation.

(5) **Artificial Intelligence for Enhancing Traceability and Labeling Efficiency:** As evidenced by complex datasets like EasyClinic (CC–CC), traditional IR methods often produce suboptimal results, with F1 scores consistently below 0.4. This underscores the demand for more effective and scalable solutions, particularly in data-scarce scenarios with limited annotations. Recent advances in LLMs, such as ChatGPT and Gemma, offer promising opportunities to enhance traceability recovery by capturing richer semantic relationships and contextual dependencies. These models can assist in key subtasks, including candidate link ranking, semantic filtering, and automatic annotation generation. While IR methods remain lightweight and interpretable, they struggle to generalize across heterogeneous software artifacts. In contrast, LLMs provide superior semantic generalization but at the cost of higher computational demands. Future research should explore hybrid approaches that combine the efficiency of IR techniques with the expressive power of LLMs, aiming to strike an optimal balance between performance, scalability, and real-world applicability in software development environments.

(6) **Enhancement of Industrial Dynamic Validation:** As summarized in Section 4.8, most current research remains at the academic validation stage, with limited industrial case studies and a lack of dynamic validation. To address these limitations, future work should strengthen collaboration with industrial partners, establishing standardized data-sharing protocols to facilitate access to industrial datasets, and integrating traceability methods into active industrial development workflows to enable

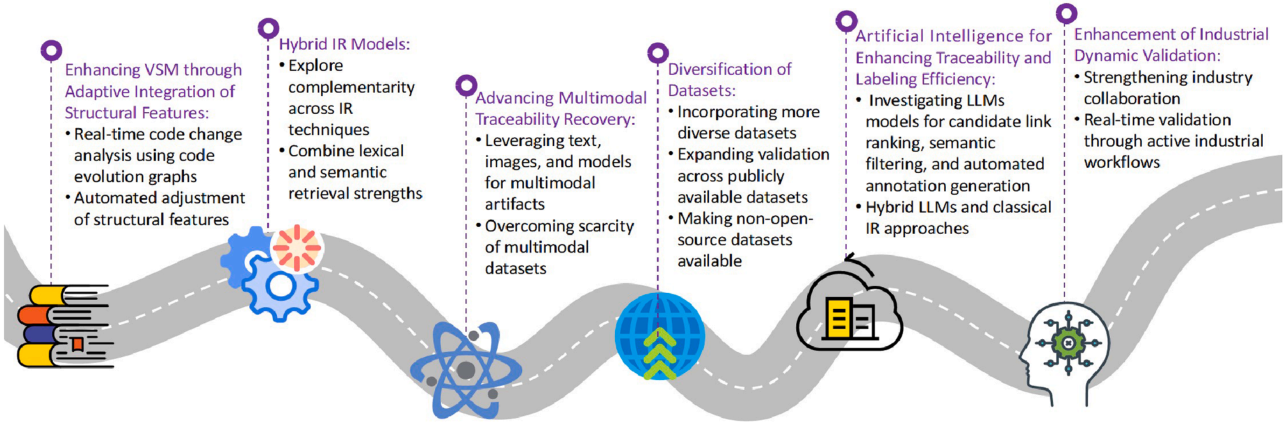

Fig. 12. Research roadmap for future directions in RT.

dynamic, real-time validation. These advancements are crucial for ensuring that IR-based RT methods demonstrate scalability, adaptability, and practical applicability across diverse and evolving industrial environments.

6. Conclusion

RT has long been recognized as a labor-intensive, time-consuming, and error-prone task (Wang et al., 2018). In the era of large models, while advanced techniques such as pre-trained models have gained prominence, IR methods continue to be crucial for automating the recovery of trace links between requirements artifacts and other software artifacts (Wang et al., 2018). This SMS offers an extensive analysis of the state-of-the-art methods that have utilized IR in automating trace link generation. Through the examination of forty primary studies, the following conclusions have been derived:

(1) Over the past decade, VSM and LSI have been the primary IR models used for retrieval tasks, followed by JS.
(2) Thirty-two enhancement strategies have been proposed to improve the performance of these models. Researchers are encouraged to employ a variety of these strategies to create a hybrid method.
(3) The CoEST repository represents the most favored, accessible, and reliable source for dataset collections. Researchers are encouraged to utilize a variety of open-source datasets to facilitate the reproducibility of experiments and validate proposed methods. Using multiple datasets can also help increase the generalizability of findings.
(4) Precision and recall are the primary measures for assessing IR techniques, with MAP being the favored secondary measure. It is recommended that researchers evaluate IR methods across various measures to obtain a comprehensive assessment of their performance.
(5) The experiments in this study are based on pure VSM, LSI, LDA, BTM and JS, using eight commonly used datasets from CoEST. The results show that VSM and LSI perform best on most datasets, providing a baseline for generating trace links using IR-based methods.
(6) The majority of studies remain at Evidence Level 3, indicating academic validation under controlled conditions. In contrast, only a small portion of the literature provides Level 4 evidence based on industrial case studies, highlighting the gap between research and practical application.

CRediT authorship contribution statement

Hongyan Wan: Writing – review & editing, Writing – original draft, Supervision, Funding acquisition. **Xinyu He:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Yang Deng:** Writing – review & editing, Supervision, Methodology. **Bangchao Wang:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT and DeepSeek in order to improve language and readability. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 62102291), and the Opening Foundation of Engineering Research Center of Hubei Province for Clothing Information (No. 2022HBCI02, No. 2022HBCI05).

Appendix A. List of excluded studies at the full-text eligibility stage

Index	Title	Exclusion Reason
1	Recovering Traceability Links between Release Notes and Related Software Artifacts	not focus on RT
2	Enhancing the Performance of Ir-Based Traceability Recovery of Requirement Artifacts Using Noun Phrases	their venues are not included in the Conference Partner list
3	Advancing Trace Recovery Evaluation-Applied Information Retrieval in a Software Engineering Context	their venues are not included in the Conference Partner list
4	Exploring Semantics of Software Artifacts to Improve Requirements Traceability Recovery: A Hybrid Approach	not focus on IR techniques
5	Towards Recovering Fault Traceability Links by Using Information Retrieval Technique	their venues are not included in the Conference Partner list
6	Estimating the number of remaining links in traceability recovery	not focus on IR techniques
7	Comparing traceability through information retrieval, commits, interaction logs, and tags	not focus on RT
8	Automating traceability link recovery through classification	not focus on IR techniques
9	Requirements Traceability Through Information Retrieval Using Dynamic Integration of Structural and Co-change Coupling	their venues are not included in the Conference Partner list
10	Requirements traceability recovery for the purpose of software reuse: an interactive genetic algorithm approach	not focus on IR techniques
11	AVIATE: Exploiting Translation Variants of Artifacts to Improve IR-based Traceability Recovery in Bilingual Software Projects	not focus on RT
12	Traceability in the Wild: Automatically Augmenting Incomplete Trace Links	not focus on RT
13	Enhancing Unsupervised Requirements Traceability with Sequential Semantics	not focus on IR techniques
14	Information retrieval versus deep learning approaches for generating traceability links in bilingual projects	not focus on RT
15	Analyzing Tools and Techniques for Evaluating Requirements Traceability	not empirical study
16	A multi-faceted roadmap of requirements traceability types adoption in SCRUM: An empirical study	their venues are not included in the Conference Partner list
17	Enhancing Traceability Link Recovery with Unlabeled Data	not focus on IR techniques
18	Traceability Support for Multi-Lingual Software Projects	not focus on RT
19	Modeling Traceability Between Requirements and Coding Using the Property Listing Task	not focus on IR techniques
20	Constructing Traceability Links between Software Requirements and Source Code Based on Neural Networks	not focus on IR techniques
21	Semi-supervised Pre-processing for Learning-Based Traceability Framework on Real-World Software Projects	not focus on RT
22	Automatic traceability link recovery via active learning	not focus on IR techniques
23	MBSE-driven visualization of requirements allocation and traceability	not focus on IR techniques
24	Search-Based Requirements Traceability Recovery	not focus on IR techniques
25	Traceability Link Recovery between Requirements and Models using an Evolutionary Algorithm Guided by a Learning to Rank Algorithm: Train Control and Management Case	not focus on IR techniques
26	Combining Machine Learning and Logical Reasoning to Improve Requirements Traceability Recovery	not focus on IR techniques
27	Enhancing Automated Requirements Traceability by Resolving Polysemy	not focus on IR techniques
28	Recovering Traceability Links Between Code and Specification Through Domain Model Extraction	not focus on IR techniques
29	Evaluation of Textual Similarity Techniques in Code Level Traceability	not focus on RT
30	Requirements Traceability: Recovering and Visualizing Traceability Links Between Requirements and Source Code of Object-oriented Software Systems	their venues are not included in the Conference Partner list
31	Multilevel Traceability Links Establishments Between SOFL Formal Specifications and Java Codes Using Multi-dimensional Similarity Measures	not focus on RT
32	A Cross-Level Requirement Trace Link Update Model Based on Bidirectional Encoder Representations from Transformers	not focus on IR techniques
33	An Improved Approach based on Balanced Keyword Weight to Traceability Recovery	their venues are not included in the Conference Partner list
34	Clustering for traceability managing in system specifications	not focus on IR techniques
35	Recovering Trace Links Between Software Documentation and Code	not focus on IR techniques
36	Cross-level Requirement Traceability: A Novel Approach Integrating Bag-of-Words and Word Embedding for Enhanced Similarity Functionality	their venues are not included in the Conference Partner list
37	Recovering Semantic Traceability between Requirements and Source Code Using Feature Representation Techniques	not focus on IR techniques
38	Semantic Recovery of Traceability Links between System Artifact	not focus on IR techniques
39	Do Information Retrieval Algorithms for Automated Traceability Perform Effectively on Issue Tracking System Data?	not focus on RT
40	How Effective Is Automated Trace Link Recovery in Model-Driven Development?	not focus on RT

Appendix B. Selected studies

Index	Title
S1	An empirical study on recovering requirement-to-code links
S2	An Empirical Study on Source Code Feature Extraction in Preprocessing of IR-Based Requirements Traceability
S3	Configuring Latent Semantic Indexing for Requirements Tracing
S4	IRRT: An Automated Software Requirements Traceability Tool based on Information Retrieval Model
S5	Analyzing closeness of code dependencies for improving IR-based Traceability Recovery
S6	An empirical study on the importance of source code entities for requirements traceability
S7	Analyzing close relations between target artifacts for improving IR-based requirement traceability recovery
S8	A Complete Traceability Methodology Between UML Diagrams and Source Code Based on Enriched Use Case Textual Description
S9	Achieving better requirements to code traceability:which refactoring should be done first?
S10	Using Consensual Biterms from Text Structures of Requirements and Code to Improve IR-Based Traceability Recovery
S11	A Semantic Approach for Traceability Link Recovery in Aerospace Requirements Management System
S12	An IR-based Artificial Bee Colony Approach for Traceability Link Recovery
S13	Propagating frugal user feedback through closeness of code dependencies to improve IR-based traceability recovery
S14	Leveraging BPMN particularities to improve traceability links recovery among requirements and BPMN models
S15	Combining VSM and BTM to improve requirements trace links generation
S16	TRIAD: Automated Traceability Recovery based on Biterm-enhanced Deduction of Transitive Links among Artifacts
S17	Visualizing Software Repositories through Requirements Trace Links
S18	Multi-Objective Information Retrieval-Based NSGA-II Optimization for Requirements Traceability Recovery
S19	Filtering of false positives from IR-based traceability links among software artifacts
S20	Quality improvements for trace links between source code and requirements
S21	Evaluation of Natural Language Processing for Requirements Traceability
S22	ATLaS: A Framework for Traceability Links Recovery Combining Information Retrieval and Semi-supervised Techniques
S23	An Automated Hybrid Approach for Generating Requirements Trace Links
S24	Evaluating the Effectiveness of Various IR Models for Requirements Traceability
S25	Search-Based Requirements Traceability Recovery: A Multi-Objective Approach
S26	Supporting Requirements to Code Traceability Creation by Code Comments
S27	SANAYOJAN A framework for traceability link recovery between use-cases in software requirement specification and regulatory documents
S28	Evolving Software Trace Links between Requirements and Source Code
S29	Interactive recovery of requirements traceability links using user feedback and configuration management logs
S30	Supporting requirements to code traceability through refactoring
S31	Recovering traceability links between requirements and source code using the configuration management log
S32	Exploiting Parts-of-Speech for Effective Automated Requirements Traceability
S33	Using Frugal User Feedback with Closeness Analysis on Code to Improve IR-Based Traceability Recovery
S34	Adaptive User Feedback for IR-Based Traceability Recovery
S35	On the role of semantics in automated requirements tracing
S36	Requirements Classification for Traceability Link Recovery
S37	Enhancing Traceability Link Recovery with Fine-Grained Query Expansion Analysis
S38	DF4RT: Deep Forest for Requirements Traceability Recovery between Use Cases and Source Code
S39	Enhancing requirements-to-code traceability with GA-XWCoDe: Integrating XGBoost, Node2Vec, and genetic algorithms for improving model performance and stability
S40	MLTracer: An Approach Based on Multi-Layered Gradient Boosting Decision Trees for Requirements Traceability Recovery

Appendix C. The results of VSM, LSI, LDA, BTM and JS for Top 8 datasets in CoEST

WARC (NFR–SRS)
GANNT

EasyClinic (CC–CC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.02	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.45%	2.27%	1.77%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.03	13.04%	13.64%	13.33%	13.04%	13.64%	13.33%	1.45%	1.52%	1.48%	13.04%	13.64%	13.33%	11.59%	12.12%	11.85%
0.04	21.74%	17.05%	19.11%	21.74%	17.05%	19.11%	1.45%	1.14%	1.27%	15.94%	12.50%	14.01%	11.59%	9.09%	10.19%
0.05	26.09%	16.36%	20.11%	26.09%	16.36%	20.11%	1.45%	0.91%	1.12%	15.94%	10.00%	12.29%	13.04%	8.18%	10.06%
0.06	28.99%	15.15%	19.90%	28.99%	15.15%	19.90%	2.90%	1.50%	1.98%	18.84%	9.77%	12.87%	15.94%	8.27%	10.89%
0.07	34.78%	15.58%	21.52%	34.78%	15.58%	21.52%	0.00%	0.00%	0.00%	17.39%	7.74%	10.71%	15.94%	7.10%	9.82%
0.08	42.03%	16.48%	23.68%	42.03%	16.48%	23.68%	2.90%	1.13%	1.63%	20.29%	7.91%	11.38%	17.39%	6.78%	9.76%
0.09	52.17%	18.18%	26.96%	52.17%	18.18%	26.96%	2.90%	1.01%	1.49%	23.19%	8.04%	11.94%	18.84%	6.53%	9.70%
0.10	56.52%	17.73%	26.99%	56.52%	17.73%	26.99%	5.80%	1.81%	2.76%	20.29%	6.33%	9.66%	24.64%	7.69%	11.72%
0.11	62.32%	17.77%	27.65%	62.32%	17.77%	27.65%	4.35%	1.23%	1.92%	24.64%	7.00%	10.90%	26.09%	7.41%	11.54%
0.12	62.32%	16.23%	25.75%	62.32%	16.23%	25.75%	8.70%	2.26%	3.59%	26.09%	6.79%	10.78%	26.09%	6.79%	10.78%
0.13	65.22%	15.68%	25.28%	65.22%	15.68%	25.28%	7.25%	1.74%	2.81%	24.64%	5.92%	9.55%	26.09%	6.27%	10.11%
0.14	68.12%	15.21%	24.87%	68.12%	15.21%	24.87%	5.80%	1.29%	2.12%	26.09%	5.83%	9.52%	28.99%	6.47%	10.58%
0.15	71.01%	14.80%	24.49%	71.01%	14.80%	24.49%	7.25%	1.51%	2.50%	28.99%	6.04%	10.00%	30.43%	6.34%	10.50%
0.16	73.91%	14.45%	24.17%	73.91%	14.45%	24.17%	4.35%	0.85%	1.42%	33.33%	6.52%	10.90%	30.43%	5.95%	9.95%
0.17	76.81%	14.13%	23.87%	76.81%	14.13%	23.87%	8.70%	1.60%	2.70%	34.78%	6.38%	10.79%	30.43%	5.59%	9.44%
0.18	78.26%	13.60%	23.17%	78.26%	13.60%	23.17%	11.59%	2.01%	3.43%	40.58%	7.04%	11.99%	31.88%	5.53%	9.42%
0.19	78.26%	12.89%	22.13%	78.26%	12.89%	22.13%	13.04%	2.14%	3.68%	37.68%	6.19%	10.63%	33.33%	5.48%	9.41%
0.20	79.71%	12.47%	21.57%	79.71%	12.47%	21.57%	15.94%	2.49%	4.31%	42.03%	6.56%	11.35%	34.78%	5.43%	9.39%
0.25	82.61%	10.33%	18.36%	82.61%	10.33%	18.36%	20.29%	2.54%	4.51%	44.93%	5.62%	9.98%	44.93%	5.62%	9.98%
0.30	85.51%	8.91%	16.14%	85.51%	8.91%	16.14%	24.64%	2.56%	4.64%	55.07%	5.73%	10.38%	44.93%	4.68%	8.47%
0.35	85.51%	7.63%	14.01%	85.51%	7.63%	14.01%	27.54%	2.46%	4.51%	57.97%	5.17%	9.50%	47.83%	4.27%	7.84%
0.40	88.41%	6.91%	12.82%	88.41%	6.91%	12.82%	31.88%	2.49%	4.62%	68.12%	5.32%	9.86%	52.17%	4.07%	7.56%
0.45	91.30%	6.34%	11.86%	91.30%	6.34%	11.86%	33.33%	2.31%	4.33%	68.12%	4.73%	8.84%	55.07%	3.82%	7.15%
0.50	91.30%	5.71%	10.75%	91.30%	5.71%	10.75%	44.93%	2.81%	5.29%	73.91%	4.62%	8.70%	57.97%	3.62%	6.82%
0.55	91.30%	5.19%	9.82%	91.30%	5.19%	9.82%	50.72%	2.88%	5.45%	79.71%	4.53%	8.57%	62.32%	3.54%	6.70%
0.60	94.20%	4.91%	9.33%	94.20%	4.91%	9.33%	55.07%	2.87%	5.45%	79.71%	4.15%	7.89%	63.77%	3.32%	6.31%
0.65	94.20%	4.53%	8.64%	94.20%	4.53%	8.64%	57.97%	2.79%	5.32%	79.71%	3.83%	7.31%	69.57%	3.34%	6.38%
0.70	94.20%	4.20%	8.04%	94.20%	4.20%	8.04%	62.32%	2.78%	5.33%	86.96%	3.88%	7.43%	72.46%	3.23%	6.19%
0.75	94.20%	3.93%	7.55%	94.20%	3.93%	7.55%	68.12%	2.84%	5.45%	88.41%	3.68%	7.07%	78.26%	3.26%	6.26%
0.80	94.20%	3.68%	7.08%	94.20%	3.68%	7.08%	82.61%	3.23%	6.21%	89.86%	3.51%	6.75%	82.61%	3.23%	6.21%
0.85	97.10%	3.57%	6.89%	97.10%	3.57%	6.89%	75.36%	2.77%	5.34%	89.86%	3.30%	6.37%	82.61%	3.04%	5.86%
0.90	97.10%	3.37%	6.51%	97.10%	3.37%	6.51%	86.96%	3.02%	5.83%	91.30%	3.17%	6.13%	89.86%	3.12%	6.03%
0.95	100.00%	3.29%	6.37%	100.00%	3.29%	6.37%	94.20%	3.10%	6.00%	91.30%	3.00%	5.81%	92.75%	3.05%	5.90%

EasyClinic (CC–TC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	10.29%	72.41%	18.02%	8.82%	62.07%	15.45%	0.00%	0.00%	0.00%	5.39%	36.67%	9.40%	8.33%	56.67%	14.53%
0.02	18.63%	64.41%	28.90%	18.14%	62.71%	28.14%	6.37%	22.03%	9.89%	7.35%	25.42%	11.41%	13.73%	47.46%	21.29%
0.03	23.04%	53.41%	32.19%	21.08%	48.86%	29.45%	2.94%	6.74%	4.10%	13.24%	30.34%	18.43%	21.57%	49.44%	30.03%
0.04	24.02%	41.53%	30.44%	25.00%	43.22%	31.68%	0.00%	0.00%	0.00%	19.12%	33.05%	24.22%	22.06%	38.14%	27.95%
0.05	27.45%	37.84%	31.82%	25.98%	35.81%	30.11%	7.35%	10.14%	8.52%	22.06%	30.41%	25.57%	23.04%	31.76%	26.70%
0.06	30.39%	35.03%	32.55%	28.43%	32.77%	30.45%	7.35%	8.43%	7.85%	23.04%	26.40%	24.61%	24.02%	27.53%	25.65%
0.07	34.80%	34.30%	34.55%	33.33%	32.85%	33.09%	8.82%	8.70%	8.76%	31.86%	31.40%	31.63%	24.51%	24.15%	24.33%
0.08	36.76%	31.78%	34.09%	36.27%	31.36%	33.64%	7.35%	6.33%	6.80%	33.82%	29.11%	31.29%	25.49%	21.94%	23.58%
0.09	40.20%	30.83%	34.90%	38.73%	29.70%	33.62%	7.35%	5.64%	6.38%	38.73%	29.70%	33.62%	25.98%	19.92%	22.55%
0.10	41.67%	28.72%	34.00%	40.20%	27.70%	32.80%	1.96%	1.35%	1.60%	41.67%	28.72%	34.00%	25.98%	17.91%	21.20%
0.11	46.57%	29.23%	35.92%	41.67%	26.15%	32.13%	22.06%	13.80%	16.98%	41.18%	25.77%	31.70%	26.96%	16.87%	20.75%
0.12	48.53%	27.89%	35.42%	45.59%	26.20%	33.28%	14.71%	8.45%	10.73%	42.65%	24.51%	31.13%	27.94%	16.06%	20.39%
0.13	50.49%	26.82%	35.03%	47.06%	25.00%	32.65%	11.76%	6.23%	8.15%	46.08%	24.42%	31.92%	28.43%	15.06%	19.69%
0.14	51.47%	25.36%	33.98%	49.02%	24.15%	32.36%	11.76%	5.78%	7.75%	49.02%	24.10%	32.31%	28.92%	14.22%	19.06%
0.15	52.94%	24.32%	33.33%	50.49%	23.20%	31.79%	5.88%	2.70%	3.70%	49.51%	22.75%	31.17%	29.41%	13.51%	18.52%
0.16	53.43%	23.04%	32.20%	52.94%	22.83%	31.90%	12.75%	5.49%	7.67%	51.96%	22.36%	31.27%	30.88%	13.29%	18.58%
0.17	54.90%	22.27%	31.69%	54.90%	22.27%	31.69%	2.94%	1.19%	1.70%	53.92%	21.87%	31.12%	31.86%	12.92%	18.39%
0.18	57.35%	21.99%	31.79%	56.37%	21.62%	31.25%	3.43%	1.31%	1.90%	52.94%	20.26%	29.31%	32.84%	12.57%	18.18%
0.19	58.82%	21.35%	31.33%	57.35%	20.82%	30.55%	15.20%	5.51%	8.08%	56.37%	20.43%	29.99%	32.84%	11.90%	17.47%
0.20	59.80%	20.61%	30.65%	58.33%	20.10%	29.90%	12.75%	4.39%	6.53%	59.31%	20.44%	30.40%	34.31%	11.82%	17.59%
0.25	64.22%	17.70%	27.75%	64.71%	17.84%	27.97%	12.75%	3.51%	5.51%	63.73%	17.57%	27.54%	40.20%	11.08%	17.37%
0.30	65.69%	15.09%	24.54%	65.20%	14.98%	24.36%	16.67%	3.83%	6.23%	69.12%	15.88%	25.82%	45.10%	10.36%	16.85%
0.35	69.61%	13.71%	22.91%	68.63%	13.51%	22.58%	2.94%	0.58%	0.97%	70.10%	13.80%	23.06%	50.00%	9.85%	16.45%
0.40	71.57%	12.33%	21.04%	72.06%	12.42%	21.19%	23.53%	4.05%	6.92%	75.49%	13.01%	22.19%	56.37%	9.71%	16.57%
0.45	78.92%	12.09%	20.97%	77.94%	11.94%	20.71%	33.33%	5.11%	8.85%	77.94%	11.94%	20.70%	58.33%	8.93%	15.49%
0.50	86.27%	11.89%	20.90%	88.24%	12.16%	21.37%	23.53%	3.24%	5.70%	79.90%	11.01%	19.36%	60.78%	8.38%	14.73%
0.55	94.12%	11.79%	20.96%	93.63%	11.73%	20.85%	24.51%	3.07%	5.46%	79.90%	10.01%	17.79%	64.71%	8.10%	14.40%
0.60	95.59%	10.98%	19.70%	96.57%	11.09%	19.90%	50.49%	5.80%	10.40%	87.25%	10.02%	17.97%	67.16%	7.71%	13.83%
0.65	100.00%	10.60%	19.17%	100.00%	10.60%	19.17%	39.71%	4.21%	7.61%	88.24%	9.35%	16.91%	69.12%	7.32%	13.25%
0.70	100.00%	9.85%	17.93%	100.00%	9.85%	17.93%	35.29%	3.47%	6.32%	87.75%	8.63%	15.72%	71.57%	7.04%	12.82%
0.75	100.00%	9.19%	16.83%	100.00%	9.19%	16.83%	46.57%	4.28%	7.84%	92.65%	8.51%	15.59%	75.98%	6.98%	12.78%
0.80	100.00%	8.61%	15.85%	100.00%	8.61%	15.85%	52.94%	4.56%	8.39%	94.12%	8.10%	14.92%	78.92%	6.80%	12.51%
0.85	100.00%	8.11%	15.00%	100.00%	8.11%	15.00%	64.22%	5.20%	9.63%	94.61%	7.67%	14.19%	81.37%	6.60%	12.20%
0.90	100.00%	7.66%	14.23%	100.00%	7.66%	14.23%	47.06%	3.60%	6.69%	100.00%	7.65%	14.22%	87.25%	6.68%	12.41%
0.95	100.00%	7.25%	13.52%	100.00%	7.25%	13.52%	83.82%	6.08%	11.34%	100.00%	7.25%	13.52%	92.16%	6.68%	12.46%
1.00	100.00%	6.89%	12.89%	100.00%	6.89%	12.89%	100.00%	6.89%	12.89%	100.00%	6.89%	12.89%	100.00%	6.89%	12.89%

EasyClinic (ID–CC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	13.04%	100.00%	23.07%	13.04%	100.00%	23.07%	0.00%	0.00%	0.00%	4.35%	33.33%	7.69%	10.14%	77.78%	17.95%
0.02	23.19%	88.89%	36.78%	21.74%	83.33%	34.48%	0.00%	0.00%	0.00%	10.14%	36.84%	15.91%	20.29%	73.68%	31.82%
0.03	34.78%	85.71%	49.48%	34.78%	85.71%	49.48%	0.00%	0.00%	0.00%	17.39%	42.86%	24.74%	24.64%	60.71%	35.05%
0.04	43.48%	81.08%	56.60%	44.93%	83.78%	58.49%	0.00%	0.00%	0.00%	27.54%	50.00%	35.51%	26.09%	47.37%	33.64%
0.05	47.83%	70.21%	56.90%	53.62%	78.72%	63.79%	4.35%	6.38%	5.17%	34.78%	51.06%	41.38%	28.99%	42.55%	34.48%
0.06	59.42%	73.21%	65.60%	59.42%	73.21%	65.60%	1.45%	1.79%	1.60%	37.68%	46.43%	41.60%	34.78%	42.86%	38.40%
0.07	66.67%	70.77%	68.66%	66.67%	70.77%	68.66%	4.35%	4.55%	4.44%	37.68%	39.39%	38.52%	37.68%	39.39%	38.52%
0.08	72.46%	66.67%	69.44%	69.57%	64.00%	66.67%	7.25%	6.67%	6.94%	47.83%	44.00%	45.83%	40.58%	37.33%	38.89%
0.09	76.81%	63.10%	69.28%	72.46%	59.52%	65.36%	1.45%	1.18%	1.30%	42.03%	34.12%	37.66%	42.03%	34.12%	37.66%
0.10	79.71%	58.51%	67.48%	75.36%	55.32%	63.80%	1.45%	1.06%	1.23%	46.38%	34.04%	39.26%	43.48%	31.91%	36.81%
0.11	81.16%	54.37%	65.12%	78.26%	52.43%	62.79%	1.45%	0.97%	1.16%	47.83%	32.04%	38.37%	43.48%	29.13%	34.88%
0.12	81.16%	50.00%	61.88%	81.16%	50.00%	61.88%	2.90%	1.77%	2.20%	53.62%	32.74%	40.66%	44.93%	27.43%	34.07%
0.13	82.61%	46.72%	59.69%	84.06%	47.54%	60.73%	7.25%	4.10%	5.24%	50.72%	28.69%	36.65%	44.93%	25.41%	32.46%
0.14	84.06%	44.27%	58.00%	84.06%	44.27%	58.00%	7.25%	3.79%	4.98%	62.32%	32.58%	42.79%	44.93%	23.48%	30.85%
0.15	86.96%	42.55%	57.14%	85.51%	41.84%	56.19%	5.80%	2.84%	3.81%	57.97%	28.37%	38.10%	46.38%	22.70%	30.48%
0.16	88.41%	40.67%	55.71%	88.41%	40.67%	55.71%	4.35%	2.00%	2.74%	57.97%	26.67%	36.53%	49.28%	22.67%	31.05%
0.17	91.30%	39.62%	55.26%	89.86%	38.99%	54.38%	10.14%	4.38%	6.11%	60.87%	26.25%	36.68%	49.28%	21.25%	29.69%
0.18	92.75%	37.87%	53.78%	92.75%	37.87%	53.78%	11.59%	4.73%	6.72%	62.32%	25.44%	36.13%	55.07%	22.49%	31.93%
0.19	92.75%	35.96%	51.83%	94.20%	36.52%	52.63%	13.04%	5.03%	7.26%	65.22%	25.14%	36.29%	56.52%	21.79%	31.45%
0.20	92.75%	34.04%	49.80%	95.65%	35.11%	51.37%	13.04%	4.79%	7.00%	60.87%	22.34%	32.68%	56.52%	20.74%	30.35%
0.25	97.10%	28.51%	44.08%	95.65%	28.09%	43.43%	17.39%	5.11%	7.89%	73.91%	21.70%	33.55%	62.32%	18.30%	28.29%
0.30	97.10%	23.76%	38.18%	97.10%	23.76%	38.18%	17.39%	4.26%	6.84%	78.26%	19.15%	30.77%	65.22%	15.96%	25.64%
0.35	97.10%	20.36%	33.66%	97.10%	20.36%	33.66%	17.39%	3.65%	6.03%	84.06%	17.63%	29.15%	68.12%	14.29%	23.62%
0.40	97.10%	17.82%	30.11%	97.10%	17.82%	30.11%	11.59%	2.13%	3.60%	85.51%	15.69%	26.52%	71.01%	13.03%	22.02%
0.45	98.55%	16.08%	27.65%	98.55%	16.08%	27.65%	24.64%	4.02%	6.91%	91.30%	14.89%	25.61%	73.91%	12.06%	20.73%
0.50	98.55%	14.47%	25.23%	98.55%	14.47%	25.23%	21.74%	3.19%	5.57%	89.86%	13.19%	23.01%	78.26%	11.49%	20.04%
0.55	98.55%	13.15%	23.20%	98.55%	13.15%	23.20%	36.23%	4.84%	8.53%	91.30%	12.19%	21.50%	81.16%	10.83%	19.11%
0.60	98.55%	12.06%	21.49%	98.55%	12.06%	21.49%	27.54%	3.37%	6.00%	92.75%	11.35%	20.22%	84.06%	10.28%	18.33%
0.65	98.55%	11.13%	20.00%	98.55%	11.13%	20.00%	44.93%	5.07%	9.12%	97.10%	10.97%	19.71%	86.96%	9.82%	17.65%
0.70	98.55%	10.33%	18.70%	98.55%	10.33%	18.70%	33.33%	3.50%	6.33%	97.10%	10.18%	18.43%	88.41%	9.27%	16.78%
0.75	98.55%	9.65%	17.58%	98.55%	9.65%	17.58%	28.99%	2.84%	5.17%	98.55%	9.65%	17.57%	91.30%	8.94%	16.28%
0.80	98.55%	9.04%	16.56%	98.55%	9.04%	16.56%	27.54%	2.53%	4.63%	97.10%	8.91%	16.32%	92.75%	8.51%	15.59%
0.85	98.55%	9.04%	16.56%	98.55%	9.04%	16.56%	46.38%	4.01%	7.37%	98.55%	8.51%	15.67%	92.75%	8.01%	14.75%
0.90	98.55%	8.51%	15.67%	100.00%	8.51%	15.69%	62.32%	5.08%	9.40%	98.55%	8.04%	14.86%	95.65%	7.80%	14.43%
0.95	100.00%	7.73%	14.35%	100.00%	7.73%	14.35%	85.51%	6.61%	12.27%	98.55%	7.61%	14.14%	97.10%	7.50%	13.93%
1.00	100.00%	7.34%	13.68%	100.00%	7.34%	13.68%	100.00%	7.34%	13.68%	100.00%	7.34%	13.68%	100.00%	7.34%	13.68%

EasyClinic (ID–ID)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.02	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.03	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.04	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.05	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	6.78%	20.00%	10.13%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.06	5.08%	12.50%	7.22%	5.08%	12.50%	7.22%	0.00%	0.00%	0.00%	5.08%	12.50%	7.23%	6.78%	16.67%	9.64%
0.07	11.86%	25.00%	16.09%	11.86%	25.00%	16.09%	3.39%	7.14%	4.60%	11.86%	25.00%	16.09%	11.86%	25.00%	16.09%
0.08	18.64%	34.38%	24.17%	18.64%	34.38%	24.17%	3.39%	6.25%	4.40%	18.64%	34.38%	24.18%	11.86%	21.88%	15.38%
0.09	25.42%	41.67%	31.58%	25.42%	41.67%	31.58%	10.17%	16.67%	12.63%	22.03%	36.11%	27.37%	18.64%	30.56%	23.16%
0.10	25.42%	37.50%	30.30%	25.42%	37.50%	30.30%	6.78%	10.00%	8.08%	25.42%	37.50%	30.30%	22.03%	32.50%	26.26%
0.11	28.81%	38.64%	33.01%	28.81%	38.64%	33.01%	10.17%	13.64%	11.65%	27.12%	36.36%	31.07%	25.42%	34.09%	29.13%
0.12	30.51%	37.50%	33.65%	30.51%	37.50%	33.65%	10.17%	12.50%	11.21%	32.20%	39.58%	35.51%	28.81%	35.42%	31.78%
0.13	35.59%	40.38%	37.83%	35.59%	40.38%	37.83%	13.56%	15.38%	14.41%	32.20%	36.54%	34.23%	35.59%	40.38%	37.84%
0.14	38.98%	41.07%	40.00%	38.98%	41.07%	40.00%	11.86%	12.50%	12.17%	37.29%	39.29%	38.26%	37.29%	39.29%	38.26%
0.15	42.37%	41.67%	42.02%	42.37%	41.67%	42.02%	22.03%	21.67%	21.85%	40.68%	40.00%	40.34%	40.68%	40.00%	40.34%
0.16	47.46%	43.75%	45.53%	47.46%	43.75%	45.53%	15.25%	14.06%	14.63%	47.46%	43.75%	45.53%	44.07%	40.62%	42.28%
0.17	54.24%	47.06%	50.40%	54.24%	47.06%	50.40%	15.25%	13.24%	14.17%	44.07%	38.24%	40.94%	47.46%	41.18%	44.09%
0.18	55.93%	45.83%	50.38%	55.93%	45.83%	50.38%	13.56%	11.11%	12.21%	54.24%	44.44%	48.85%	47.46%	38.89%	42.75%
0.19	55.93%	43.42%	48.89%	55.93%	43.42%	48.89%	27.12%	21.05%	23.70%	61.02%	47.37%	53.33%	54.24%	42.11%	47.41%
0.20	55.93%	41.25%	47.48%	55.93%	41.25%	47.48%	22.03%	16.25%	18.71%	55.93%	41.25%	47.48%	54.24%	40.00%	46.04%
0.25	69.49%	41.00%	51.57%	69.49%	41.00%	51.57%	25.42%	15.00%	18.87%	74.58%	44.00%	55.35%	62.71%	37.00%	46.54%
0.30	74.58%	36.67%	49.17%	74.58%	36.67%	49.17%	37.29%	18.33%	24.58%	88.14%	43.33%	58.10%	66.10%	32.50%	43.58%
0.35	81.36%	34.29%	48.25%	81.36%	34.29%	48.25%	23.73%	10.00%	14.07%	93.22%	39.29%	55.28%	69.49%	29.29%	41.21%
0.40	84.75%	31.25%	45.66%	84.75%	31.25%	45.66%	22.03%	8.12%	11.87%	98.31%	36.25%	52.97%	72.88%	26.88%	39.27%
0.45	84.75%	27.78%	41.84%	84.75%	27.78%	41.84%	35.59%	11.67%	17.57%	98.31%	32.22%	48.54%	74.58%	24.44%	36.82%
0.50	88.14%	26.00%	40.15%	88.14%	26.00%	40.15%	37.29%	11.00%	16.99%	98.31%	29.00%	44.79%	84.75%	25.00%	38.61%
0.55	91.53%	24.55%	38.72%	91.53%	24.55%	38.72%	30.51%	8.18%	12.90%	100.00%	26.82%	42.29%	88.14%	23.64%	37.28%
0.60	91.53%	22.50%	36.12%	91.53%	22.50%	36.12%	35.59%	8.75%	14.05%	100.00%	24.58%	39.46%	88.14%	21.67%	34.78%
0.65	94.92%	21.54%	35.11%	94.92%	21.54%	35.11%	47.46%	10.77%	17.55%	100.00%	22.69%	36.99%	88.14%	20.00%	32.60%
0.70	94.92%	20.00%	33.04%	94.92%	20.00%	33.04%	37.29%	7.86%	12.98%	100.00%	21.07%	34.81%	88.14%	18.57%	30.68%
0.75	96.61%	19.00%	31.75%	96.61%	19.00%	31.75%	35.59%	7.00%	11.70%	100.00%	19.67%	32.87%	88.14%	17.33%	28.97%
0.80	100.00%	18.44%	31.14%	100.00%	18.44%	31.14%	62.71%	11.56%	19.53%	100.00%	18.44%	31.13%	93.22%	17.19%	29.02%
0.85	100.00%	17.35%	29.57%	100.00%	17.35%	29.57%	81.36%	14.12%	24.06%	100.00%	17.35%	29.57%	93.22%	16.18%	27.57%
0.90	100.00%	16.39%	28.16%	100.00%	16.39%	28.16%	93.22%	15.28%	26.25%	100.00%	16.39%	28.16%	93.22%	15.28%	26.25%
0.95	100.00%	15.53%	26.88%	100.00%	15.53%	26.88%	91.53%	14.21%	24.60%	100.00%	15.53%	26.88%	98.31%	15.26%	26.42%
1.00	100.00%	14.75%	25.71%	100.00%	14.75%	25.71%	100.00%	14.75%	25.71%	100.00%	14.75%	25.71%	100.00%	14.75%	25.71%

EasyClinic (ID–TC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	6.86%	48.28%	12.01%	7.84%	55.17%	13.73%	0.49%	3.33%	0.85%	4.41%	30.00%	7.69%	8.33%	56.67%	14.53%
0.02	9.80%	33.90%	15.20%	15.69%	54.24%	24.34%	0.00%	0.00%	0.00%	12.75%	44.07%	19.77%	13.73%	47.46%	21.29%
0.03	18.63%	43.18%	26.03%	19.61%	45.45%	27.40%	0.00%	0.00%	0.00%	16.67%	38.20%	23.21%	21.57%	49.44%	30.03%
0.04	25.98%	44.92%	32.92%	25.98%	44.92%	32.92%	7.35%	12.71%	9.32%	18.63%	32.20%	23.60%	22.06%	38.14%	27.95%
0.05	32.35%	44.59%	37.50%	33.33%	45.95%	38.64%	3.43%	4.73%	3.98%	18.63%	25.68%	21.59%	23.04%	31.76%	26.70%
0.06	40.20%	46.33%	43.05%	40.20%	46.33%	43.05%	2.45%	2.81%	2.62%	24.51%	28.09%	26.18%	24.02%	27.53%	25.65%
0.07	45.59%	44.93%	45.26%	45.59%	44.93%	45.26%	3.43%	3.38%	3.41%	30.39%	29.95%	30.17%	24.51%	24.15%	24.33%
0.08	49.51%	42.80%	45.91%	49.02%	42.37%	45.45%	5.39%	4.64%	4.99%	34.31%	29.54%	31.75%	25.49%	21.94%	23.58%
0.09	53.43%	40.98%	46.38%	52.94%	40.60%	45.96%	10.78%	8.27%	9.36%	37.25%	28.57%	32.34%	25.98%	19.92%	22.55%
0.10	57.35%	39.53%	46.80%	55.39%	38.18%	45.20%	11.76%	8.11%	9.60%	40.69%	28.04%	33.20%	25.98%	17.91%	21.20%
0.11	62.25%	39.08%	48.02%	58.33%	36.62%	44.99%	14.22%	8.90%	10.94%	43.14%	26.99%	33.21%	26.96%	16.87%	20.75%
0.12	66.67%	38.31%	48.66%	60.78%	34.93%	44.36%	23.04%	13.24%	16.82%	45.59%	26.20%	33.27%	27.94%	16.06%	20.39%
0.13	67.65%	35.94%	46.94%	64.71%	34.38%	44.90%	23.53%	12.47%	16.30%	48.04%	25.45%	33.28%	28.43%	15.06%	19.69%
0.14	70.59%	34.78%	46.60%	67.16%	33.09%	44.34%	22.55%	11.08%	14.86%	50.98%	25.06%	33.60%	28.92%	14.22%	19.06%
0.15	71.57%	32.88%	45.06%	68.63%	31.53%	43.21%	30.39%	13.96%	19.14%	50.98%	23.42%	32.10%	29.41%	13.51%	18.52%
0.16	72.55%	31.29%	43.72%	68.63%	29.60%	41.36%	23.53%	10.13%	14.16%	49.51%	21.31%	29.79%	30.88%	13.29%	18.58%
0.17	74.51%	30.22%	43.00%	71.08%	28.83%	41.02%	25.98%	10.54%	14.99%	50.98%	20.68%	29.42%	31.86%	12.92%	18.39%
0.18	75.00%	28.76%	41.58%	71.57%	27.44%	39.67%	16.67%	6.38%	9.23%	53.43%	20.45%	29.58%	32.84%	12.57%	18.18%
0.19	75.49%	27.40%	40.21%	74.51%	27.05%	39.69%	30.39%	11.01%	16.17%	57.35%	20.78%	30.51%	32.84%	11.90%	17.47%
0.20	75.49%	26.01%	38.69%	75.49%	26.01%	38.69%	24.02%	8.28%	12.31%	57.35%	19.76%	29.40%	34.31%	11.82%	17.59%
0.25	81.37%	22.43%	35.17%	79.41%	21.89%	34.32%	29.41%	8.11%	12.71%	62.25%	17.16%	26.91%	40.20%	11.08%	17.37%
0.30	86.76%	19.93%	32.41%	84.31%	19.37%	31.50%	35.29%	8.11%	13.19%	68.14%	15.65%	25.46%	45.10%	10.36%	16.85%
0.35	91.67%	18.05%	30.16%	89.22%	17.57%	29.36%	30.88%	6.08%	10.16%	74.02%	14.58%	24.35%	50.00%	9.85%	16.45%
0.40	91.67%	15.79%	26.94%	91.67%	15.79%	26.94%	41.67%	7.18%	12.25%	76.47%	13.18%	22.48%	56.37%	9.71%	16.57%
0.45	91.67%	14.04%	24.35%	91.67%	14.04%	24.35%	57.35%	8.78%	15.23%	76.96%	11.79%	20.44%	58.33%	8.93%	15.49%
0.50	92.16%	12.70%	22.32%	92.16%	12.70%	22.32%	44.61%	6.15%	10.81%	77.94%	10.74%	18.88%	60.78%	8.38%	14.73%
0.55	92.65%	11.61%	20.63%	92.65%	11.61%	20.63%	36.27%	4.54%	8.07%	80.39%	10.07%	17.89%	64.71%	8.10%	14.40%
0.60	93.63%	10.75%	19.29%	95.59%	10.98%	19.70%	45.10%	5.18%	9.29%	86.76%	9.96%	17.87%	67.16%	7.71%	13.83%
0.65	97.55%	10.34%	18.70%	100.00%	10.60%	19.17%	42.65%	4.52%	8.17%	87.25%	9.25%	16.72%	69.12%	7.32%	13.25%
0.70	100.00%	9.85%	17.93%	100.00%	9.85%	17.93%	44.61%	4.39%	7.99%	88.24%	8.68%	15.81%	71.57%	7.04%	12.82%
0.75	100.00%	9.19%	16.83%	100.00%	9.19%	16.83%	34.80%	3.20%	5.86%	92.16%	8.46%	15.51%	75.98%	6.98%	12.78%
0.80	100.00%	8.61%	15.85%	100.00%	8.61%	15.85%	70.10%	6.04%	11.12%	94.12%	8.10%	14.92%	78.92%	6.80%	12.51%
0.85	100.00%	8.11%	15.00%	100.00%	8.11%	15.00%	65.20%	5.28%	9.78%	97.55%	7.91%	14.63%	81.37%	6.60%	12.20%
0.90	100.00%	7.66%	14.23%	100.00%	7.66%	14.23%	69.12%	5.29%	9.83%	100.00%	7.65%	14.22%	87.25%	6.68%	12.41%
0.95	100.00%	7.25%	13.52%	100.00%	7.25%	13.52%	87.25%	6.33%	11.80%	100.00%	7.25%	13.52%	92.16%	6.68%	12.46%
1.00	100.00%	6.89%	12.89%	100.00%	6.89%	12.89%	100.00%	6.89%	12.89%	100.00%	6.89%	12.89%	100.00%	6.89%	12.89%

EasyClinic (ID–UC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	4.82%	33.33%	8.42%	4.82%	33.33%	8.42%	0.00%	0.00%	0.00%	4.82%	30.77%	8.33%	12.05%	76.92%	20.83%
0.02	13.25%	44.00%	20.37%	13.25%	44.00%	20.37%	0.00%	0.00%	0.00%	7.23%	24.00%	11.11%	15.66%	52.00%	24.07%
0.03	20.48%	45.95%	28.33%	20.48%	45.95%	28.33%	0.00%	0.00%	0.00%	12.05%	26.32%	16.53%	16.87%	36.84%	23.14%
0.04	26.51%	44.00%	33.09%	26.51%	44.00%	33.09%	0.00%	0.00%	0.00%	19.28%	32.00%	24.06%	16.87%	28.00%	21.05%
0.05	33.73%	44.44%	38.35%	33.73%	44.44%	38.35%	0.00%	0.00%	0.00%	24.10%	31.75%	27.40%	16.87%	22.22%	19.18%
0.06	42.17%	46.67%	44.31%	42.17%	46.67%	44.31%	2.41%	2.63%	2.52%	36.14%	39.47%	37.74%	18.07%	19.74%	18.87%
0.07	45.78%	43.18%	44.44%	45.78%	43.18%	44.44%	0.00%	0.00%	0.00%	37.35%	35.23%	36.26%	19.28%	18.18%	18.71%
0.08	50.60%	42.00%	45.90%	50.60%	42.00%	45.90%	4.82%	3.96%	4.35%	45.78%	37.62%	41.30%	21.69%	17.82%	19.57%
0.09	54.22%	39.82%	45.92%	54.22%	39.82%	45.92%	1.20%	0.88%	1.02%	51.81%	38.05%	43.88%	21.69%	15.93%	18.37%
0.10	63.86%	42.06%	50.72%	63.86%	42.06%	50.72%	1.20%	0.79%	0.96%	53.01%	34.92%	42.11%	21.69%	14.29%	17.22%
0.11	67.47%	40.58%	50.68%	67.47%	40.58%	50.68%	4.82%	2.88%	3.60%	53.01%	31.65%	39.64%	21.69%	12.95%	16.22%
0.12	72.29%	39.74%	51.29%	72.29%	39.74%	51.29%	0.00%	0.00%	0.00%	57.83%	31.79%	41.03%	21.69%	11.92%	15.38%
0.13	73.49%	37.42%	49.59%	73.49%	37.42%	49.59%	6.02%	3.05%	4.05%	65.06%	32.93%	43.72%	22.89%	11.59%	15.38%
0.14	74.70%	35.23%	47.88%	74.70%	35.23%	47.88%	0.00%	0.00%	0.00%	65.06%	30.68%	41.70%	22.89%	10.80%	14.67%
0.15	77.11%	33.86%	47.06%	77.11%	33.86%	47.06%	6.02%	2.65%	3.68%	66.27%	29.10%	40.44%	24.10%	10.58%	14.71%
0.16	81.93%	33.83%	47.89%	81.93%	33.83%	47.89%	7.23%	2.97%	4.21%	69.88%	28.71%	40.70%	25.30%	10.40%	14.74%
0.17	81.93%	31.78%	45.80%	81.93%	31.78%	45.80%	1.20%	0.47%	0.67%	74.70%	28.97%	41.75%	26.51%	10.28%	14.81%
0.18	83.13%	30.53%	44.66%	83.13%	30.53%	44.66%	0.00%	0.00%	0.00%	67.47%	24.67%	36.13%	27.71%	10.13%	14.84%
0.19	86.75%	30.13%	44.73%	86.75%	30.13%	44.73%	0.00%	0.00%	0.00%	72.29%	25.10%	37.27%	30.12%	10.46%	15.53%
0.20	87.95%	28.97%	43.58%	87.95%	28.97%	43.58%	1.20%	0.40%	0.60%	84.34%	27.78%	41.79%	30.12%	9.92%	14.93%
0.25	90.36%	23.81%	37.69%	90.36%	23.81%	37.69%	1.20%	0.32%	0.50%	86.75%	22.86%	36.18%	32.53%	8.57%	13.57%
0.30	91.57%	20.11%	32.98%	91.57%	20.11%	32.98%	1.20%	0.26%	0.43%	87.95%	19.31%	31.67%	34.94%	7.67%	12.58%
0.35	92.77%	17.46%	29.39%	92.77%	17.46%	29.39%	0.00%	0.00%	0.00%	96.39%	18.14%	30.53%	38.55%	7.26%	12.21%
0.40	92.77%	15.28%	26.24%	92.77%	15.28%	26.24%	4.82%	0.79%	1.36%	100.00%	16.47%	28.28%	39.76%	6.55%	11.24%
0.45	95.18%	13.93%	24.30%	95.18%	13.93%	24.30%	15.66%	2.29%	4.00%	100.00%	14.64%	25.54%	43.37%	6.35%	11.08%
0.50	95.18%	12.54%	22.16%	95.18%	12.54%	22.16%	13.25%	1.75%	3.09%	100.00%	13.17%	23.28%	49.40%	6.51%	11.50%
0.55	95.18%	11.40%	20.36%	95.18%	11.40%	20.36%	0.00%	0.00%	0.00%	100.00%	11.98%	21.39%	54.22%	6.49%	11.60%
0.60	97.59%	10.71%	19.30%	97.59%	10.71%	19.30%	25.30%	2.78%	5.01%	100.00%	10.98%	19.79%	57.83%	6.35%	11.44%
0.65	97.59%	9.89%	17.96%	97.59%	9.89%	17.96%	24.10%	2.44%	4.43%	100.00%	10.13%	18.40%	60.24%	6.11%	11.09%
0.70	100.00%	9.41%	17.20%	100.00%	9.41%	17.20%	28.92%	2.72%	4.97%	100.00%	9.41%	17.20%	68.67%	6.46%	11.81%
0.75	100.00%	8.78%	16.14%	100.00%	8.78%	16.14%	22.89%	2.01%	3.70%	100.00%	8.78%	16.15%	69.88%	6.14%	11.28%
0.80	100.00%	8.23%	15.21%	100.00%	8.23%	15.21%	21.69%	1.79%	3.30%	100.00%	8.23%	15.22%	69.88%	5.75%	10.63%
0.85	100.00%	7.75%	14.39%	100.00%	7.75%	14.39%	16.87%	1.31%	2.43%	100.00%	7.75%	14.38%	71.08%	5.51%	10.23%
0.90	100.00%	7.32%	13.64%	100.00%	7.32%	13.64%	61.45%	4.50%	8.38%	100.00%	7.32%	13.64%	74.70%	5.47%	10.19%
0.95	100.00%	6.93%	12.96%	100.00%	6.93%	12.96%	62.65%	4.34%	8.12%	100.00%	6.93%	12.97%	85.54%	5.93%	11.09%
1.00	100.00%	6.59%	12.37%	100.00%	6.59%	12.37%	100.00%	6.59%	12.36%	100.00%	6.59%	12.36%	100.00%	6.59%	12.36%

EasyClinic (TC–CC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	15.38%	66.67%	24.99%	19.23%	83.33%	31.25%	0.00%	0.00%	0.00%	7.69%	33.33%	12.50%	11.54%	50.00%	18.75%
0.02	23.08%	50.00%	31.58%	30.77%	66.67%	42.11%	0.00%	0.00%	0.00%	23.08%	50.00%	31.58%	15.38%	33.33%	21.05%
0.03	30.77%	44.44%	36.36%	38.46%	55.56%	45.45%	0.00%	0.00%	0.00%	26.92%	38.89%	31.82%	23.08%	33.33%	27.27%
0.04	34.62%	37.50%	36.00%	38.46%	41.67%	40.00%	0.00%	0.00%	0.00%	38.46%	41.67%	40.00%	30.77%	33.33%	32.00%
0.05	38.46%	33.33%	35.71%	38.46%	33.33%	35.71%	0.00%	0.00%	0.00%	42.31%	36.67%	39.29%	34.62%	30.00%	32.14%
0.06	42.31%	30.56%	35.49%	50.00%	36.11%	41.93%	0.00%	0.00%	0.00%	42.31%	30.56%	35.48%	38.46%	27.78%	32.26%
0.07	46.15%	28.57%	35.29%	50.00%	30.95%	38.23%	0.00%	0.00%	0.00%	42.31%	26.19%	32.35%	38.46%	23.81%	29.41%
0.08	57.69%	31.25%	40.54%	57.69%	31.25%	40.54%	0.00%	0.00%	0.00%	53.85%	29.17%	37.84%	46.15%	25.00%	32.43%
0.09	61.54%	29.63%	40.00%	57.69%	27.78%	37.50%	0.00%	0.00%	0.00%	53.85%	25.93%	35.00%	46.15%	22.22%	30.00%
0.10	69.23%	30.00%	41.86%	65.38%	28.33%	39.53%	0.00%	0.00%	0.00%	57.69%	25.00%	34.88%	50.00%	21.67%	30.23%
0.11	73.08%	28.79%	41.31%	73.08%	28.79%	41.31%	0.00%	0.00%	0.00%	61.54%	24.24%	34.78%	50.00%	19.70%	28.26%
0.12	73.08%	26.39%	38.78%	76.92%	27.78%	40.82%	0.00%	0.00%	0.00%	61.54%	22.22%	32.65%	53.85%	19.44%	28.57%
0.13	76.92%	25.64%	38.46%	76.92%	25.64%	38.46%	0.00%	0.00%	0.00%	65.38%	21.79%	32.69%	53.85%	17.95%	26.92%
0.14	80.77%	25.00%	38.18%	76.92%	23.81%	36.36%	0.00%	0.00%	0.00%	69.23%	21.43%	32.73%	57.69%	17.86%	27.27%
0.15	80.77%	23.33%	36.20%	80.77%	23.33%	36.20%	0.00%	0.00%	0.00%	73.08%	21.11%	32.76%	57.69%	16.67%	25.86%
0.16	80.77%	21.88%	34.43%	80.77%	21.88%	34.43%	0.00%	0.00%	0.00%	76.92%	20.83%	32.79%	57.69%	15.62%	24.59%
0.17	84.62%	21.57%	34.38%	84.62%	21.57%	34.38%	3.85%	0.98%	1.56%	84.62%	21.57%	34.38%	57.69%	14.71%	23.44%
0.18	84.62%	20.37%	32.84%	88.46%	21.30%	34.33%	0.00%	0.00%	0.00%	84.62%	20.37%	32.84%	57.69%	13.89%	22.39%
0.19	84.62%	19.30%	31.43%	88.46%	20.18%	32.86%	0.00%	0.00%	0.00%	80.77%	18.42%	30.00%	57.69%	13.16%	21.43%
0.20	84.62%	18.33%	30.13%	88.46%	19.17%	31.51%	0.00%	0.00%	0.00%	80.77%	17.50%	28.77%	61.54%	13.33%	21.92%
0.25	88.46%	15.33%	26.13%	92.31%	16.00%	27.27%	0.00%	0.00%	0.00%	88.46%	15.33%	26.14%	69.23%	12.00%	20.45%
0.30	96.15%	13.89%	24.27%	96.15%	13.89%	24.27%	7.69%	1.11%	1.94%	92.31%	13.33%	23.30%	73.08%	10.56%	18.45%
0.35	100.00%	12.38%	22.03%	100.00%	12.38%	22.03%	0.00%	0.00%	0.00%	96.15%	11.90%	21.19%	76.92%	9.52%	16.95%
0.40	100.00%	10.83%	19.54%	100.00%	10.83%	19.54%	15.38%	1.67%	3.01%	96.15%	10.42%	18.80%	80.77%	8.75%	15.79%
0.45	100.00%	9.63%	17.57%	100.00%	9.63%	17.57%	7.69%	0.74%	1.35%	100.00%	9.63%	17.57%	80.77%	7.78%	14.19%
0.50	100.00%	8.67%	15.96%	100.00%	8.67%	15.96%	15.38%	1.33%	2.45%	96.15%	8.33%	15.34%	84.62%	7.33%	13.50%
0.55	100.00%	7.88%	14.61%	100.00%	7.88%	14.61%	7.69%	0.61%	1.12%	100.00%	7.88%	14.61%	88.46%	6.97%	12.92%
0.60	100.00%	7.22%	13.47%	100.00%	7.22%	13.47%	19.23%	1.39%	2.59%	100.00%	7.22%	13.47%	88.46%	6.39%	11.92%
0.65	100.00%	6.67%	12.51%	100.00%	6.67%	12.51%	11.54%	0.77%	1.44%	100.00%	6.67%	12.50%	88.46%	5.90%	11.06%
0.70	100.00%	6.19%	11.66%	100.00%	6.19%	11.66%	19.23%	1.19%	2.24%	100.00%	6.19%	11.66%	92.31%	5.71%	10.76%
0.75	100.00%	5.78%	10.93%	100.00%	5.78%	10.93%	34.62%	2.00%	3.78%	100.00%	5.78%	10.92%	92.31%	5.33%	10.08%
0.80	100.00%	5.42%	10.28%	100.00%	5.42%	10.28%	50.00%	2.71%	5.14%	100.00%	5.42%	10.28%	96.15%	5.21%	9.88%
0.85	100.00%	5.10%	9.71%	100.00%	5.10%	9.71%	23.08%	1.18%	2.24%	100.00%	5.10%	9.70%	100.00%	5.10%	9.70%
0.90	100.00%	4.81%	9.18%	100.00%	4.81%	9.18%	50.00%	2.41%	4.59%	100.00%	4.81%	9.19%	100.00%	4.81%	9.19%
0.95	100.00%	4.56%	8.72%	100.00%	4.56%	8.72%	69.23%	3.16%	6.04%	100.00%	4.56%	8.72%	100.00%	4.56%	8.72%
1.00	100.00%	4.33%	8.30%	100.00%	4.33%	8.30%	100.00%	4.33%	8.31%	100.00%	4.33%	8.31%	100.00%	4.33%	8.31%

EasyClinic (UC–CC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	12.90%	85.71%	22.42%	12.90%	85.71%	22.42%	1.08%	7.14%	1.87%	3.23%	21.43%	5.61%	6.45%	42.86%	11.21%
0.02	22.58%	75.00%	34.71%	25.81%	85.71%	39.67%	2.15%	7.14%	3.31%	7.53%	25.00%	11.57%	9.68%	32.14%	14.88%
0.03	32.26%	71.43%	44.45%	33.33%	73.81%	45.92%	0.00%	0.00%	0.00%	8.60%	19.05%	11.85%	13.98%	30.95%	19.26%
0.04	41.94%	69.64%	52.35%	43.01%	71.43%	53.69%	0.00%	0.00%	0.00%	12.90%	21.43%	16.11%	19.35%	32.14%	24.16%
0.05	50.54%	67.14%	57.67%	49.46%	65.71%	56.44%	0.00%	0.00%	0.00%	13.98%	18.57%	15.95%	22.58%	30.00%	25.77%
0.06	58.06%	64.29%	61.02%	54.84%	60.71%	57.63%	2.15%	2.35%	2.25%	18.28%	20.00%	19.10%	25.81%	28.24%	26.97%
0.07	65.59%	62.24%	63.87%	62.37%	59.18%	60.73%	0.00%	0.00%	0.00%	21.51%	20.20%	20.83%	26.88%	25.25%	26.04%
0.08	68.82%	57.14%	62.44%	68.82%	57.14%	62.44%	4.30%	3.54%	3.88%	24.73%	20.35%	22.33%	26.88%	22.12%	24.27%
0.09	70.97%	52.38%	60.27%	72.04%	53.17%	61.18%	0.00%	0.00%	0.00%	29.03%	21.26%	24.55%	26.88%	19.69%	22.73%
0.10	74.19%	48.94%	58.98%	74.19%	48.94%	58.98%	4.30%	2.84%	3.42%	25.81%	17.02%	20.51%	27.96%	18.44%	22.22%
0.11	75.27%	45.16%	56.45%	78.49%	47.10%	58.87%	3.23%	1.94%	2.42%	27.96%	16.77%	20.97%	29.03%	17.42%	21.77%
0.12	76.34%	42.01%	54.20%	80.65%	44.38%	57.25%	2.15%	1.18%	1.53%	32.26%	17.75%	22.90%	30.11%	16.57%	21.37%
0.13	78.49%	39.89%	52.90%	81.72%	41.53%	55.07%	1.08%	0.55%	0.72%	33.33%	16.94%	22.46%	32.26%	16.39%	21.74%
0.14	81.72%	38.58%	52.41%	82.80%	39.09%	53.11%	4.30%	2.03%	2.76%	32.26%	15.23%	20.69%	35.48%	16.75%	22.76%
0.15	81.72%	36.02%	50.00%	82.80%	36.49%	50.66%	9.68%	4.25%	5.90%	35.48%	15.57%	21.64%	37.63%	16.51%	22.95%
0.16	82.80%	34.22%	48.43%	83.87%	34.67%	49.06%	4.30%	1.77%	2.51%	39.78%	16.37%	23.20%	37.63%	15.49%	21.94%
0.17	83.87%	32.64%	46.99%	83.87%	32.64%	46.99%	4.30%	1.67%	2.40%	39.78%	15.42%	22.22%	39.78%	15.42%	22.22%
0.18	86.02%	31.62%	46.24%	84.95%	31.23%	45.67%	6.45%	2.36%	3.46%	40.86%	14.96%	21.90%	39.78%	14.57%	21.33%
0.19	86.02%	29.96%	44.44%	84.95%	29.59%	43.89%	5.38%	1.87%	2.77%	43.01%	14.93%	22.16%	40.86%	14.18%	21.05%
0.20	88.17%	29.08%	43.74%	84.95%	28.01%	42.13%	9.68%	3.19%	4.80%	47.31%	15.60%	23.47%	43.01%	14.18%	21.33%
0.25	91.40%	24.15%	38.21%	90.32%	23.86%	37.75%	9.68%	2.56%	4.04%	54.84%	14.49%	22.92%	49.46%	13.07%	20.67%
0.30	93.55%	20.57%	33.72%	92.47%	20.33%	33.33%	24.73%	5.44%	8.91%	58.06%	12.77%	20.93%	52.69%	11.58%	18.99%
0.35	94.62%	17.85%	30.03%	93.55%	17.65%	29.70%	25.81%	4.87%	8.19%	63.44%	11.97%	20.14%	58.06%	10.95%	18.43%
0.40	94.62%	15.60%	26.78%	94.62%	15.60%	26.78%	24.73%	4.08%	7.00%	66.67%	10.99%	18.87%	60.22%	9.93%	17.05%
0.45	95.70%	14.04%	24.49%	94.62%	13.88%	24.21%	36.56%	5.36%	9.35%	72.04%	10.57%	18.43%	62.37%	9.15%	15.96%
0.50	96.77%	12.77%	22.56%	96.77%	12.77%	22.56%	23.66%	3.12%	5.51%	78.49%	10.35%	18.30%	64.52%	8.51%	15.04%
0.55	96.77%	11.61%	20.73%	97.85%	11.74%	20.96%	29.03%	3.48%	6.21%	79.57%	9.54%	17.03%	69.89%	8.38%	14.96%
0.60	97.85%	10.76%	19.39%	97.85%	10.76%	19.39%	29.03%	3.19%	5.75%	90.32%	9.93%	17.89%	70.97%	7.80%	14.06%
0.65	97.85%	9.93%	18.03%	97.85%	9.93%	18.03%	29.03%	2.95%	5.35%	94.62%	9.61%	17.44%	75.27%	7.64%	13.88%
0.70	97.85%	9.23%	16.87%	97.85%	9.23%	16.87%	26.88%	2.53%	4.63%	94.62%	8.92%	16.30%	78.49%	7.40%	13.52%
0.75	98.92%	8.70%	15.99%	98.92%	8.70%	15.99%	40.86%	3.59%	6.60%	97.85%	8.60%	15.81%	82.80%	7.28%	13.38%
0.80	100.00%	8.24%	15.23%	100.00%	8.24%	15.23%	44.09%	3.63%	6.72%	97.85%	8.07%	14.91%	84.95%	7.00%	12.94%
0.85	100.00%	7.76%	14.40%	100.00%	7.76%	14.40%	52.69%	4.09%	7.59%	100.00%	7.76%	14.41%	91.40%	7.10%	13.17%
0.90	100.00%	7.33%	13.66%	100.00%	7.33%	13.66%	58.06%	4.26%	7.93%	100.00%	7.33%	13.66%	95.70%	7.01%	13.07%
0.95	100.00%	6.95%	13.00%	100.00%	6.95%	13.00%	81.72%	5.67%	10.61%	100.00%	6.94%	12.98%	98.92%	6.87%	12.84%
1.00	100.00%	6.60%	12.38%	100.00%	6.60%	12.38%	100.00%	6.60%	12.38%	100.00%	6.60%	12.38%	100.00%	6.60%	12.38%

EasyClinic (UC–CC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	11.54%	50.00%	18.75%	11.54%	50.00%	18.75%	0.00%	0.00%	0.00%	3.85%	16.67%	6.25%	11.54%	50.00%	18.75%
0.02	15.38%	33.33%	21.05%	19.23%	41.67%	26.32%	0.00%	0.00%	0.00%	26.92%	58.33%	36.84%	15.38%	33.33%	21.05%
0.03	15.38%	22.22%	18.18%	30.77%	44.44%	36.36%	0.00%	0.00%	0.00%	34.62%	50.00%	40.91%	23.08%	33.33%	27.27%
0.04	30.77%	33.33%	32.00%	34.62%	37.50%	36.00%	0.00%	0.00%	0.00%	38.46%	41.67%	40.00%	30.77%	33.33%	32.00%
0.05	50.00%	43.33%	46.43%	46.15%	40.00%	42.86%	0.00%	0.00%	0.00%	42.31%	36.67%	39.29%	34.62%	30.00%	32.14%
0.06	57.69%	41.67%	48.39%	50.00%	36.11%	41.93%	0.00%	0.00%	0.00%	42.31%	30.56%	35.48%	38.46%	27.78%	32.26%
0.07	57.69%	35.71%	44.11%	65.38%	40.48%	50.00%	3.85%	2.38%	2.94%	57.69%	35.71%	44.12%	38.46%	23.81%	29.41%
0.08	65.38%	35.42%	45.95%	73.08%	39.58%	51.35%	3.85%	2.08%	2.70%	50.00%	27.08%	35.14%	46.15%	25.00%	32.43%
0.09	69.23%	33.33%	45.00%	80.77%	38.89%	52.50%	7.69%	3.70%	5.00%	57.69%	27.78%	37.50%	46.15%	22.22%	30.00%
0.10	69.23%	30.00%	41.86%	80.77%	35.00%	48.84%	0.00%	0.00%	0.00%	61.54%	26.67%	37.21%	50.00%	21.67%	30.23%
0.11	76.92%	30.30%	43.47%	80.77%	31.82%	45.65%	0.00%	0.00%	0.00%	61.54%	24.24%	34.78%	50.00%	19.70%	28.26%
0.12	80.77%	29.17%	42.86%	84.62%	30.56%	44.90%	3.85%	1.39%	2.04%	65.38%	23.61%	34.69%	53.85%	19.44%	28.57%
0.13	84.62%	28.21%	42.31%	84.62%	28.21%	42.31%	11.54%	3.85%	5.77%	61.54%	20.51%	30.77%	53.85%	17.95%	26.92%
0.14	84.62%	26.19%	40.00%	84.62%	26.19%	40.00%	3.85%	1.19%	1.82%	69.23%	21.43%	32.73%	57.69%	17.86%	27.27%
0.15	84.62%	24.44%	37.93%	84.62%	24.44%	37.93%	3.85%	1.11%	1.72%	69.23%	20.00%	31.03%	57.69%	16.67%	25.86%
0.16	84.62%	22.92%	36.07%	88.46%	23.96%	37.71%	0.00%	0.00%	0.00%	73.08%	19.79%	31.15%	57.69%	15.62%	24.59%
0.17	84.62%	21.57%	34.38%	88.46%	22.55%	35.94%	11.54%	2.94%	4.69%	84.62%	21.57%	34.38%	57.69%	14.71%	23.44%
0.18	84.62%	20.37%	32.84%	88.46%	21.30%	34.33%	11.54%	2.78%	4.48%	76.92%	18.52%	29.85%	57.69%	13.89%	22.39%
0.19	84.62%	19.30%	31.43%	88.46%	20.18%	32.86%	0.00%	0.00%	0.00%	80.77%	18.42%	30.00%	57.69%	13.16%	21.43%
0.20	84.62%	18.33%	30.13%	88.46%	19.17%	31.51%	7.69%	1.67%	2.74%	84.62%	18.33%	30.14%	61.54%	13.33%	21.92%
0.25	92.31%	16.00%	27.27%	96.15%	16.67%	28.41%	3.85%	0.67%	1.14%	88.46%	15.33%	26.14%	69.23%	12.00%	20.45%
0.30	96.15%	13.89%	24.27%	96.15%	13.89%	24.27%	7.69%	1.11%	1.94%	96.15%	13.89%	24.27%	73.08%	10.56%	18.45%
0.35	96.15%	11.90%	21.18%	100.00%	12.38%	22.03%	0.00%	0.00%	0.00%	96.15%	11.90%	21.19%	76.92%	9.52%	16.95%
0.40	100.00%	10.83%	19.54%	100.00%	10.83%	19.54%	15.38%	1.67%	3.01%	100.00%	10.83%	19.55%	80.77%	8.75%	15.79%
0.45	100.00%	9.63%	17.57%	100.00%	9.63%	17.57%	7.69%	0.74%	1.35%	100.00%	9.63%	17.57%	80.77%	7.78%	14.19%
0.50	100.00%	8.67%	15.96%	100.00%	8.67%	15.96%	3.85%	0.33%	0.61%	100.00%	8.67%	15.95%	84.62%	7.33%	13.50%
0.55	100.00%	7.88%	14.61%	100.00%	7.88%	14.61%	19.23%	1.52%	2.81%	100.00%	7.88%	14.61%	88.46%	6.97%	12.92%
0.60	100.00%	7.22%	13.47%	100.00%	7.22%	13.47%	19.23%	1.39%	2.59%	100.00%	7.22%	13.47%	88.46%	6.39%	11.92%
0.65	100.00%	6.67%	12.51%	100.00%	6.67%	12.51%	19.23%	1.28%	2.40%	100.00%	6.67%	12.50%	88.46%	5.90%	11.06%
0.70	100.00%	6.19%	11.66%	100.00%	6.19%	11.66%	26.92%	1.67%	3.14%	100.00%	6.19%	11.66%	92.31%	5.71%	10.76%
0.75	100.00%	5.78%	10.93%	100.00%	5.78%	10.93%	19.23%	1.11%	2.10%	100.00%	5.78%	10.92%	92.31%	5.33%	10.08%
0.80	100.00%	5.42%	10.28%	100.00%	5.42%	10.28%	30.77%	1.67%	3.16%	100.00%	5.42%	10.28%	96.15%	5.21%	9.88%
0.85	100.00%	5.10%	9.71%	100.00%	5.10%	9.71%	50.00%	2.55%	4.85%	100.00%	5.10%	9.70%	100.00%	5.10%	9.70%
0.90	100.00%	4.81%	9.18%	100.00%	4.81%	9.18%	50.00%	2.41%	4.59%	100.00%	4.81%	9.19%	100.00%	4.81%	9.19%
0.95	100.00%	4.56%	8.72%	100.00%	4.56%	8.72%	53.85%	2.46%	4.70%	100.00%	4.56%	8.72%	100.00%	4.56%	8.72%
1.00	100.00%	4.33%	8.30%	100.00%	4.33%	8.30%	100.00%	4.33%	8.31%	100.00%	4.33%	8.31%	100.00%	4.33%	8.31%

EasyClinic (UC–ID)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	15.87%	55.56%	24.69%	6.35%	22.22%	9.88%	0.00%	0.00%	0.00%	17.46%	57.89%	26.83%	6.35%	21.05%	9.76%
0.02	28.57%	48.65%	36.00%	23.81%	40.54%	30.00%	0.00%	0.00%	0.00%	28.57%	47.37%	35.64%	12.70%	21.05%	15.84%
0.03	36.51%	41.07%	38.66%	28.57%	32.14%	30.25%	0.00%	0.00%	0.00%	46.03%	50.88%	48.33%	12.70%	14.04%	13.33%
0.04	44.44%	37.33%	40.58%	33.33%	28.00%	30.43%	0.00%	0.00%	0.00%	57.14%	47.37%	51.80%	14.29%	11.84%	12.95%
0.05	49.21%	32.98%	39.49%	39.68%	26.60%	31.85%	0.00%	0.00%	0.00%	71.43%	47.87%	57.32%	14.29%	9.57%	11.46%
0.06	50.79%	28.32%	36.36%	46.03%	25.66%	32.95%	0.00%	0.00%	0.00%	77.78%	43.36%	55.68%	14.29%	7.96%	10.23%
0.07	55.56%	26.52%	35.90%	47.62%	22.73%	30.77%	0.00%	0.00%	0.00%	77.78%	37.12%	50.26%	14.29%	6.82%	9.23%
0.08	57.14%	23.84%	33.64%	52.38%	21.85%	30.84%	0.00%	0.00%	0.00%	87.30%	36.42%	51.40%	14.29%	5.96%	8.41%
0.09	60.32%	22.35%	32.62%	53.97%	20.00%	29.18%	0.00%	0.00%	0.00%	88.89%	32.94%	48.07%	17.46%	6.47%	9.44%
0.10	65.08%	21.69%	32.54%	55.56%	18.52%	27.78%	1.59%	0.53%	0.79%	95.24%	31.75%	47.62%	17.46%	5.82%	8.73%
0.11	68.25%	20.77%	31.85%	60.32%	18.36%	28.15%	1.59%	0.48%	0.74%	95.24%	28.85%	44.28%	17.46%	5.29%	8.12%
0.12	71.43%	19.91%	31.14%	60.32%	16.81%	26.29%	1.59%	0.44%	0.69%	98.41%	27.31%	42.76%	17.46%	4.85%	7.59%
0.13	73.02%	18.78%	29.88%	63.49%	16.33%	25.98%	0.00%	0.00%	0.00%	98.41%	25.20%	40.13%	19.05%	4.88%	7.77%
0.14	73.02%	17.42%	28.13%	65.08%	15.53%	25.08%	0.00%	0.00%	0.00%	98.41%	23.40%	37.80%	19.05%	4.53%	7.32%
0.15	79.37%	17.67%	28.90%	65.08%	14.49%	23.70%	0.00%	0.00%	0.00%	100.00%	22.18%	36.31%	19.05%	4.23%	6.92%
0.16	79.37%	16.56%	27.40%	69.84%	14.57%	24.11%	0.00%	0.00%	0.00%	100.00%	20.86%	34.52%	19.05%	3.97%	6.58%
0.17	80.95%	15.89%	26.57%	73.02%	14.33%	23.96%	0.00%	0.00%	0.00%	100.00%	19.63%	32.81%	19.05%	3.74%	6.25%
0.18	80.95%	15.00%	25.31%	73.02%	13.53%	22.83%	0.00%	0.00%	0.00%	100.00%	18.53%	31.27%	20.63%	3.82%	6.45%
0.19	82.54%	14.48%	24.64%	73.02%	12.81%	21.80%	0.00%	0.00%	0.00%	100.00%	17.55%	29.86%	20.63%	3.62%	6.16%
0.20	84.13%	14.02%	24.03%	76.19%	12.70%	21.77%	0.00%	0.00%	0.00%	100.00%	16.67%	28.57%	20.63%	3.44%	5.90%
0.25	87.30%	11.65%	20.56%	84.13%	11.23%	19.82%	0.00%	0.00%	0.00%	100.00%	13.35%	23.55%	26.98%	3.60%	6.36%
0.30	88.89%	9.88%	17.78%	88.89%	9.88%	17.78%	0.00%	0.00%	0.00%	100.00%	11.11%	20.00%	30.16%	3.35%	6.03%
0.35	92.06%	8.77%	16.01%	90.48%	8.62%	15.74%	4.76%	0.45%	0.83%	100.00%	9.52%	17.38%	39.68%	3.78%	6.90%
0.40	96.83%	8.07%	14.90%	93.65%	7.80%	14.40%	17.46%	1.46%	2.69%	100.00%	8.33%	15.38%	52.38%	4.37%	8.06%
0.45	100.00%	7.41%	13.80%	100.00%	7.41%	13.80%	7.94%	0.59%	1.10%	100.00%	7.41%	13.80%	60.32%	4.47%	8.32%
0.50	100.00%	6.67%	12.51%	100.00%	6.67%	12.51%	1.59%	0.11%	0.20%	100.00%	6.67%	12.50%	71.43%	4.76%	8.93%
0.55	100.00%	6.06%	11.43%	100.00%	6.06%	11.43%	1.59%	0.10%	0.18%	100.00%	6.06%	11.42%	74.60%	4.52%	8.52%
0.60	100.00%	5.56%	10.53%	100.00%	5.56%	10.53%	15.87%	0.88%	1.67%	100.00%	5.56%	10.53%	74.60%	4.14%	7.85%
0.65	100.00%	5.13%	9.76%	100.00%	5.13%	9.76%	9.52%	0.49%	0.93%	100.00%	5.13%	9.76%	76.19%	3.91%	7.44%
0.70	100.00%	4.76%	9.09%	100.00%	4.76%	9.09%	4.76%	0.23%	0.43%	100.00%	4.76%	9.09%	80.95%	3.85%	7.36%
0.75	100.00%	4.45%	8.52%	100.00%	4.45%	8.52%	17.46%	0.78%	1.49%	100.00%	4.44%	8.51%	80.95%	3.60%	6.89%
0.80	100.00%	4.17%	8.01%	100.00%	4.17%	8.01%	28.57%	1.19%	2.29%	100.00%	4.17%	8.00%	80.95%	3.37%	6.48%
0.85	100.00%	3.92%	7.54%	100.00%	3.92%	7.54%	9.52%	0.37%	0.72%	100.00%	3.92%	7.55%	82.54%	3.24%	6.23%
0.90	100.00%	3.70%	7.14%	100.00%	3.70%	7.14%	68.25%	2.53%	4.88%	100.00%	3.70%	7.14%	82.54%	3.06%	5.90%
0.95	100.00%	3.51%	6.78%	100.00%	3.51%	6.78%	50.79%	1.78%	3.44%	100.00%	3.51%	6.78%	95.24%	3.34%	6.46%
1.00	100.00%	3.33%	6.45%	100.00%	3.33%	6.45%	100.00%	3.33%	6.45%	100.00%	3.33%	6.45%	100.00%	3.33%	6.45%

EasyClinic (UC–TC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.02	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.39%	11.11%	2.47%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.03	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.39%	7.41%	2.34%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
0.04	4.17%	16.67%	6.67%	4.17%	16.67%	6.67%	4.17%	16.67%	6.67%	4.17%	16.67%	6.67%	4.17%	16.67%	6.67%
0.05	10.42%	33.33%	15.88%	10.42%	33.33%	15.88%	3.47%	11.11%	5.29%	10.42%	33.33%	15.87%	10.42%	33.33%	15.87%
0.06	15.28%	40.74%	22.22%	15.28%	40.74%	22.22%	2.08%	5.56%	3.03%	15.28%	40.74%	22.22%	15.28%	40.74%	22.22%
0.07	18.75%	42.86%	26.09%	18.75%	42.86%	26.09%	5.56%	12.70%	7.73%	21.53%	49.21%	29.95%	21.53%	49.21%	29.95%
0.08	22.22%	44.44%	29.63%	22.22%	44.44%	29.63%	11.11%	22.22%	14.81%	25.00%	50.00%	33.33%	26.39%	52.78%	35.19%
0.09	27.08%	48.15%	34.66%	27.08%	48.15%	34.66%	0.69%	1.23%	0.89%	31.25%	55.56%	40.00%	30.56%	54.32%	39.11%
0.10	30.56%	48.89%	37.61%	30.56%	48.89%	37.61%	10.42%	16.67%	12.82%	37.50%	60.00%	46.15%	33.33%	53.33%	41.03%
0.11	34.72%	50.51%	41.15%	34.72%	50.51%	41.15%	13.19%	19.19%	15.64%	40.97%	59.60%	48.56%	36.11%	52.53%	42.80%
0.12	39.58%	52.78%	45.24%	39.58%	52.78%	45.24%	10.42%	13.89%	11.90%	45.14%	60.19%	51.59%	39.58%	52.78%	45.24%
0.13	44.44%	54.70%	49.04%	44.44%	54.70%	49.04%	10.42%	12.82%	11.49%	50.00%	61.54%	55.17%	41.67%	51.28%	45.98%
0.14	49.31%	56.35%	52.60%	49.31%	56.35%	52.60%	10.42%	11.90%	11.11%	52.78%	60.32%	56.30%	47.92%	54.76%	51.11%
0.15	52.08%	55.56%	53.76%	52.08%	55.56%	53.76%	9.72%	10.37%	10.04%	56.25%	60.00%	58.06%	52.08%	55.56%	53.76%
0.16	57.64%	57.64%	57.64%	57.64%	57.64%	57.64%	20.14%	20.14%	20.14%	61.11%	61.11%	61.11%	54.17%	54.17%	54.17%
0.17	61.11%	57.52%	59.26%	61.11%	57.52%	59.26%	11.11%	10.46%	10.77%	61.81%	58.17%	59.93%	54.86%	51.63%	53.20%
0.18	63.19%	56.17%	59.47%	63.19%	56.17%	59.47%	21.53%	19.14%	20.26%	63.89%	56.79%	60.13%	55.56%	49.38%	52.29%
0.19	64.58%	54.39%	59.05%	64.58%	54.39%	59.05%	14.58%	12.28%	13.33%	65.97%	55.56%	60.32%	55.56%	46.78%	50.79%
0.20	66.67%	53.33%	59.26%	66.67%	53.33%	59.26%	18.75%	15.00%	16.67%	65.97%	52.78%	58.64%	56.94%	45.56%	50.62%
0.25	77.08%	49.33%	60.16%	77.08%	49.33%	60.16%	25.69%	16.44%	20.05%	72.22%	46.22%	56.37%	63.89%	40.89%	49.86%
0.30	84.03%	44.81%	58.45%	84.03%	44.81%	58.45%	36.11%	19.26%	25.12%	79.86%	42.59%	55.56%	66.67%	35.56%	46.38%
0.35	93.06%	42.54%	58.39%	93.06%	42.54%	58.39%	33.33%	15.24%	20.92%	86.81%	39.68%	54.47%	68.75%	31.43%	43.14%
0.40	95.83%	38.33%	54.76%	95.83%	38.33%	54.76%	42.36%	16.94%	24.21%	90.28%	36.11%	51.59%	75.00%	30.00%	42.86%
0.45	95.83%	34.07%	50.27%	95.83%	34.07%	50.27%	31.94%	11.36%	16.76%	91.67%	32.59%	48.09%	76.39%	27.16%	40.07%
0.50	97.22%	31.11%	47.14%	97.22%	31.11%	47.14%	40.97%	13.11%	19.87%	93.75%	30.00%	45.45%	79.17%	25.33%	38.38%
0.55	100.00%	29.09%	45.07%	100.00%	29.09%	45.07%	38.89%	11.31%	17.53%	95.14%	27.68%	42.88%	82.64%	24.04%	37.25%
0.60	100.00%	26.67%	42.11%	100.00%	26.67%	42.11%	30.56%	8.15%	12.87%	97.22%	25.93%	40.94%	85.42%	22.78%	35.96%
0.65	100.00%	24.62%	39.51%	100.00%	24.62%	39.51%	55.56%	13.68%	21.95%	99.31%	24.44%	39.23%	87.50%	21.54%	34.57%
0.70	100.00%	22.86%	37.21%	100.00%	22.86%	37.21%	54.17%	12.38%	20.16%	100.00%	22.86%	37.21%	87.50%	20.00%	32.56%
0.75	100.00%	21.33%	35.16%	100.00%	21.33%	35.16%	52.78%	11.26%	18.56%	100.00%	21.33%	35.16%	88.89%	18.96%	31.26%
0.80	100.00%	20.00%	33.33%	100.00%	20.00%	33.33%	56.25%	11.25%	18.75%	100.00%	20.00%	33.33%	89.58%	17.92%	29.86%
0.85	100.00%	18.82%	31.68%	100.00%	18.82%	31.68%	65.28%	12.29%	20.68%	100.00%	18.82%	31.68%	90.97%	17.12%	28.82%
0.90	100.00%	17.78%	30.19%	100.00%	17.78%	30.19%	72.22%	12.84%	21.80%	100.00%	17.78%	30.19%	94.44%	16.79%	28.51%
0.95	100.00%	16.84%	28.83%	100.00%	16.84%	28.83%	89.58%	15.09%	25.83%	100.00%	16.84%	28.83%	95.83%	16.14%	27.63%
1.00	100.00%	16.00%	27.59%	100.00%	16.00%	27.59%	100.00%	16.00%	27.59%	100.00%	16.00%	27.59%	100.00%	16.00%	27.59%

EasyClinic (UC–UC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	22.22%	90.91%	35.71%	17.78%	72.73%	28.57%	0.00%	0.00%	0.00%	15.56%	58.33%	24.56%	11.11%	41.67%	17.54%
0.02	26.67%	52.17%	35.30%	28.89%	56.52%	38.24%	0.00%	0.00%	0.00%	24.44%	47.83%	32.35%	11.11%	21.74%	14.71%
0.03	31.11%	41.18%	35.44%	37.78%	50.00%	43.04%	0.00%	0.00%	0.00%	33.33%	42.86%	37.50%	13.33%	17.14%	15.00%
0.04	35.56%	34.78%	35.17%	40.00%	39.13%	39.56%	4.44%	4.26%	4.35%	33.33%	31.91%	32.61%	13.33%	12.77%	13.04%
0.05	37.78%	29.31%	33.01%	40.00%	31.03%	34.95%	2.22%	1.72%	1.94%	35.56%	27.59%	31.07%	17.78%	13.79%	15.53%
0.06	40.00%	26.09%	31.58%	42.22%	27.54%	33.34%	4.44%	2.86%	3.48%	35.56%	22.86%	27.83%	17.78%	11.43%	13.91%
0.07	42.22%	23.46%	30.16%	46.67%	25.93%	33.34%	2.22%	1.22%	1.57%	42.22%	23.17%	29.92%	17.78%	9.76%	12.60%
0.08	46.67%	22.58%	30.43%	48.89%	23.66%	31.89%	2.22%	1.08%	1.45%	44.44%	21.51%	28.99%	17.78%	8.60%	11.59%
0.09	51.11%	22.12%	30.88%	53.33%	23.08%	32.22%	2.22%	0.95%	1.33%	48.89%	20.95%	29.33%	20.00%	8.57%	12.00%
0.10	53.33%	20.69%	29.81%	53.33%	20.69%	29.81%	2.22%	0.85%	1.23%	46.67%	17.95%	25.93%	20.00%	7.69%	11.11%
0.11	57.78%	20.31%	30.06%	57.78%	20.31%	30.06%	2.22%	0.78%	1.16%	48.89%	17.19%	25.43%	20.00%	7.03%	10.40%
0.12	60.00%	19.42%	29.34%	60.00%	19.42%	29.34%	2.22%	0.71%	1.08%	53.33%	17.14%	25.95%	20.00%	6.43%	9.73%
0.13	62.22%	18.54%	28.57%	62.22%	18.54%	28.57%	2.22%	0.66%	1.02%	60.00%	17.76%	27.41%	20.00%	5.92%	9.14%
0.14	64.44%	17.79%	27.88%	64.44%	17.79%	27.88%	2.22%	0.61%	0.96%	64.44%	17.79%	27.88%	20.00%	5.52%	8.65%
0.15	64.44%	16.67%	26.49%	64.44%	16.67%	26.49%	4.44%	1.14%	1.82%	64.44%	16.57%	26.36%	20.00%	5.14%	8.18%
0.16	68.89%	16.67%	26.84%	68.89%	16.67%	26.84%	6.67%	1.60%	2.59%	55.56%	13.37%	21.55%	20.00%	4.81%	7.76%
0.17	68.89%	15.66%	25.52%	71.11%	16.16%	26.34%	4.44%	1.01%	1.65%	55.56%	12.63%	20.58%	22.22%	5.05%	8.23%
0.18	73.33%	15.79%	25.98%	75.56%	16.27%	26.77%	2.22%	0.48%	0.78%	57.78%	12.38%	20.39%	22.22%	4.76%	7.84%
0.19	75.56%	15.38%	25.56%	82.22%	16.74%	27.82%	6.67%	1.35%	2.25%	66.67%	13.51%	22.47%	26.67%	5.41%	8.99%
0.20	80.00%	15.45%	25.90%	84.44%	16.31%	27.34%	6.67%	1.29%	2.16%	68.89%	13.30%	22.30%	26.67%	5.15%	8.63%
0.25	84.44%	13.06%	22.62%	86.67%	13.40%	23.21%	8.89%	1.37%	2.37%	68.89%	10.62%	18.40%	28.89%	4.45%	7.72%
0.30	86.67%	11.17%	19.79%	88.89%	11.46%	20.30%	11.11%	1.43%	2.53%	77.78%	10.00%	17.72%	31.11%	4.00%	7.09%
0.35	91.11%	10.05%	18.10%	91.11%	10.05%	18.10%	13.33%	1.47%	2.65%	80.00%	8.82%	15.89%	33.33%	3.68%	6.62%
0.40	95.56%	9.23%	16.83%	95.56%	9.23%	16.83%	17.78%	1.72%	3.13%	91.11%	8.80%	16.05%	37.78%	3.65%	6.65%
0.45	95.56%	8.21%	15.12%	95.56%	8.21%	15.12%	17.78%	1.52%	2.81%	93.33%	8.00%	14.74%	44.44%	3.81%	7.02%
0.50	97.78%	7.55%	14.02%	97.78%	7.55%	14.02%	11.11%	0.86%	1.59%	88.89%	6.86%	12.74%	46.67%	3.60%	6.69%
0.55	97.78%	6.86%	12.82%	97.78%	6.86%	12.82%	20.00%	1.40%	2.62%	91.11%	6.40%	11.95%	53.33%	3.74%	7.00%
0.60	97.78%	6.29%	11.82%	97.78%	6.29%	11.82%	17.78%	1.14%	2.15%	93.33%	6.00%	11.28%	55.56%	3.57%	6.71%
0.65	97.78%	5.81%	10.97%	97.78%	5.81%	10.97%	11.11%	0.66%	1.25%	95.56%	5.67%	10.71%	57.78%	3.43%	6.48%
0.70	100.00%	5.51%	10.44%	97.78%	5.39%	10.22%	26.67%	1.47%	2.79%	95.56%	5.27%	9.99%	71.11%	3.92%	7.43%
0.75	100.00%	5.15%	9.80%	97.78%	5.03%	9.57%	24.44%	1.26%	2.39%	97.78%	5.03%	9.58%	75.56%	3.89%	7.40%
0.80	100.00%	4.83%	9.21%	100.00%	4.83%	9.21%	42.22%	2.04%	3.89%	100.00%	4.82%	9.20%	82.22%	3.97%	7.57%
0.85	100.00%	4.54%	8.69%	100.00%	4.54%	8.69%	42.22%	1.92%	3.67%	100.00%	4.54%	8.69%	86.67%	3.94%	7.53%
0.90	100.00%	4.29%	8.23%	100.00%	4.29%	8.23%	57.78%	2.48%	4.75%	100.00%	4.29%	8.23%	88.89%	3.81%	7.31%
0.95	100.00%	4.07%	7.82%	100.00%	4.07%	7.82%	66.67%	2.71%	5.20%	100.00%	4.06%	7.81%	93.33%	3.79%	7.29%
1.00	100.00%	3.86%	7.43%	100.00%	3.86%	7.43%	100.00%	3.86%	7.43%	100.00%	3.86%	7.43%	100.00%	3.86%	7.43%

CM-1

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	13.73%	70.00%	22.96%	17.65%	90.00%	29.51%	0.00%	0.00%	0.00%	13.73%	70.00%	22.95%	15.69%	80.00%	26.23%
0.02	27.45%	70.00%	39.44%	29.41%	75.00%	42.25%	3.92%	10.00%	5.63%	15.69%	40.00%	22.54%	23.53%	60.00%	33.80%
0.03	37.25%	63.33%	46.91%	43.14%	73.33%	54.32%	3.92%	6.45%	4.88%	27.45%	45.16%	34.15%	33.33%	54.84%	41.46%
0.04	45.10%	56.10%	50.00%	50.98%	63.41%	56.52%	5.88%	7.32%	6.52%	29.41%	36.59%	32.61%	35.29%	43.90%	39.13%
0.05	50.98%	50.98%	50.98%	60.78%	60.78%	60.78%	3.92%	3.92%	3.92%	37.25%	37.25%	37.25%	41.18%	41.18%	41.18%
0.06	56.86%	47.54%	51.78%	64.71%	54.10%	58.93%	5.88%	4.84%	5.31%	47.06%	38.71%	42.48%	43.14%	35.48%	38.94%
0.07	60.78%	43.66%	50.82%	66.67%	47.89%	55.74%	3.92%	2.78%	3.25%	50.98%	36.11%	42.28%	45.10%	31.94%	37.40%
0.08	60.78%	37.80%	46.61%	66.67%	41.46%	51.13%	7.84%	4.88%	6.02%	50.98%	31.71%	39.10%	45.10%	28.05%	34.59%
0.09	64.71%	35.87%	46.16%	68.63%	38.04%	48.95%	5.88%	3.26%	4.20%	54.90%	30.43%	39.16%	45.10%	25.00%	32.17%
0.10	68.63%	34.31%	45.75%	70.59%	35.29%	47.06%	5.88%	2.94%	3.92%	56.86%	28.43%	37.91%	45.10%	22.55%	30.07%
0.11	74.51%	33.93%	46.63%	74.51%	33.93%	46.63%	7.84%	3.54%	4.88%	58.82%	26.55%	36.59%	47.06%	21.24%	29.27%
0.12	74.51%	30.89%	43.67%	74.51%	30.89%	43.67%	7.84%	3.25%	4.60%	62.75%	26.02%	36.78%	47.06%	19.51%	27.59%
0.13	78.43%	30.08%	43.48%	74.51%	28.57%	41.30%	5.88%	2.26%	3.26%	68.63%	26.32%	38.04%	47.06%	18.05%	26.09%
0.14	78.43%	27.97%	41.23%	74.51%	26.57%	39.17%	7.84%	2.78%	4.10%	74.51%	26.39%	38.97%	47.06%	16.67%	24.62%
0.15	82.35%	27.45%	41.18%	76.47%	25.49%	38.24%	7.84%	2.60%	3.90%	74.51%	24.68%	37.07%	47.06%	15.58%	23.41%
0.16	82.35%	25.61%	39.07%	80.39%	25.00%	38.14%	7.84%	2.44%	3.72%	76.47%	23.78%	36.28%	47.06%	14.63%	22.33%
0.17	82.35%	24.14%	37.34%	84.31%	24.71%	38.22%	5.88%	1.72%	2.67%	76.47%	22.41%	34.67%	47.06%	13.79%	21.33%
0.18	82.35%	22.83%	35.75%	86.27%	23.91%	37.44%	5.88%	1.63%	2.55%	74.51%	20.65%	32.34%	47.06%	13.04%	20.43%
0.19	82.35%	21.65%	34.29%	86.27%	22.68%	35.92%	5.88%	1.54%	2.44%	74.51%	19.49%	30.89%	47.06%	12.31%	19.51%
0.20	82.35%	20.49%	32.82%	86.27%	21.46%	34.37%	5.88%	1.46%	2.34%	76.47%	19.02%	30.47%	47.06%	11.71%	18.75%
0.25	88.24%	17.58%	29.32%	92.16%	18.36%	30.62%	9.80%	1.95%	3.26%	80.39%	16.02%	26.71%	52.94%	10.55%	17.59%
0.30	88.24%	14.66%	25.14%	94.12%	15.64%	26.82%	9.80%	1.62%	2.79%	82.35%	13.64%	23.40%	56.86%	9.42%	16.16%
0.35	96.08%	13.69%	23.97%	98.04%	13.97%	24.46%	13.73%	1.95%	3.41%	82.35%	11.70%	20.49%	64.71%	9.19%	16.10%
0.40	96.08%	11.95%	21.26%	98.04%	12.20%	21.70%	13.73%	1.71%	3.04%	92.16%	11.46%	20.39%	68.63%	8.54%	15.18%
0.45	96.08%	10.63%	19.14%	98.04%	10.85%	19.54%	15.69%	1.74%	3.12%	96.08%	10.63%	19.14%	68.63%	7.59%	13.67%
0.50	98.04%	9.77%	17.77%	98.04%	9.77%	17.77%	15.69%	1.56%	2.84%	98.04%	9.77%	17.76%	70.59%	7.03%	12.79%
0.55	98.04%	8.88%	16.28%	98.04%	8.88%	16.28%	17.65%	1.60%	2.93%	98.04%	8.87%	16.26%	76.47%	6.91%	12.68%
0.60	98.04%	8.13%	15.01%	98.04%	8.13%	15.01%	17.65%	1.46%	2.70%	98.04%	8.13%	15.02%	78.43%	6.50%	12.01%
0.65	98.04%	7.51%	13.95%	100.00%	7.66%	14.23%	15.69%	1.20%	2.23%	100.00%	7.66%	14.23%	78.43%	6.01%	11.16%
0.70	100.00%	7.11%	13.28%	100.00%	7.11%	13.28%	21.57%	1.53%	2.86%	100.00%	7.10%	13.26%	82.35%	5.85%	10.92%
0.75	100.00%	6.64%	12.45%	100.00%	6.64%	12.45%	25.49%	1.69%	3.17%	100.00%	6.63%	12.44%	90.20%	5.98%	11.22%
0.80	100.00%	6.22%	11.71%	100.00%	6.22%	11.71%	35.29%	2.20%	4.13%	100.00%	6.22%	11.71%	90.20%	5.61%	10.56%
0.85	100.00%	5.86%	11.07%	100.00%	5.86%	11.07%	25.49%	1.49%	2.82%	100.00%	5.86%	11.06%	92.16%	5.40%	10.20%
0.90	100.00%	5.53%	10.48%	100.00%	5.53%	10.48%	52.94%	2.93%	5.55%	100.00%	5.53%	10.48%	94.12%	5.21%	9.87%
0.95	100.00%	5.24%	9.96%	100.00%	5.24%	9.96%	58.82%	3.08%	5.85%	100.00%	5.24%	9.95%	98.04%	5.13%	9.76%
1.00	100.00%	4.98%	9.49%	100.00%	4.98%	9.49%	100.00%	4.98%	9.48%	100.00%	4.98%	9.48%	100.00%	4.98%	9.48%

EBT subset (RE–TC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	24.36%	50.00%	32.76%	25.64%	52.63%	34.48%	0.00%	0.00%	0.00%	19.23%	40.54%	26.09%	26.92%	56.76%	36.52%
0.02	33.33%	34.21%	33.76%	34.62%	35.53%	35.07%	0.00%	0.00%	0.00%	34.62%	36.00%	35.29%	37.18%	38.67%	37.91%
0.03	46.15%	31.58%	37.50%	43.59%	29.82%	35.41%	1.28%	0.89%	1.05%	42.31%	29.46%	34.74%	42.31%	29.46%	34.74%
0.04	56.41%	28.76%	38.10%	56.41%	28.76%	38.10%	1.28%	0.67%	0.88%	50.00%	26.00%	34.21%	48.72%	25.33%	33.33%
0.05	60.26%	24.61%	34.95%	58.97%	24.08%	34.20%	1.28%	0.53%	0.75%	56.41%	23.53%	33.21%	52.56%	21.93%	30.94%
0.06	66.67%	22.71%	33.88%	62.82%	21.40%	31.92%	1.28%	0.45%	0.66%	60.26%	20.98%	31.13%	55.13%	19.20%	28.48%
0.07	71.79%	20.97%	32.46%	65.38%	19.10%	29.56%	2.56%	0.76%	1.18%	61.54%	18.32%	28.24%	56.41%	16.79%	25.88%
0.08	73.08%	18.63%	29.69%	74.36%	18.95%	30.20%	2.56%	0.67%	1.06%	62.82%	16.39%	25.99%	57.69%	15.05%	23.87%
0.09	76.92%	17.44%	28.43%	78.21%	17.73%	28.91%	2.56%	0.60%	0.97%	65.38%	15.18%	24.64%	61.54%	14.29%	23.19%
0.10	79.49%	16.23%	26.96%	80.77%	16.49%	27.39%	5.13%	1.07%	1.77%	70.51%	14.71%	24.34%	61.54%	12.83%	21.24%
0.11	80.77%	15.00%	25.30%	83.33%	15.48%	26.11%	3.85%	0.73%	1.23%	75.64%	14.36%	24.13%	61.54%	11.68%	19.63%
0.12	85.90%	14.60%	24.96%	84.62%	14.38%	24.58%	3.85%	0.67%	1.14%	80.77%	14.03%	23.91%	61.54%	10.69%	18.22%
0.13	85.90%	13.48%	23.30%	87.18%	13.68%	23.65%	1.28%	0.21%	0.35%	82.05%	13.17%	22.70%	61.54%	9.88%	17.02%
0.14	85.90%	12.52%	21.85%	88.46%	12.90%	22.52%	3.85%	0.57%	1.00%	75.64%	11.28%	19.63%	61.54%	9.18%	15.97%
0.15	87.18%	11.85%	20.86%	88.46%	12.02%	21.16%	5.13%	0.71%	1.25%	76.92%	10.70%	18.78%	61.54%	8.56%	15.02%
0.16	88.46%	11.27%	19.99%	89.74%	11.44%	20.29%	5.13%	0.67%	1.18%	78.21%	10.20%	18.05%	62.82%	8.19%	14.50%
0.17	89.74%	10.77%	19.23%	89.74%	10.77%	19.23%	5.13%	0.63%	1.12%	78.21%	9.61%	17.11%	62.82%	7.72%	13.74%
0.18	89.74%	10.17%	18.27%	91.03%	10.32%	18.54%	6.41%	0.74%	1.33%	79.49%	9.21%	16.51%	62.82%	7.28%	13.05%
0.19	91.03%	9.77%	17.65%	91.03%	9.77%	17.65%	6.41%	0.70%	1.27%	80.77%	8.87%	15.99%	62.82%	6.90%	12.44%
0.20	91.03%	9.28%	16.84%	91.03%	9.28%	16.84%	5.13%	0.53%	0.97%	82.05%	8.56%	15.50%	62.82%	6.55%	11.86%
0.25	93.59%	7.64%	14.13%	93.59%	7.64%	14.13%	10.26%	0.86%	1.58%	87.18%	7.28%	13.44%	64.10%	5.35%	9.88%
0.30	96.15%	6.53%	12.23%	96.15%	6.53%	12.23%	11.54%	0.80%	1.50%	88.46%	6.16%	11.51%	66.67%	4.64%	8.67%
0.35	96.15%	5.60%	10.58%	96.15%	5.60%	10.58%	14.10%	0.84%	1.59%	91.03%	5.43%	10.25%	70.51%	4.20%	7.94%
0.40	96.15%	4.90%	9.32%	96.15%	4.90%	9.32%	14.10%	0.74%	1.40%	93.59%	4.88%	9.28%	73.08%	3.81%	7.25%
0.45	97.44%	4.41%	8.44%	97.44%	4.41%	8.44%	16.67%	0.77%	1.48%	94.87%	4.40%	8.41%	75.64%	3.51%	6.70%
0.50	98.72%	4.03%	7.74%	97.44%	3.97%	7.63%	20.51%	0.86%	1.64%	96.15%	4.01%	7.70%	75.64%	3.16%	6.06%
0.55	100.00%	3.71%	7.15%	97.44%	3.61%	6.96%	26.92%	1.02%	1.97%	96.15%	3.65%	7.03%	78.21%	2.97%	5.72%
0.60	100.00%	3.40%	6.58%	97.44%	3.31%	6.40%	16.67%	0.58%	1.12%	96.15%	3.34%	6.46%	79.49%	2.76%	5.34%
0.65	100.00%	3.14%	6.09%	97.44%	3.06%	5.93%	19.23%	0.62%	1.20%	98.72%	3.17%	6.14%	82.05%	2.63%	5.10%
0.70	100.00%	2.91%	5.66%	100.00%	2.91%	5.66%	20.51%	0.61%	1.19%	98.72%	2.94%	5.71%	82.05%	2.45%	4.75%
0.75	100.00%	2.72%	5.30%	100.00%	2.72%	5.30%	24.36%	0.68%	1.32%	100.00%	2.78%	5.41%	88.46%	2.46%	4.79%
0.80	100.00%	2.55%	4.97%	100.00%	2.55%	4.97%	20.51%	0.54%	1.04%	100.00%	2.61%	5.08%	93.59%	2.44%	4.76%
0.85	100.00%	2.40%	4.69%	100.00%	2.40%	4.69%	23.08%	0.57%	1.11%	100.00%	2.46%	4.79%	97.44%	2.39%	4.67%
0.90	100.00%	2.26%	4.42%	100.00%	2.26%	4.42%	42.31%	0.98%	1.92%	100.00%	2.32%	4.53%	98.72%	2.29%	4.47%
0.95	100.00%	2.15%	4.21%	100.00%	2.15%	4.21%	53.85%	1.18%	2.31%	100.00%	2.20%	4.30%	100.00%	2.20%	4.30%
1.00	100.00%	2.04%	4.00%	100.00%	2.04%	4.00%	100.00%	2.09%	4.09%	100.00%	2.09%	4.09%	100.00%	2.09%	4.09%

EBT subset (RE–TC)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	24.14%	77.78%	36.84%	24.14%	77.78%	36.84%	0.00%	0.00%	0.00%	22.41%	68.42%	33.77%	18.97%	57.89%	28.57%
0.02	37.93%	59.46%	46.32%	37.93%	59.46%	46.32%	1.72%	2.70%	2.11%	37.93%	59.46%	46.32%	27.59%	43.24%	33.68%
0.03	43.10%	44.64%	43.86%	41.38%	42.86%	42.11%	0.00%	0.00%	0.00%	46.55%	48.21%	47.37%	34.48%	35.71%	35.09%
0.04	48.28%	37.84%	42.43%	46.55%	36.49%	40.91%	0.00%	0.00%	0.00%	58.62%	45.33%	51.13%	37.93%	29.33%	33.08%
0.05	51.72%	32.26%	39.74%	53.45%	33.33%	41.06%	0.00%	0.00%	0.00%	63.79%	39.78%	49.01%	41.38%	25.81%	31.79%
0.06	58.62%	30.36%	40.00%	60.34%	31.25%	41.18%	0.00%	0.00%	0.00%	63.79%	33.04%	43.53%	43.10%	22.32%	29.41%
0.07	63.79%	28.46%	39.36%	62.07%	27.69%	38.30%	3.45%	1.53%	2.12%	67.24%	29.77%	41.27%	46.55%	20.61%	28.57%
0.08	70.69%	27.52%	39.62%	65.52%	25.50%	36.71%	6.90%	2.67%	3.85%	72.41%	28.00%	40.38%	48.28%	18.67%	26.92%
0.09	72.41%	25.00%	37.17%	67.24%	23.21%	34.51%	6.90%	2.38%	3.54%	77.59%	26.79%	39.82%	50.00%	17.26%	25.66%
0.10	77.59%	24.19%	36.88%	67.24%	20.97%	31.97%	8.62%	2.67%	4.08%	77.59%	24.06%	36.73%	50.00%	15.51%	23.67%
0.11	79.31%	22.44%	34.98%	68.97%	19.51%	30.42%	8.62%	2.43%	3.79%	77.59%	21.84%	34.09%	50.00%	14.08%	21.97%
0.12	79.31%	20.54%	32.63%	74.14%	19.20%	30.50%	12.07%	3.12%	4.96%	77.59%	20.09%	31.91%	53.45%	13.84%	21.99%
0.13	82.76%	19.83%	31.99%	77.59%	18.60%	30.01%	12.07%	2.88%	4.65%	79.31%	18.93%	30.56%	53.45%	12.76%	20.60%
0.14	82.76%	18.39%	30.09%	82.76%	18.39%	30.09%	13.79%	3.05%	5.00%	81.03%	17.94%	29.38%	53.45%	11.83%	19.38%
0.15	84.48%	17.50%	28.99%	82.76%	17.14%	28.40%	12.07%	2.50%	4.14%	82.76%	17.14%	28.40%	53.45%	11.07%	18.34%
0.16	86.21%	16.72%	28.01%	87.93%	17.06%	28.58%	15.52%	3.01%	5.04%	84.48%	16.39%	27.45%	55.17%	10.70%	17.93%
0.17	87.93%	16.09%	27.20%	87.93%	16.09%	27.20%	17.24%	3.14%	5.32%	84.48%	15.41%	26.06%	55.17%	10.06%	17.02%
0.18	87.93%	15.18%	25.89%	87.93%	15.18%	25.89%	18.97%	3.27%	5.58%	84.48%	14.58%	24.87%	55.17%	9.52%	16.24%
0.19	87.93%	14.37%	24.70%	87.93%	14.37%	24.70%	20.69%	3.38%	5.81%	84.48%	13.80%	23.73%	55.17%	9.01%	15.50%
0.20	89.66%	13.94%	24.13%	87.93%	13.67%	23.66%	20.69%	3.21%	5.56%	86.21%	13.37%	23.15%	55.17%	8.56%	14.81%
0.25	89.66%	11.13%	19.80%	89.66%	11.13%	19.80%	20.69%	2.57%	4.57%	91.38%	11.35%	20.19%	55.17%	6.85%	12.19%
0.30	89.66%	9.29%	16.84%	89.66%	9.29%	16.84%	27.59%	2.85%	5.17%	91.38%	9.45%	17.12%	56.90%	5.88%	10.66%
0.35	89.66%	7.95%	14.60%	89.66%	7.95%	14.60%	29.31%	2.60%	4.78%	93.10%	8.26%	15.17%	58.62%	5.20%	9.55%
0.40	89.66%	6.96%	12.92%	89.66%	6.96%	12.92%	24.14%	1.87%	3.47%	94.83%	7.35%	13.65%	62.07%	4.81%	8.93%
0.45	89.66%	6.18%	11.56%	89.66%	6.18%	11.56%	29.31%	2.02%	3.78%	93.10%	6.42%	12.01%	63.79%	4.40%	8.23%
0.50	93.10%	5.78%	10.88%	89.66%	5.57%	10.49%	32.76%	2.03%	3.83%	93.10%	5.78%	10.89%	72.41%	4.50%	8.47%
0.55	96.55%	5.45%	10.32%	89.66%	5.06%	9.58%	36.21%	2.04%	3.87%	93.10%	5.25%	9.94%	75.86%	4.28%	8.10%
0.60	96.55%	5.00%	9.51%	89.66%	4.64%	8.82%	32.76%	1.69%	3.22%	94.83%	4.91%	9.33%	79.31%	4.10%	7.80%
0.65	96.55%	4.61%	8.80%	89.66%	4.28%	8.17%	37.93%	1.81%	3.46%	96.55%	4.61%	8.80%	82.76%	3.95%	7.54%
0.70	96.55%	4.28%	8.20%	91.38%	4.05%	7.76%	34.48%	1.53%	2.93%	100.00%	4.43%	8.49%	86.21%	3.82%	7.32%
0.75	100.00%	4.14%	7.95%	93.10%	3.85%	7.39%	36.21%	1.50%	2.88%	100.00%	4.14%	7.95%	86.21%	3.57%	6.85%
0.80	100.00%	3.88%	7.47%	93.10%	3.61%	6.95%	37.93%	1.47%	2.83%	100.00%	3.88%	7.47%	89.66%	3.48%	6.70%
0.85	100.00%	3.65%	7.04%	93.10%	3.46%	6.67%	44.83%	1.64%	3.16%	100.00%	3.65%	7.04%	93.10%	3.40%	6.56%
0.90	100.00%	3.45%	6.67%	96.55%	3.33%	6.44%	51.72%	1.78%	3.45%	100.00%	3.45%	6.67%	94.83%	3.27%	6.32%
0.95	100.00%	3.27%	6.33%	98.28%	3.21%	6.22%	55.17%	1.80%	3.49%	100.00%	3.27%	6.32%	98.28%	3.21%	6.22%
1.00	100.00%	3.10%	6.01%	100.00%	3.10%	6.01%	100.00%	3.10%	6.02%	100.00%	3.10%	6.02%	100.00%	3.10%	6.02%

WARC (FRS–SRS)

Selectivity	VSM			LSI			LDA			BTM			JS		
	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure	Recall	Precision	F-Measure
0.01	7.35%	41.67%	12.50%	11.76%	66.67%	19.99%	0.00%	0.00%	0.00%	4.41%	25.00%	7.50%	8.82%	50.00%	15.00%
0.02	10.29%	29.17%	15.21%	11.76%	33.33%	17.39%	0.00%	0.00%	0.00%	8.82%	26.09%	13.19%	16.18%	47.83%	24.18%
0.03	20.59%	37.84%	26.67%	22.06%	40.54%	28.57%	0.00%	0.00%	0.00%	16.18%	31.43%	21.36%	17.65%	34.29%	23.30%
0.04	27.94%	38.78%	32.48%	27.94%	38.78%	32.48%	1.47%	2.13%	1.74%	25.00%	36.17%	29.57%	20.59%	29.79%	24.35%
0.05	32.35%	35.48%	33.84%	35.29%	38.71%	36.92%	1.47%	1.69%	1.57%	27.94%	32.20%	29.92%	26.47%	30.51%	28.35%
0.06	42.65%	39.19%	40.85%	42.65%	39.19%	40.85%	1.47%	1.43%	1.45%	30.88%	30.00%	30.43%	29.41%	28.57%	28.99%
0.07	42.65%	33.72%	37.66%	42.65%	33.72%	37.66%	1.47%	1.22%	1.33%	38.24%	31.71%	34.67%	32.35%	26.83%	29.33%
0.08	45.59%	31.31%	37.12%	44.12%	30.30%	35.93%	2.94%	2.13%	2.47%	48.53%	35.11%	40.74%	32.35%	23.40%	27.16%
0.09	47.06%	28.83%	35.76%	47.06%	28.83%	35.76%	2.94%	1.89%	2.30%	50.00%	32.08%	39.08%	38.24%	24.53%	29.89%
0.10	48.53%	26.61%	34.37%	50.00%	27.42%	35.42%	2.94%	1.71%	2.16%	51.47%	29.91%	37.84%	41.18%	23.93%	30.27%
0.11	54.41%	27.21%	36.28%	52.94%	26.47%	35.29%	4.41%	2.33%	3.05%	52.94%	27.91%	36.55%	41.18%	21.71%	28.43%
0.12	57.35%	26.17%	35.94%	55.88%	25.50%	35.02%	2.94%	1.42%	1.91%	55.88%	26.95%	36.36%	41.18%	19.86%	26.79%
0.13	58.82%	24.84%	34.93%	57.35%	24.22%	34.06%	5.88%	2.63%	3.64%	63.24%	28.29%	39.09%	41.18%	18.42%	25.45%
0.14	60.29%	23.70%	34.02%	61.76%	24.28%	34.86%	5.88%	2.44%	3.45%	63.24%	26.22%	37.07%	41.18%	17.07%	24.14%
0.15	60.29%	22.04%	32.28%	64.71%	23.66%	34.65%	7.35%	2.84%	4.10%	66.18%	25.57%	36.89%	41.18%	15.91%	22.95%
0.16	66.18%	22.73%	33.84%	66.18%	22.73%	33.84%	5.88%	2.13%	3.12%	66.18%	23.94%	35.16%	42.65%	15.43%	22.66%
0.17	70.59%	22.75%	34.41%	70.59%	22.75%	34.41%	5.88%	2.01%	3.00%	70.59%	24.12%	35.96%	42.65%	14.57%	21.72%
0.18	70.59%	21.52%	32.98%	73.53%	22.42%	34.36%	4.41%	1.42%	2.15%	73.53%	23.70%	35.84%	44.12%	14.22%	21.51%
0.19	73.53%	21.28%	33.01%	73.53%	21.28%	33.01%	10.29%	3.14%	4.81%	73.53%	22.42%	34.36%	44.12%	13.45%	20.62%
0.20	75.00%	20.56%	32.27%	75.00%	20.56%	32.27%	7.35%	2.13%	3.30%	75.00%	21.70%	33.66%	44.12%	12.77%	19.80%
0.25	82.35%	18.06%	29.62%	82.35%	18.06%	29.62%	11.76%	2.73%	4.43%	83.82%	19.45%	31.58%	47.06%	10.92%	17.73%
0.30	88.24%	16.13%	27.27%	88.24%	16.13%	27.27%	14.71%	2.84%	4.76%	85.29%	16.48%	27.62%	48.53%	9.38%	15.71%
0.35	89.71%	14.06%	24.31%	89.71%	14.06%	24.31%	14.71%	2.43%	4.18%	91.18%	15.09%	25.89%	54.41%	9.00%	15.45%
0.40	91.18%	12.50%	21.99%	92.65%	12.70%	22.34%	19.12%	2.77%	4.84%	92.65%	13.43%	23.46%	55.88%	8.10%	14.15%
0.45	95.59%	11.65%	20.77%	95.59%	11.65%	20.77%	19.12%	2.46%	4.36%	94.12%	12.12%	21.48%	58.82%	7.58%	13.42%
0.50	97.06%	10.63%	19.16%	97.06%	10.63%	19.16%	22.06%	2.56%	4.59%	94.12%	10.92%	19.57%	61.76%	7.17%	12.84%
0.55	97.06%	9.66%	17.57%	97.06%	9.66%	17.57%	23.53%	2.48%	4.49%	94.12%	9.92%	17.95%	64.71%	6.82%	12.34%
0.60	97.06%	8.86%	16.24%	97.06%	8.86%	16.24%	26.47%	2.56%	4.66%	94.12%	9.09%	16.58%	73.53%	7.10%	12.95%
0.65	100.00%	8.43%	15.55%	98.53%	8.30%	15.31%	30.88%	2.76%	5.06%	97.06%	8.66%	15.90%	76.47%	6.82%	12.53%
0.70	100.00%	7.83%	14.52%	100.00%	7.83%	14.52%	33.82%	2.80%	5.17%	98.53%	8.16%	15.07%	80.88%	6.70%	12.37%
0.75	100.00%	7.30%	13.61%	100.00%	7.30%	13.61%	38.24%	2.95%	5.49%	100.00%	7.73%	14.35%	83.82%	6.48%	12.03%
0.80	100.00%	6.85%	12.82%	100.00%	6.85%	12.82%	33.82%	2.45%	4.57%	100.00%	7.25%	13.52%	85.29%	6.18%	11.53%
0.85	100.00%	6.45%	12.12%	100.00%	6.45%	12.12%	41.18%	2.81%	5.26%	100.00%	6.82%	12.77%	91.18%	6.22%	11.64%
0.90	100.00%	6.09%	11.48%	100.00%	6.09%	11.48%	50.00%	3.22%	6.05%	100.00%	6.44%	12.10%	95.59%	6.16%	11.57%
0.95	100.00%	5.77%	10.91%	100.00%	5.77%	10.91%	72.06%	4.40%	8.29%	100.00%	6.10%	11.51%	95.59%	5.83%	11.00%
1.00	100.00%	5.48%	10.39%	100.00%	5.48%	10.39%	100.00%	5.80%	10.96%	100.00%	5.80%	10.96%	100.00%	5.80%	10.96%

Data availability

The data that has been used is confidential.

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